



Illuminating
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TECHNICAL MEMORANDUM: MEASUREMENT UNCERTAINTY FOR LIGHTING EQUIPMENT CALIBRATION USING INTEGRATING SPHERES

AN AMERICAN NATIONAL STANDARD



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Publication of this Technical Memorandum
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by
The IES Testing Procedures Committee**



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Approved by the IES Standards Committee, February 23, 2018, as a Transaction of the Illuminating Engineering Society.

Approved February 25, 2020 as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN# 978-0-87995-397-3

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1.0 Introduction and Scope

1.1 Introduction

Uncertainty is inherent in every measurement. To truly understand the value of a measurement, the uncertainty of such should be analyzed. A critical step of this analysis is to know the uncertainty contributed by the instruments used to make the measurement. This Technical Memorandum provides a detailed and step-by-step method for determining uncertainty of certain calibration procedures common to instruments used in lumen measurement laboratories. This document uses the structure proposed in the *Guide to the Expression of Uncertainty in Measurement* (GUM) for its recommendations. For each calibration procedure presented in this document, a mathematical model is developed, estimates of each input quantity are discussed, and guidance is provided for the evaluation of the standard uncertainty of each input estimate. With this information, the combined standard uncertainty is calculated, effective degrees of freedom are calculated, and finally, the expanded uncertainty is reported.

The intent of this document is to provide a common approach to the uncertainty analysis for calibration of lumen measuring equipment (integrating spheres) with standard incandescent lamps, including halogen, that have been assigned values of total luminous flux and/or total spectral radiant flux. In addition, uncertainty budget templates are provided for calibration of incandescent (including halogen) working standard lamps that are virtually identical to the standard lamps they are compared against. These templates are starting points for uncertainty analysis. A laboratory applying these templates is expected to estimate each input quantity and the standard uncertainty of each input quantity, except where industry standard values are specifically noted in this document. Additional input quantities may be required based on specific situations that arise in the uniqueness of each laboratory measurement system. Future documents will cover topics such as uncertainty determination for colorimetric calibrations, and photometric and colorimetric measurements for testing general lighting products.

1.2 Scope

This document provides templates for the analysis of measurement uncertainty for the photometric calibration of integrating sphere systems using standard incandescent lamps, as well as the creation of working standard lamps using integrating sphere systems that are effectively identical to the primary standard lamps. It does not provide templates for the analysis of measurement uncertainty for colorimetric calibration of working standard lamps or for the testing of general lighting products.

2.0 Normative References

JCGM 100-2008, *Evaluation of Measurement Data – Guide to the Expression of Uncertainty in Measurement* (GUM). Bureau International des Poids et Mesures (BIPM); Sep 2008. Free download: www.bipm.org/en/publications/guides/gum.html.

ANSI/IES LS-1-20, *Nomenclature and Definitions for Illuminating Engineering*. New York: Illuminating Engineering Society; 2017. Free access from www.ies.org/standards/definitions/

3.0 Definitions

3.1 array spectrometer

In this document, an instrument that simultaneously detects and disperses light with respect to wavelength. The CIE S 017/E:2011 (ILV)¹ definition for *array detector* (17-57: “physical arrangement of detector elements that are able to sense the output e.g., of a polychromator. *NOTE:* In some cases, optical components may form a part of the array, as in, for example, a fiber optics faceplate coupled array detector”) is close to the definition used in this document.

3.2 combined standard uncertainty

The standard uncertainty of the result of a measurement when that result is obtained from the values of a number of other quantities; it is equal to the positive square root of a sum of terms, the term being the variances or covariances of these quantities weighted according to how the measurement result varies with changes in these quantities. [VIM,² Section 2]

3.3 coverage factor

A numerical factor used as a multiplier of the combined standard uncertainty in order to obtain an expanded uncertainty. [2.3.6, VIM²]

3.4 expanded uncertainty

The quantity defining an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand. [2.3.5, VIM²]

Note 1: The fraction may be viewed as the coverage probability or level of confidence of the interval.

Note 2: To associate a specific level of confidence with the interval defined by the expanded uncertainty requires explicit or implicit assumptions regarding the probability distribution characterized by the measurement result and its combined standard uncertainty. The level of confidence that may be attributed to this interval can be known only to the extent to which such assumptions may be justified.

3.5 photometer

An instrument for measuring photometric quantities such as luminance, luminous intensity, luminous flux or illuminance [8.4.1, LS-1-20]. In this document, *photometer* will refer to a physical photometer with reversible changes – an instrument containing a physical receptor and associated filters that is calibrated so as to read photometric quantities directly [8.4.1.3, LS-1-20; see **Section 2 Normative References**]. A *physical receptor* is defined as a device in which incident radiation produces an observable physical change, including but not limited to the generation of electrical current or voltage and the darkening of photographic film. Reversible changes are used in measuring flux and flux-related quantities. Irreversible changes are used in measuring exposure [8.4.1.2, LS-1-20]. The CIE ILV¹ definitions are similar: *photometer* (17-909) – “instrument for measuring photometric quantities”; *detector (of optical radiation)* (17-296) – “device in which incident optical radiation produces a measurable physical effect”; and *output (of a detector of optical radiation)* (17-856) – “physical quantity yielded by a detector in response to an optical input. *NOTE:* This quantity, usually electrical, may for example be current, voltage or a change in resistance; the output may also be chemical, as in a photographic film or an actinometer, or mechanical as in a Golay detector.”

3.6 primary standard (lamp)

A light source by which the unit of light is established and from which the values of other standards are derived. This order of standard also is designated as the *national standard* [8.1.1, LS-1-20]. This is very similar to the CIE ILV¹ definition

for *standard lamp* (17-1256): “lamp used as a reference in photometric or spectroradiometric measurements for which the calibration is traceable to a primary photometric or spectroradiometric standard.”

3.7 relative uncertainty

The uncertainty of measurement divided by the result of a measurement.

3.8 secondary standard (lamp)

A stable light source calibrated directly or indirectly by comparison with a primary standard. This order of standard also is designated as a *reference standard* [8.1.2, LS-1-20]. This is very similar to the CIE ILV¹ definition for *secondary standard lamp* (17-1149): “lamp intended to be used as secondary photometric standard.”

3.9 standard deviation (experimental)

For a series of n measurements of the same measurand, the quantity $s(q_k)$ characterizing the dispersion of the results and given by the formula:

$$s(q_k) = \sqrt{\frac{\sum_{j=1}^n (q_j - \bar{q})^2}{n - 1}},$$

with q_k being the result of the k th measurement and $\bar{q}$ being the arithmetic mean of the n results considered. [3.8, VIM²]

Note 1: Considering the series of n values as a sample of a distribution, $\bar{q}$ is an unbiased estimate of the mean μ_q , and $s(q_k)^2$ is an unbiased estimate of the variance, σ^2 , of that distribution.

Note 2: The expression $s(q_k)/\sqrt{n}$ is an estimate of the standard deviation of the distribution of $\bar{q}$ and is called the *experimental standard deviation of the mean*.

3.10 standard uncertainty

The uncertainty of the result of a measurement expressed as a standard deviation. [2.3.1, VIM²]

3.11 uncertainty (of measurement)

The parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand.

Note 1: The parameter may be, for example, a standard deviation (or a given multiple of it), or the half-width of an interval having a stated level of confidence.

Note 2: Uncertainty of measurement comprises, in general, many components. Some of these components may be evaluated from the statistical distribution of the results of series of measurements and can be characterized by experimental standard deviations. The other components, which can also be characterized by standard deviations, are evaluated from assumed probability distributions based on experience or other information.

Note 3: It is understood that the result of the measurement is the best estimate of the value of the measurand, and that all components of uncertainty, including those arising from systematic effects, such as components associated with corrections and reference standards, contribute to the dispersion.

3.12 working standard (lamp)

A standardized light source for regular use in photometry [8.1.3, LS-1-20]. This is very similar to the CIE ILV¹ definition for *working standard lamp* (17-1444): “lamp intended to be used as a working photometric standard.”

4.0 Guide to the Expression of Uncertainty in Measurement (GUM): A Summary

4.1 Introduction

The *Guide to the Expression of Uncertainty in Measurement* (GUM; see **Section 2.0 Normative References**) provides a procedural method for the analysis of uncertainty. These concepts will be used in this document for the photometric calibration of integrating spheres using standard incandescent lamps, as well as the calibration using integrating spheres of *working standard lamps* that are effectively identical to the standard lamps. The procedure has eight steps:

1. Define the measurand, Y , and the model (mathematical equation) to calculate it.
2. Determine an estimate, x_i , for each input parameter, X_i .
3. Evaluate the standard uncertainty, $u(x_i)$, for each input parameter, X_i .
4. Calculate an estimate, y , of the measurand, Y .
5. Determine the sensitivity coefficient, $\partial y / \partial x_i$, for each input parameter, X_i .
6. Determine the combined standard uncertainty, $u_c(y)$.
7. Calculate the coverage factor, k , based on the degrees of freedom, ν , and determine the expanded uncertainty, $U(y)$.
8. Report the measurand estimate, y , the expanded uncertainty, $U(y)$, and the coverage factor, k .

These steps will be contextually expounded on in the following sections.

4.2 Description of the Model Equation and Input Parameter Estimation

Per the GUM, a mathematical equation is required to describe the measurement process, considering the various factors that contribute uncertainty to the measurement. The calculated quantity or the result of the mathematical equation is the “measurand” or output variable. For the examples presented in this document, the measurand is the calibration factor required for the various photometric integrating spheres.

The relationship between the measurand, Y , and the input quantities, X_i , on which Y depends, is written mathematically as: $Y = f(X_1, X_2, \dots, X_N)$. The function f should contain every quantity needed to calculate the measurand, including all correction factors that can contribute a significant component of uncertainty to the result of the measurement. An estimated value, x_i , is determined for each input quantity, X_i , either based on the statistical analysis of a series of observations or by other means, which may include data from a calibration certificate, manufacturer specification, or published literature, among other sources of information.

4.3 Standard Uncertainty Evaluation

For each input estimate, x_i , an evaluation of the standard uncertainty, $u(x_i)$, is required. Per the GUM, the evaluation methods used to determine the magnitude of the standard uncertainty may be grouped into two categories: those that are evaluated by statistical methods, and those that are evaluated by other means. The evaluation of standard uncertainty by the statistical analysis of a series of observations is termed a *Type A* evaluation of uncertainty. The evaluation of uncertainty by means other than the statistical analysis of a series of observations is termed a *Type B* evaluation of uncertainty.

4.3.1 Type A Uncertainty Evaluation. A Type A standard uncertainty evaluation is obtained from a probability density function derived from an observed frequency distribution. A Type A analysis involves making repeated direct measurements of a quantity, such as a voltage or a current from a photometer. The best available estimate, x_i , for an input quantity that varies randomly and where n independent observations, $X_{i,k}$, have been measured under the same conditions, is the arithmetic mean, $\bar{X}_i$:

$$x_i = \bar{X}_i = \frac{1}{n} \sum_{k=1}^n X_{i,k} \quad (4.1)$$

The standard uncertainty, $u(x_i)$, to be associated with x_i is the estimated standard deviation of the mean:

$$u(x_i) = s(\bar{X}_i) = \sqrt{\frac{1}{n(n-1)} \sum_{k=1}^n (X_{i,k} - \bar{X}_i)^2} \quad (4.2)$$

A least squares fit to a set of measured data to determine the parameters of the function and the standard deviation of the parameters is a Type A uncertainty. For example, the calibration of a 6.5-digit voltmeter with a calibrated 8.5-digit voltmeter and precision voltage supply involves a linear fit, which is described in **Annex A**. In addition, **Annex A** discusses the uncertainty associated with using the function resulting from the fit.

4.3.2 Type B Uncertainty Evaluation. Type B uncertainty evaluations are determined by other than statistical methods. In these cases, the standard uncertainty is often provided in calibration certificates, published literature, and manufacturer specifications or best estimates based on experience. Type B standard uncertainty evaluations may also occur when a Type A uncertainty evaluation cannot be calculated due to insufficient information.

For example, an input quantity where the value is estimated from a rectangular or uniform probability distribution with a lower limit of a_- and an upper limit of a_+ may be considered. A digital display where the last digit does not vary in time is an example of a rectangular probability distribution for the standard uncertainty. Variation in the measurement exists, but the display does not have enough digits to resolve this variation. In this case, the input quantity is estimated by:

$$x_i = (a_- + a_+)/2, \quad (4.3)$$

and the standard uncertainty, $u(x_i)$, associated with x_i is:

$$u(x_i) = (a_+ - a_-)/(2\sqrt{3}), \quad (4.4)$$

where a_+ is equal to the display plus half the last digit of resolution, and a_- is equal to the display minus half the last digit of resolution. The GUM provides other examples of Type B evaluations, based on other probability distributions.

Converting the tolerance of an AC-power analyzer to standard uncertainty is another example that uses a uniform distribution. The specifications of a typical AC-power analyzer states that the tolerance for AC voltage measurements at 60 Hz is 0.1% of the reading plus 0.1% of the range plus 10 mV. Measuring a 120-V lamp on a range of 150 V gives a tolerance of ± 0.28 V for a reading of 120.00 V. The tolerance is modeled by a uniform distribution, where $a_- = 120.00$ V - 0.28 V = 119.72 V, and $a_+ = 120.00$ V + 0.28 V = 120.28 V; therefore, $u(x_i) = 0.16$ V.

4.4 Sensitivity Coefficients and Combined Standard Uncertainty

To determine the contribution of each component to the combined standard uncertainty, $u_c(y)$, the standard uncertainty, $u(x_i)$, is multiplied by the sensitivity coefficient, $\frac{\partial y}{\partial x_i}$. The sensitivity coefficient is described as the influence of the uncertainty component on the measurand, y ; therefore, if a component changes in value, the calculation of y is affected. The partial derivative of the model equation for each component of uncertainty describes

this dependence. (**Annex B** provides a brief explanation of the determination of partial derivatives.) Since the partial derivative is the slope of the model equation with a given set of input estimates, another method of determining the sensitivity coefficient is to determine the change in the measurand with respect to the uncertainty component. Using this method, a partial derivative, or sensitivity coefficient, can be calculated by using:

$$\frac{\partial y}{\partial x_i} = \frac{f[x_1, x_2, \dots, x_i + u(x_i), \dots, x_n] - f[x_1, x_2, \dots, x_i - u(x_i), \dots, x_n]}{2[u(x_i)]} \quad (4.5)$$

Annex C provides a demonstration of using **Equation 4.5** instead of partial derivatives.

The combined standard uncertainty $u_c(y)$, is determined using:

$$u_c(y) = \sqrt{\sum_{i=1}^m \left(\frac{\partial y}{\partial x_i} \right)^2 u^2(x_i) + 2 \sum_{i=1}^{m-1} \sum_{j=i+1}^m \frac{\partial y}{\partial x_i} \frac{\partial y}{\partial x_j} u(x_i) u(x_j) r(x_i, x_j)} \quad (4.6)$$

The combined standard uncertainty may also be expressed as a relative combined standard uncertainty by dividing $u_c(y)$ by y . The correlation coefficient in **Equation 4.6**, $r(x_i, x_j)$, is the mathematical dependency between two components of uncertainty, and ranges from -1 to +1. As an example, a correlation exists between the increase in luminous flux and the correlated color temperature when the lamp current is increased. The templates provided in this document have been designed to eliminate correlations, such that the contributions due to correlations are zero; i.e., $r(x_i, x_j) = 0$ in **Equation 4.6**; therefore, **Equation 4.6** reduces to the square root of the sum of the squares:

$$u_c(y) = \sqrt{\sum_{i=1}^m \left(\frac{\partial y}{\partial x_i} \right)^2 u^2(x_i)} \quad (4.7)$$

Calibrating a sphere's responsivity using four lamps purchased from a national metrology institute is an example of correlation. A laboratory may use four lamps instead of one lamp to calibrate the measurement system because the average result has a tendency to track the scale better. (**Section 5.1.9** shows a more complete example.) The model equation for using four lamps to calibrate an integrating sphere is:

$$y = C_\phi = \frac{C_1 + C_2 + C_3 + C_4}{4}, \quad (4.8)$$

where C_1 , C_2 , C_3 and C_4 are the calibration factors determined for each lamp. Without considering correlation, the combined standard uncertainty might be calculated as:

$$u_c(C_\phi) = \sqrt{\left(\frac{1}{4}\right)^2 u^2(C_1) + \left(\frac{1}{4}\right)^2 u^2(C_2) + \left(\frac{1}{4}\right)^2 u^2(C_3) + \left(\frac{1}{4}\right)^2 u^2(C_4)}, \quad (4.9)$$

where $u(C_1)$, $u(C_2)$, $u(C_3)$, and $u(C_4)$ are the standard uncertainty for each calibration factor determined with each lamp. For example, if, $u(C_1) = u(C_2) = u(C_3) = u(C_4) = 0.50\%$, then the combined standard uncertainty would be calculated as $u_c(C_\phi) = 0.25\%$, which would be incorrect. The four lamps are highly correlated; therefore, $r = 1$ between the lamps. The uncertainty budget is predominately Type B uncertainty components, which cannot be averaged or reduced with multiple determinations. The proper combined standard uncertainty calculation for this situation is **Equation 4.10**, for which $u_c(C_\phi) = 0.50\%$.

$$u_c(C_\phi) = \sqrt{\begin{aligned} &\left(\frac{1}{4}\right)^2 u^2(C_1) + \left(\frac{1}{4}\right)^2 u^2(C_2) + \left(\frac{1}{4}\right)^2 u^2(C_3) + \left(\frac{1}{4}\right)^2 u^2(C_4) + \\ &2 \frac{\partial C_\phi}{\partial C_1} \frac{\partial C_\phi}{\partial C_2} u(C_1) u(C_2) r(C_1, C_2) + 2 \frac{\partial C_\phi}{\partial C_1} \frac{\partial C_\phi}{\partial C_3} u(C_1) u(C_3) r(C_1, C_3) + \\ &2 \frac{\partial C_\phi}{\partial C_1} \frac{\partial C_\phi}{\partial C_4} u(C_1) u(C_4) r(C_1, C_4) + 2 \frac{\partial C_\phi}{\partial C_2} \frac{\partial C_\phi}{\partial C_3} u(C_2) u(C_3) r(C_2, C_3) + \\ &2 \frac{\partial C_\phi}{\partial C_2} \frac{\partial C_\phi}{\partial C_4} u(C_2) u(C_4) r(C_2, C_4) + 2 \frac{\partial C_\phi}{\partial C_3} \frac{\partial C_\phi}{\partial C_4} u(C_3) u(C_4) r(C_3, C_4) \end{aligned}} \quad (4.10)$$

4.5 Expanded Uncertainty and Degrees of Freedom

For many measurements in photometry, a normal or Gaussian probability distribution characterizes the measurement result, y , and its combined standard uncertainty, $u_c(y)$; therefore, $u_c(y)$ defines an interval, $y - u_c(y)$ to $y + u_c(y)$, about the measurement result, y , where the value of the measurand, Y , estimated by y , is believed to lie, with an approximate level of confidence of 68.3%, which is one standard deviation for a normal probability distribution. This is the case when the conditions of the Central Limit Theorem are met. Since the estimate, y , is not directly measured and is obtained from the estimate of many input quantities that have well-behaved probability distributions, the Central Limit Theorem holds true to an acceptable level.

Traditionally, industry has required a more encompassing measure of uncertainty that defines an interval about the measurement result, y , where the value of the measurand, Y , is confidently believed to lie. This coverage probability, or level of confidence, is 95% and used as a normative value in this document. The measure of uncertainty intended to meet this requirement is termed expanded uncertainty, U . The *expanded uncertainty*, U , is obtained by multiplying the combined standard uncertainty, $u_c(y)$, by a coverage factor, k .

Conventionally, a coverage factor of $k = 2$ is used in industry. This coverage factor assumes a large number of degrees of freedom for the final probability distribution associated with the combined standard uncertainty, $u_c(y)$. In order to validate that a large number of degrees of freedom exist, the effective degrees of freedom, v_{eff} , are calculated using the Welch-Satterthwaite formula:

$$v_{\text{eff}} = \frac{u_c^4(y)}{\sum_{i=1}^N \frac{\left[\frac{\partial y}{\partial x_i} u(x_i) \right]^4}{v_i}}, \quad (4.11)$$

where $u_c(y)$ is the combined standard uncertainty, $\frac{\partial y}{\partial x_i}$ is the sensitivity coefficient for each uncertainty component, $u(x_i)$ is the standard uncertainty for each uncertainty component, and v_i is the degrees of freedom for each uncertainty component. For a Type A evaluation, the degrees of freedom are the number of data points minus the parameters used to describe the data. If n data points are estimated by a mean $u(x_i)$, the degrees of freedom is $v_i = n - 1$. If n data points are fit by a curve with m parameters, the degrees of freedom are $v_i = n - m$. The degrees of freedom for uncertainty components obtained using Type B evaluation ranges from v_i to infinity. The coverage factor, k , is approximated using the Student's t distribution, $t_p(v_{\text{eff}})$, for the required level of confidence from a table of values of $t_p(v)$, as shown in **Table 4.1**.³ **Table 4.1** only shows the $t_p(v)$ values for $p = 95\%$. The coverage factor may also be approximated using:

$$k_{0.95} = 2 \sqrt{0.95 + \frac{3.3}{(v_{\text{eff}} - 0.92)}}, \quad (4.12)$$

which is a fit to a series of results for the Student's t distribution with different degrees of freedom at a 95% confidence level, or using the EXCEL® function T.INV.2T(1- p , v).

Using this information and the final uncertainty budget, a laboratory can then determine the ideal number of measurement repetitions for a Type A uncertainty component. One does not need to average any more data than is required to make the uncertainty component very small (on the order of 10% or less) compared to the combined standard uncertainty.

Table 4.1. Value of t_p (ν) from the Student's t Distribution, for Degrees of Freedom ν and $p = 95\%$.

Degrees of Freedom, ν	$t_{0.95}(\nu)$	Degrees of Freedom, ν	$t_{0.95}(\nu)$	Degrees of Freedom, ν	$t_{0.95}(\nu)$
1	12.71		2.20		2.06
2	4.30	12	2.18	30	2.04
3	3.18	13	2.16	35	2.03
4	2.78	14	2.14	40	2.02
5	2.57	15	2.13	45	2.01
6	2.45	16	2.12	50	2.01
7	2.36	17	2.11		
8	2.31	18	2.10	100	1.984
9	2.26	19	2.09		
10	2.23	20	2.09	∞	1.960

5.0 Calibration of Integrating Spheres

5.1 Calibration Using a Total Luminous Flux Standard Lamp

5.1.1 Model for an Integrating Sphere with Photometer. The scope of this uncertainty budget is the calibration of an integrating sphere measurement system utilizing a photometer as the detection device and an incandescent (including halogen) lamp as the standard lamp calibrated for total luminous flux (lumens) at a particular operating current. The quantity to be determined is the photometer calibration factor, C_ϕ , in units of signal per lumen, and its associated expanded uncertainty, $U(C_\phi)$. In the example presented in this section, the photometer calibration factor is referenced to an empty sphere (sphere with the standard lamp removed) and Illuminant A as the source correlated color temperature.

The topic of calibrating secondary or working standard lamps is discussed in **Section 6**.

The model equation describing how to calculate, and the uncertainty components affecting the determination of the sphere calibration factor, C_ϕ , is written as:

$$C_\phi = \frac{y_r - y_{r_d}}{\Phi_r} \cdot \left(\frac{J_r \cdot R}{U_r \cdot c_U} \right)^{m_{J_r}} \cdot \frac{y_E - y_{E_d}}{y_{X_r} - y_{X_{r_d}}} \cdot A_r \cdot C_{f_r}, \quad (5.1)$$

where subscript **r** refers to a standard lamp property, and:

C_ϕ **sphere calibration factor:** The measurand, which is in units of signal (volt, amps, or lumens) per lumen.

Φ_r **total luminous flux of the standard lamp:** The total luminous flux (in lumens) for the standard lamp is provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 2,800.0 lm, with an expanded uncertainty ($k = 2$) of 16.4 lm. Because the coverage factor is equal to 2, the standard uncertainty is the expanded uncertainty divided by 2, or 8.2 lm. These values are recorded in Line 1 of **Table 5.2**.

y_r **sphere photometer signal:** The photometer signal is a photocurrent, which may be measured using various types of equipment. The photocurrent may be measured with a calibrated ammeter. The repeatability and the calibration uncertainty of the ammeter become the standard uncertainty for this parameter.

A second method uses a transimpedance amplifier to convert the photocurrent to a voltage. The transimpedance amplifier and the voltmeter are calibrated and have certificates with stated uncertainties that need to be included along with the repeatability of the measured signal.

When the user does not have access to the measurement process, a photometer with software reporting a lumen value as the raw signal is an acceptable method. The repeatability of the measurement becomes the standard uncertainty.

To determine the signal and standard uncertainty for this component, 10 measurements are typically made of the illuminated standard lamp. For this example, the signal is the average of the 10 measurements, which is $y_r = 1.7652$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_r) = 0.0007$ nA. This is a Type A uncertainty, with degrees of freedom equal to $10 - 1 = 9$. The signal and standard uncertainty are recorded in Line 2 of **Table 5.2** (see **Section 5.1.4**).

For integrating sphere systems using an array spectrometer as the detection device, the reader is referred to **Section 5.1.9** for a detailed discussion of the uncertainty of the signal component.

y_{r_d} **sphere photometer dark signal:** The photometer signal measured when the standard lamp is mounted in the sphere but not illuminated. To determine the dark signal and standard uncertainty for this component, a user would typically run 10 measurements of the non-illuminated standard lamp. For this example, the signal is the average of the ten measurements, which is $y_{r_d} = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{r_d}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 3 of **Table 5.2**.

U_r **lamp current measurement:** The lamp current measurement for this model is done by measuring the voltage across a calibrated shunt resistor, R . This is a Type A uncertainty because several measurements are collected. In this example, 30 measurements are made, integrated over 10 power line cycles. The average of the 30 measurements is 0.582743 V and the standard deviation is 0.000015 V, which is the standard uncertainty, $u(U_r)$. There are 29 degrees of freedom for this measurement ($30 - 1$). The average, standard uncertainty and degrees of freedom are recorded in Line 4 of **Table 5.2**.

c_U **voltmeter calibration factor:** The voltmeter has a center value close to the nominal reading, with a given uncertainty provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 1.000032 with an expanded uncertainty ($k = 2$) of 0.000008. The standard uncertainty is the expanded uncertainty divided by 2, or 0.000004. These values are recorded in line 5 of **Table 5.2**.

J_r **reference current for the standard lamp:** The reference current (in amps, A) for the standard lamp is provided on the calibration certificate. This component does not have uncertainty, because it is stated as a given value on the calibration certificate.

In this example, the calibration certificate states 5.825 A. This value is recorded in Line 6 of **Table 5.2**.

R **resistance of the shunt resistor:** The resistor value and uncertainty are listed on the resistor calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 0.100045 ohms (Ω) with an expanded uncertainty ($k = 2$) of 0.000006 Ω . The standard uncertainty is the expanded uncertainty divided by 2, or 0.000003 Ω . These values are recorded in Line 7 of **Table 5.2**.

m_{J_r} **sensitivity of luminous flux with respect to current:** For incandescent lamps, including halogen, the change in lumen output with current can be expressed as a ratio of the actual lamp current divided by the desired lamp current, and the ratio then raised to an exponent of 6.50, which has a standard uncertainty of

0.02. These are typical industry standard values. This is a Type B uncertainty because it is based on industry knowledge. These values are recorded in Line 8 of **Table 5.2**.

y_{x_r} **self-absorption signal for the standard lamp:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall back onto the lamp, thus reducing the responsivity of the integrating-sphere photometer. The recommended method for correcting this error relies on the use of an auxiliary lamp to determine the magnitude of the self-absorption.

The self-absorption signal for the standard lamp is the photometer signal measured when the auxiliary lamp is on and the standard lamp is mounted in the sphere but not illuminated. For this example, the signal is the average of the 10 measurements, which is $y_{x_r} = 0.5243$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{x_r}) = 0.0004$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and the standard uncertainty are recorded in Line 9 of **Table 5.2**.

For integrating sphere systems using an array spectrometer as the detection device, the reader is referred to **Section 5.1.9** for a detailed discussion of the uncertainty of the self-absorption signal component.

$y_{x_{rd}}$ **self-absorption dark signal for the standard lamp:** The photometer signal measured when the auxiliary lamp is off and the standard lamp is mounted in the sphere. For this example, the signal is the average of the 10 measurements, which is $y(y_{x_{rd}}) = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{x_{rd}}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 10 of **Table 5.2**.

y_E **self-absorption factor for “empty” sphere:** This measurement is the photometer signal measured when the auxiliary lamp is on and no standard lamp is present in the sphere. A reference condition is required. In this example, it is the sphere with the standard lamp removed; i.e., empty sphere. Otherwise, a self-absorption measurement of the standard lamp would be required with every measurement in order to account for changes in the sphere over time. For this example, the signal is the average of the 10 measurements, which is $y_E = 0.5437$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_E) = 0.0004$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 11 of **Table 5.2**.

y_{Ed} **self-absorption dark signal for “empty” sphere:** The photometer signal measured when the auxiliary lamp is off and no standard lamp is present in the sphere. For this example, the signal is the average of the 10 measurements, which is $y_{Ed} = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{Ed}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 12 of **Table 5.2**.

A_r **standard lamp age or burning time:** The operation of an incandescent (including halogen) standard lamp typically reduces the lumen output by 0.4% over a period of 24 hours. Not all lamps meet this condition, and very few are predictable. Many calibration laboratories will provide a statement approximating this dependency on the calibration certificates for the standard lamps they issue. If this information is not available, the decay rate is assumed to be 0.4% per 24 hours. This is a Type B uncertainty because it is based on industry knowledge or data from a calibration certificate.

The value of A_r is set to 1. It is difficult and risky to apply a correction factor for the aging of the standard lamp without assuming knowledge. Increasing the standard uncertainty with the operation of the standard lamp is a more conservative approach. For this example, assuming the lamp has been used for 2 hours, the standard uncertainty is estimated at 0.4% divided by 24 hours, then multiplied by the 2 hours of use, resulting in a value of 0.00033. These values are recorded in Line 13 of **Table 5.2**.

C_{fr} **additional uncertainty components:** Discussed in **Section 5.1.2**.

5.1.2 Additional Uncertainty Components. In order to simplify the presentation of the model equation (**Equation 5.1**), a separate table of uncertainty components is presented, where the components do not change the magnitude of the measurand, C_{ϕ} , but do add uncertainty, and the contribution of which is relatively small (less than 1/5 that of the largest contributor). If the contribution is large, it should be listed in the model equation as a separate factor. **Table 5.1** lists a few examples. The uncertainty contributions are estimated, and the combined standard uncertainty is estimated by the square root of the sum of the squares. These are all Type B uncertainties. The relative standard uncertainty contributions presented in **Table 5.1** are reasonable estimates for industry standard uncertainties. The estimation of C_{f_r} is equal to 1, and the standard uncertainty for C_{f_r} is the combined standard uncertainty of the components listed in **Table 5.1**. The estimation of C_{f_r} and the standard uncertainty for C_{f_r} are recorded in Line 14 of **Table 5.2**.

Table 5.1. Additional Uncertainty Components for the Model of an Integrating Sphere with Photometer.

Uncertainty Component	Relative Standard Uncertainty Contribution
Spectral mismatch correction factor	0.10 %
Photometer temperature	0.06 %
Near-field absorption	0.05 %
Lamp and sphere ambient temperature	0.05 %
Stray light from the room	0.02 %
Combined standard uncertainty	0.14 %

5.1.2.1 Spectral Mismatch Correction Factor. For a standard lamp, the spectral responsivity of the integrating sphere photometer cannot be perfectly matched to the $V(\lambda)$ function. The photometer responsivity is referenced to a correlated color temperature (CCT), which is usually (and in this example) 2856 K, the CCT of Illuminant A. When the integrating sphere and photometer are calibrated by standard lamps operating at other correlated color temperatures, a spectral mismatch error occurs. (See ANSI/IES LM-78-20,⁴ Section 7.3.2.) Standard lamps typically have correlated color temperatures ranging from 2700 K to 2950 K. A commercially available system should not deviate more than 0.25% over this range of CCTs. For this example, the CCT is close to 2856 K, so the uncertainty due to spectral mismatch correction was approximated to be 0.10%.

5.1.2.2 Photometer Temperature. A photometer changes responsivity with temperature. In particular, the filter commonly used to cut off blue wavelength light changes wavelength cutoff with temperature, resulting in a sensitivity of 0.1% per degree Celsius. It is important to note the temperature at which the photometer was calibrated. If the photometer was calibrated at 25 °C and is used in a laboratory maintaining a constant temperature of 25 °C $\pm$ 1 °C, the standard uncertainty would be estimated by a uniform probability function: $1\text{ °C} - [(-1\text{ °C})/2\sqrt{3}](0.1\%/1\text{ °C}) = 0.06\%$. Photometers are often temperature controlled, or temperature monitored and corrected, which reduces this contribution of uncertainty to zero.

5.1.2.3 Near-Field Absorption. Near-field absorption is described in ANSI/IES LM-78-20 Section 7.3.4. A well-designed system should have a small contribution from near-field absorption. As the sphere diameter decreases, self-absorption of the standard lamp becomes near-field absorption, which is not corrected by the auxiliary lamp measurement. For spheres 1 m in diameter or larger, the near-field absorption should be small. For this example, the standard uncertainty due to near-field absorption is approximated as 0.05%. For spheres smaller than 1 m, the near-field absorption error can be large and difficult to assess. A series of calibrated lamps differing in size (specifically, surface area) may be used to approximate the near-field absorption for smaller spheres, with respect to the surface area of the standard lamp.

5.1.2.4 Lamp and Sphere Ambient Temperature. For an incandescent lamp (including halogen lamps), the ambient temperature does not normally affect the operation unless the sphere is unusually small. If the sphere is small enough to affect the operation of the lamp, then the sphere coating reflectance most likely will be affected by the ambient temperature. (Refer to ANSI/IES LM-78-20 Section 5.3 and Section 7.3.5 for more details concerning sphere coating reflectivity and ambient temperature.) For typical measurement systems (and this example), the standard uncertainty due to lamp and sphere ambient temperature is approximately 0.05%.

5.1.2.5 Stray Light from the Room. Stray light from the room is typically very small or non-existent. Possible sources of room light include open ports in the sphere, the seam where the sphere halves close, and gaps around detectors or a wire feed-through. If the measurements are normally taken with the room lights on, then the stray light contribution can be approximated by measuring a dark signal with the lights on and off. For typical measurement systems and this example, the standard uncertainty due to room stray light is approximately 0.02%.

5.1.3 Integrating Sphere with Photometer Sensitivity Coefficients. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 5.1 (Section 5.1.1)** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**. The equations for the partial derivatives follow, along with the respective line number in **Table 5.2** where the result is recorded. It should be noted that if the ratio of currents is exactly 1, Line 8 goes to zero; therefore, a small factor that is approximately one-tenth of the resolution should be added to this ratio.

$\frac{\partial C_\phi}{\partial \Phi_r} = \frac{-C_\phi}{\Phi_r}$	Line 1	$\frac{\partial C_\phi}{\partial y_r} = \frac{C_\phi}{y_r - y_{r_d}}$	Line 2	$\frac{\partial C}{\partial y_{r_d}} = \frac{-C}{y_r - y_{r_d}}$	Line 3
$\frac{\partial C_\phi}{\partial U_r} = \frac{-m_{J_r} \cdot C_\phi}{U_r}$	Line 4	$\frac{\partial C_\phi}{\partial c_U} = \frac{-m_{J_r} \cdot C_\phi}{c_U}$	Line 5	$\frac{\partial C_\phi}{\partial r} = \frac{m_{J_r} \cdot C_\phi}{R}$	Line 7
$\frac{\partial C_\phi}{\partial m_{J_r}} = C_\phi \cdot \ln\left(\frac{J_r \cdot R}{U_r \cdot c_U}\right)$	Line 8	$\frac{\partial C_\phi}{\partial y_{x_r}} = \frac{-C_\phi}{y_{x_r} - y_{x_{r_d}}}$	Line 9	$\frac{\partial C_\phi}{\partial y_{x_{r_d}}} = \frac{C_\phi}{y_{x_r} - y_{x_{r_d}}}$	Line 10
$\frac{\partial C_\phi}{\partial y_E} = \frac{C_\phi}{y_E - y_{E_d}}$	Line 11	$\frac{\partial C_\phi}{\partial y_{E_d}} = \frac{-C_\phi}{y_E - y_{E_d}}$	Line 12	$\frac{\partial C_\phi}{\partial A_r} = \frac{C_\phi}{A_r}$	Line 13
$\frac{\partial C_\phi}{\partial C_{f_r}} = \frac{C_\phi}{C_{f_r}}$	Line 14				

5.1.4 Example Uncertainty Budget. Provided in **Table 5.2** is an uncertainty budget using **Equation 5.1** as the model and typical values (see **Section 5.1.1**) for a high-quality sphere system with a photometer detector using a high-quality luminous flux standard lamp.

5.1.5 Measurand and Combined Standard Uncertainty. Using **Equation 5.1** and the estimated values for x_i provided in **Table 5.2**, the measurand is calculated: $C_\phi = 6.537 \times 10^{-4}$ nA/lm.

The combined standard uncertainty is the square root of the sum of squares, $u_c(C_\phi) = 2.3 \times 10^{-6}$ nA/lm, or 0.35% as a relative combined standard uncertainty. The combined standard uncertainty in this example is dominated by the standard uncertainty of the standard lamp. An increase in the number of measurements for the Type A components would not significantly reduce the combined standard uncertainty.

Table 5.2. Uncertainty Budget for an Integrating Sphere Using a Photometer Detector and a Total Luminous Flux Standard Lamp.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	Φ_r	2,800.0	lm	8.2	B	∞	-2.335×10^{-7}	-1.914×10^{-6}
2	y_r	1.7652	nA	0.0007	A	9	3.704×10^{-4}	2.593×10^{-7}
3	y_{rd}	0.0002	nA	0.0002	A	9	-3.704×10^{-4}	-7.407×10^{-8}
4	U_r	0.582743	V	0.000015	A	29	-7.291×10^{-3}	-1.094×10^{-7}
5	c_U	1.000032		0.000004	B	∞	-4.249×10^{-3}	-1.700×10^{-8}
6	J_r	5.8250	A					
7	R	0.100045	Ω	0.000003	B	∞	4.247×10^{-2}	1.274×10^{-7}
8	m_{Jr}	6.50		0.02	B	∞	5.353×10^{-10}	1.071×10^{-11}
9	y_{xr}	0.5243	nA	0.0004	A	9	-1.247×10^{-3}	-4.989×10^{-7}
10	y_{xrd}	0.0002	nA	0.0002	A	9	1.247×10^{-3}	2.495×10^{-7}
11	y_E	0.5437	nA	0.0004	A	9	1.203×10^{-3}	4.811×10^{-7}
12	y_{Ed}	0.0002	nA	0.0002	A	9	-1.203×10^{-3}	-2.405×10^{-7}
13	A_r	1.0000		0.00033	B	∞	-6.537×10^{-4}	2.157×10^{-7}
14	C_{fr}	1.0000		0.0014	B	∞	6.537×10^{-4}	9.152×10^{-7}
	C_Φ	6.5369×10^{-4}	nA/lm				$u_c(C_\Phi)$	2.29×10^{-6}

5.1.6 Degrees of Freedom and Expanded Uncertainty. Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), the effective degrees of freedom for this uncertainty budget, ν_{eff} , are 1,950. Using an effective degrees of freedom of 1,950 and the Student's t approximation for coverage factor (**Equation 4.12**, presented in **Section 4.5**), the coverage factor is approximately 2, resulting in an expanded uncertainty of $u(C_\Phi) = 4.6 \times 10^{-6}$ nA/lm, or 0.70% as a relative expanded uncertainty. The uncertainty community typically rounds the coverage factor to 2.00 if the calculated value is within 0.10 of 2.00. The final statement of uncertainty would be:

The luminous flux responsivity of the integrating sphere system on [date] was determined, traceable to the candela SI unit, with a magnitude of 6.537×10^{-4} nA/lm and an associated expanded uncertainty of 4.6×10^{-6} nA/lm, or a relative expanded uncertainty of 0.70%. The stated uncertainty was calculated according to the GUM by a multiplication of the combined standard uncertainty with a coverage factor of $k = 2$.

5.1.7 Model Modifications. The model **Equation 5.1** is a starting point. Based on the equipment used by the laboratory, the model equation should be modified to represent the laboratory. For example, if the laboratory has a multimeter or power supply measuring the current directly, the model equation would be altered to:

$$C_\Phi = \frac{(y_r - y_{rd})}{\Phi_r} \cdot \left(\frac{J_r}{J \cdot c_J} \right)^{m_{Jr}} \cdot \frac{(y_E - y_{Ed})}{(y_{Ar} - y_{Ar_d})} \cdot A_r \cdot C_{fr} , \quad (5.2)$$

where J , the measured current, and c_J , the ammeter calibration, replace the voltmeter measurement, U , voltmeter calibration, c_U , and shunt resistor calibration, R .

The changes to the uncertainty analysis based on **Equation 5.2**, where the current reading comes from the power supply display instead of measured across a shunt, are straightforward. For this example, the current reading, J , is 5.825 A. The

standard uncertainty of the current reading is a Type B uncertainty determined using **Equation 4.4**. Since the last digit in the display does not vary, the actual current is within the range 5.8245 to 5.8255 amps. The standard uncertainty is $0.001/2\sqrt{3} = 0.00029$ A. The display has a calibration factor $c_J = 1.000$ for this example, with a standard uncertainty of 0.1% of the signal: $u(c_J) = 0.00017$.

In **Table 5.2**, Lines 4 and 5 are replaced with the lines shown in **Table 5.3**.

Table 5.3. Replacement Lines in Table 5.2 for Modified Model Equation.

4	J	5.825	A	0.00029	B	∞	-7.294×10^{-4}	-2.115×10^{-7}
5	c_J	1.00001		0.00017	B	∞	-4.249×10^{-3}	-7.223×10^{-7}

The sensitivity coefficients for these two parameters are:

$$\frac{\partial C_\phi}{\partial J} = \frac{-m_{J_r} \cdot C_\phi}{J} \quad \text{and} \quad \frac{\partial C_\phi}{\partial c_J} = \frac{-m_{J_r} \cdot C_\phi}{c_J}$$

The measurand, C_ϕ , is 6.536×10^{-4} nA/lm, and the combined standard uncertainty is 2.4×10^{-6} nA/lm. The degrees of freedom are 2,370, so the coverage factor is approximately 2. The expanded uncertainty is 4.8×10^{-6} nA/lm.

Another possible modification is where the laboratory does not reference to an “empty” sphere and would like to use a working reference lamp every day before starting measurements; therefore, the model equation for the photometer calibration factor would be:

$$C_\phi = \frac{(y_r - y_{r,d})}{\Phi_r} \cdot \left(\frac{J_r \cdot R}{U_r \cdot c_U} \right)^{m_{J_r}} \cdot A_r \cdot C_{f_r} \quad (5.3)$$

5.1.8 Multiple Standard Lamps. Many laboratories will not rely on a single standard lamp to calibrate their integrating sphere measurement system. This section describes the uncertainty analysis when three standard lamps are used to calibrate the system. Lamps age differently and tend to operate slightly differently on different days; therefore, the average of three standard lamps tends to produce a more consistent calibration factor over time. However, as discussed in **Section 4.4**, standard lamps are highly correlated, whether they come from the same or different calibration laboratories. The only factor common between the three standard lamp measurements is the determination of the empty sphere self-absorption factor; therefore, the measurement equation is:

$$C_\phi = \left(\frac{C_1 + C_2 + C_3}{3} \right) (y_E - y_{E,d}) , \quad (5.4)$$

where

$$C_i = \frac{(y_{r,i} - y_{r,i,d})}{\Phi_{r,i}} \cdot \left(\frac{J_{r,i} \cdot R}{U_{r,i} \cdot c_U} \right)^{m_{J_r}} \cdot \frac{1}{(y_{X_{r,i}} - y_{X_{r,i,d}})} \cdot A_{r,i} \cdot C_{f_r} \quad (5.5)$$

The standard lamp described in **Table 5.2** would now have an uncertainty budget for C_1 , and the combined standard uncertainty, $u_c(C_1)$, shown in **Table 5.4**. **Table 5.5** is the uncertainty budget for a second standard lamp, with a higher wattage. **Table 5.6** is the uncertainty budget for a third standard lamp, of lower wattage. All three lamps produce a correlated color temperature near 2856 K and have very similar luminous intensity angular distributions. It is important to notice that the relative standard uncertainty of the total luminous flux for each of the standard lamps is slightly different: $u(\Phi_{r1}) = 0.29\%$, $u(\Phi_{r2}) = 0.41\%$, and $u(\Phi_{r3}) = 0.36\%$.

Table 5.4. Uncertainty Budget for Sphere Responsivity Using Standard Lamp 1

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity $"c_i$	Contribution $u_i(y)$
1	Φ_{r2}	2,800.0	lm	8.2	B	∞	-4.296×10^{-7}	-3.522×10^{-6}
2	y_{r2}	1.7652	nA	0.0007	A	9	6.814×10^{-4}	4.770×10^{-7}
3	y_{r2d}	0.0002	nA	0.0002	A	9	-6.814×10^{-4}	-1.363×10^{-7}
4	U_{r2}	0.582743	V	0.000015	A	29	-1.342×10^{-2}	-2.012×10^{-7}
5	c_U	1.000032		0.000004	B	∞	-7.818×10^{-3}	-3.127×10^{-8}
6	J_{r2}	5.825	A					
7	R	0.100045	Ω	0.000003	B	∞	7.814×10^{-2}	2.344×10^{-7}
8	m_{Jr}	6.5		0.2	B	∞	9.849×10^{-10}	1.970×10^{-10}
9	$y_{x_{r2}}$	0.5243	nA	0.0004	A	9	-2.295×10^{-3}	-9.180×10^{-7}
10	$y_{x_{r2d}}$	0.0002	nA	0.0002	A	9	2.295×10^{-3}	4.590×10^{-7}
13	y_E	1.0000		0.00033	B	∞	1.203×10^{-3}	3.969×10^{-7}
14	y_{Ed}	1.0000		0.0014	B	∞	1.203×10^{-3}	1.684×10^{-6}
	$C_{\phi 1}$	1.2027×10^{-3}	nA/lm			36,052	$u_c(C_{\phi 1})$	4.10×10^{-6}

Since standard lamp 3 is a lower wattage, the gain was changed on the transimpedance amplifier. A silicon-based photometer is inherently very linear (orders of magnitude). The non-linearity of the signal comes from the methods used to determine the photocurrent. Transimpedance amplifiers may be non-linear and, more important, can have large differences as gain ranges are changed. For this example, a calibrated transimpedance amplifier is used, which has a small standard uncertainty, $U(G)$, equal to 0.05%. Uncalibrated transimpedance amplifiers may have gain range differences as large as 1%, which can be the most significant uncertainty component. If lamps of different electrical wattages are measured, then the gain factors need to be addressed in the uncertainty estimation, which is accounted for in Line 13 of **Table 5.6**.

The combined standard uncertainty for the average sphere responsivity, C_ϕ , is calculated using **Equation 5.6**, which is based on **Equation 4.6**. All of the correlation coefficients are taken to be equal to 1. **Table 5.7** summarizes the uncertainty budget for the average sphere responsivity, C_ϕ , using three standard lamps.

$$u_c(C_\phi) = \sqrt{\left(\frac{\partial C_\phi}{\partial C_{\phi 1}}\right)^2 u^2(C_{\phi 1}) + \left(\frac{\partial C_\phi}{\partial C_{\phi 2}}\right)^2 u^2(C_{\phi 2}) + \left(\frac{\partial C_\phi}{\partial C_{\phi 3}}\right)^2 u^2(C_{\phi 3}) + \left(\frac{\partial C_\phi}{\partial y_E}\right)^2 u^2(y_E) + \left(\frac{\partial C_\phi}{\partial y_{Ed}}\right)^2 u^2(y_{Ed}) + 2 \frac{\partial C_\phi}{\partial C_{\phi 1}} \frac{\partial C_\phi}{\partial C_{\phi 2}} u(C_{\phi 1}) u(C_{\phi 2}) r(C_{\phi 1}, C_{\phi 2}) + 2 \frac{\partial C_\phi}{\partial C_{\phi 1}} \frac{\partial C_\phi}{\partial C_{\phi 3}} u(C_{\phi 1}) u(C_{\phi 3}) r(C_{\phi 1}, C_{\phi 3}) + 2 \frac{\partial C_\phi}{\partial C_{\phi 2}} \frac{\partial C_\phi}{\partial C_{\phi 3}} u(C_{\phi 2}) u(C_{\phi 3}) r(C_{\phi 2}, C_{\phi 3})} \quad (5.6)$$

Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), the effective degrees of freedom, ν_{eff} , for this uncertainty budget are 7,431, which leads to a coverage factor approximately equal to 2, resulting in an expanded uncertainty of $U(C_\phi) = 5.3 \times 10^{-6}$ nA/lm, or 0.81% as a relative expanded uncertainty. The final statement of uncertainty would be:

Table 5.5. Uncertainty Budget for Sphere Responsivity Using Standard Lamp 2.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	Φ_{r2}	13,353.5	lm	54.7	B	∞	-9.004×10^{-8}	-4.925×10^{-6}
2	y_{r2}	8.72930	nA	0.00007	A	9	1.377×10^{-4}	9.642×10^{-9}
3	y_{r2d}	0.0002	nA	0.0002	A	9	-1.377×10^{-4}	-2.755×10^{-8}
4	U_{r2}	0.733000	V	0.000009	A	29	-1.066×10^{-2}	-9.596×10^{-8}
5	c_U	1.000032		0.000004	B	∞	-7.815×10^{-3}	-3.126×10^{-8}
6	J_{r2}	7.321	A					
7	R	0.100045	Ω	0.000003	B	∞	7.814×10^{-2}	2.344×10^{-7}
8	m_{Jr}	6.5		0.2	B	∞	-9.748×10^{-7}	-1.950×10^{-7}
9	$y_{x_{r2}}$	0.5410	nA	0.0004	A	9	-2.223×10^{-3}	-8.893×10^{-7}
10	$y_{x_{r2d}}$	0.0002	nA	0.0002	A	9	2.223×10^{-3}	4.447×10^{-7}
13	y_E	1.0000		0.00033	B	∞	1.202×10^{-3}	3.968×10^{-7}
14	y_{Ed}	1.0000		0.0014	B	∞	1.202×10^{-3}	1.683×10^{-6}
	$C_{\phi 2}$	1.2024×10^{-3}	nA/lm			10,891	$u_c(C_{\phi 2})$	2.29×10^{-6}

Table 5.6. Uncertainty Budget for Sphere Responsivity Using Standard Lamp 3.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	Φ_{r3}	280.0	lm	1.00	B	∞	-4.297×10^{-6}	-4.927×10^{-6}
2	y_{r3}	0.18303	nA	0.00009	A	9	6.575×10^{-3}	5.918×10^{-7}
3	y_{r3d}	0.00005	nA	0.00002	A	9	-6.575×10^{-3}	-1.315×10^{-7}
4	U_{r3}	0.167000	V	0.000009	A	29	-4.683×10^{-2}	-4.215×10^{-7}
5	c_U	1.000032		0.000004	B	∞	-7.820×10^{-3}	-3.128×10^{-8}
6	J_{r3}	1.667	A					
7	R	0.100045	Ω	0.000003	B	∞	7.817×10^{-2}	2.345×10^{-7}
8	m_{Jr}	6.5		0.2	B	∞	-1.661×10^{-6}	-3.321×10^{-7}
9	$y_{x_{r3}}$	0.5385	nA	0.0004	A	9	-2.235×10^{-3}	-8.940×10^{-7}
10	$y_{x_{r3d}}$	0.0002	nA	0.0002	A	9	2.235×10^{-3}	4.470×10^{-7}
11	y_E	1.0000		0.00033	B	∞	1.203×10^{-3}	3.970×10^{-7}
12	C_{fr}	1.0000		0.0014	B	∞	1.203×10^{-3}	1.684×10^{-6}
13	G	1.0000		0.0005	B	∞	1.203×10^{-3}	6.016×10^{-7}
	$C_{\phi 3}$	1.2032×10^{-3}	nA/lm			6,215	$u_c(C_{\phi 3})$	4.85×10^{-6}

Table 5.7. Uncertainty Budget for Average Sphere Responsivity Using Three Standard Lamps.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	$C_{\phi 1}$	1.2027×10^{-3}	lm-1	4.098×10^{-6}	B	36,052	1.812×10^{-1}	7.425×10^{-7}
2	$C_{\phi 2}$	1.2024×10^{-3}	lm-1	5.320×10^{-6}	B	10,891	1.812×10^{-1}	9.638×10^{-7}
3	$C_{\phi 3}$	1.2032×10^{-3}	lm-1	4.830×10^{-6}	B	6,215	1.812×10^{-1}	8.750×10^{-7}
4	y_E	0.5437	nA	0.0004	A	9	1.203×10^{-3}	4.811×10^{-7}
5	y_{Ed}	0.0002	nA	0.0002	A	9	-1.203×10^{-3}	-2.406×10^{-7}
	C_ϕ	6.5371×10^{-4}	nA/lm			7,431	$u_c(C_\phi)$	2.64×10^{-6}

The luminous flux responsivity of the integrating sphere system on [date] was determined, traceable to the candela SI unit, with a magnitude of 6.537×10^{-4} nA/lm and an associated expanded uncertainty of 5.3×10^{-6} nA/lm, or a relative expanded uncertainty of 0.81%. The stated uncertainty was calculated per the GUM by a multiplication of the standard uncertainty with a coverage factor of $k = 2$.

5.1.9 Describing the Model Equation in Relative Terms. An uncertainty value expressed as a fraction or percentage of the measurand or parameter is relative standard uncertainty, as opposed to absolute standard uncertainty. Using relative standard uncertainty leads to simplification of the combined standard uncertainty equation, which is desirable, but it is not always applicable. By using relative standard uncertainty, the sensitivity coefficients are equal to an integer value, which in most cases is 1. The difficulty is that this is only applicable for uncorrelated variables. Many labs have developed uncertainty budgets based on relative standard uncertainty. Therefore, it is important to show how relative standard uncertainty values are developed from the above process based on absolute standard uncertainty.

The combined standard uncertainty as given in **Equation 4.7** will be solved term-by-term, using the terms from the model expressed in **Equation 5.1**, except the relative standard uncertainty will be substituted for the absolute standard uncertainty. For convenience, **Equation 4.7** is repeated here.

$$u_c(y) = \sqrt{\sum_{i=1}^m \left(\frac{\partial y}{\partial x_i} \right)^2 u^2(x_i)} \quad (4.7)$$

It should be noted that **Equation 4.12** implies there is no correlation between any of the variables given in **Equation 5.1**. The sensitivity coefficients needed for **Equation 4.7** have all been defined in lines 1 through 14 of **Section 5.1.3**.

The subscript *rel* is used to denote the relative standard uncertainty. First, the relationship between the standard uncertainty, $u(x_i)$, and relative standard uncertainty, $u_{rel}(x_i)$, for the measurand, y , and the variables, x_i , are given by:

$$u_{rel}(y) = u(y)/y, \text{ or} \quad (5.7a)$$

$$u_{rel}(x_i) = u(x_i)/x_i \quad (5.7b)$$

For the sum of standard uncertainty of the measurand, the relative standard uncertainty from **Equation 5.7a** is substituted. For the right-hand side of **Equation 4.7**, each of the actual parameters shown in **Equation 5.1** is substituted with the value of the corresponding sensitivity coefficient from **Section 5.1.3** multiplied by the relative standard uncertainty for the relevant parameter as defined in **Equation 5.7b**. This will be done for each term in **Equation 5.1**. There are three computational situations in **Equation 5.1**, and examples for each computational situation are described below.

(1) Consider the substitution process for first term, Φ_r .

$$u_{rel}^2(C_\phi) = \left(\frac{u(C_\phi)}{C_\phi} \right)^2 = \left(\frac{1}{C_\phi} \right)^2 \left(\frac{\partial C_\phi}{\partial (\Phi_r)} \right)^2 u_{rel}^2(\Phi_r) \cdot (\Phi_r)^2 + \dots$$

Substituting the value of the sensitivity coefficient from **Section 5.1.3** gives:

$$u_{rel}^2(C_\phi) = \left(\frac{1}{C_\phi} \right)^2 \left(\frac{-C_\phi}{\Phi_r} \right)^2 u_{rel}^2(\Phi_r) \cdot (\Phi_r)^2 + \dots$$

This reduces to:

$$u_{rel}^2(C_\phi) = u_{rel}^2(\Phi_r) + \dots \quad (5.8)$$

It should be noted that in terms of the relative standard uncertainty, the sensitivity coefficient for the Φ_r term is 1.

(2) Using the same mathematical substitution procedure for the other terms on the right-hand side of **Equation 5.1**, this gives for the $(y_r - y_{rd})$ term:

$$u_{rel}^2(C_\phi) = \left(\frac{1}{C_\phi} \right)^2 \left(\frac{C_\phi}{(y_r - y_{rd})} \right)^2 u_{rel}^2(y_r) \cdot (y_r)^2 + \left(\frac{1}{C_\phi} \right)^2 \left(\frac{-C_\phi}{(y_r - y_{rd})} \right)^2 u_{rel}^2(y_{rd}) \cdot (y_{rd})^2 + \dots$$

After substitution of the sensitivity coefficients from **Section 5.1.3**, this becomes:

$$u_{rel}^2(C_\phi) = \dots + \left(\frac{y_r}{(y_r - y_{rd})} \right)^2 u_{rel}^2(y_r) + \left(\frac{y_{rd}}{(y_r - y_{rd})} \right)^2 u_{rel}^2(y_{rd}) + \dots$$

Defining $\varepsilon = y_{rd}/y_r$ and inserting into the above equation gives:

$$u_{rel}^2(C_\phi) = \dots + \left(\frac{1}{1 - \varepsilon} \right)^2 u_{rel}^2(y_r) + \left(\frac{\varepsilon}{1 - \varepsilon} \right)^2 u_{rel}^2(y_{rd}) + \dots \quad (5.9)$$

Thus, if the dark current is much smaller than the actual light signal – i.e., ε is small, as it normally is – then the sensitivity coefficient for the y_r term is near 1 (slightly above), but the sensitivity coefficient for the subtracted term, y_{rd} , is close to the ratio of the dark current to the signal. As a rough approximation to the standard uncertainty, one can set the dark current term to zero and only use the signal. In this case, the sensitivity coefficient is 1.

(3) This computational situation considers the $\left(\frac{J_r \cdot R}{U_R \cdot c_U} \right)^{m_{J_r}}$ terms. Specifically, one begins with the resistor, R . Then the uncertainty equation for this term is:

$$u_{rel}^2(C_\phi) = \dots + \left(\frac{1}{C_\phi} \right)^2 \left(\frac{\partial C_\phi}{\partial (R)} \right)^2 u_{rel}^2(R) \cdot R^2 + \dots$$

With insertion of the value for the sensitivity coefficient from **Section 5.1.3**, this becomes:

$$u_{rel}^2(C_\phi) = \dots + (m_{J_r})^2 u_{rel}^2(R) + \dots \quad (5.10)$$

In other words, the relative standard uncertainty of the combined standard uncertainty, due to factors that are modeled as a power law, is the exponent of a power-law function multiplied by the relative standard uncertainty of the factor. This holds true for factors either in the numerator or denominator of the model function. Thus, the contribution to the combined standard uncertainty in the calibration factor, due to c_U and U_R , are the relative standard uncertainty factors multiplied by the exponent, m_{J_r} . (Note: J_r is assumed to have no uncertainty.)

That leaves the uncertainty in the exponent to be evaluated.

$$u_{rel}^2(C_\phi) = \dots + \left(\frac{1}{C_\phi}\right)^2 \left(\frac{\partial C_\phi}{\partial (m_{J_r})}\right)^2 u_{rel}^2(m_{J_r}) \cdot m_{J_r}^2 + \dots$$

Again, substituting the value for the relevant sensitivity coefficient from **Section 5.1.3** gives:

$$u_{rel}^2(C_\phi) = \dots + \left[\ln\left(\frac{J_r \cdot R}{U_r \cdot C_U}\right) \cdot m_{J_r} \right]^2 u_{rel}^2(m_{J_r}) + \dots \quad (5.11)$$

Normally, the variation of the bracketed term $\left(\frac{J_r \cdot R}{U_r \cdot C_U}\right)$ varies from one-by-one on the order of 0.1%. If this is the case, then the natural log of that term multiplied by the exponent is much less than 1, particularly when squared; therefore, under normal conditions the contribution to the combined standard uncertainty that is due to the uncertainty of the exponent value can be ignored; however, the relative standard uncertainty in the exponent value is likely to be rather high. If one is not certain of the contribution, **Equation 5.11** can be used to evaluate specific cases.

All the other variables in **Equation 5.1** are similar to others described; i.e., A_r and C_{fr} are similar to ϕ_r and give a sensitivity coefficient of 1, as shown in **Equation 5.8.1**, and $(y_E - y_{Ed})$ and $(y_{x_r} - y_{x_{rd}})$ are similar to $(y_r - y_{rd})$, as shown in **Equation 5.9**.

In conclusion, use of relative standard uncertainties of the variables to obtain the relative combined standard uncertainty only works if the variables are not correlated to each other. It does not work when correlation exists; thus, it has limited application but is useful because the mathematics becomes much simpler.

If the modeled variable is multiplicative, then the sensitivity coefficient is exactly 1 when using relative standard uncertainties for evaluation. If the variable has a subtractive (or additive) term, then the main variable will be slightly different than 1, and the uncertainty of the subtracted (added) term can usually be ignored. If the variable is modeled as a power law, then the relative standard uncertainty needs to be multiplied by the exponent to give the correct uncertainty weighting. These three cases cover all the combinations shown in **Equation 5.1**. By using the above method and derived formulas in **Equations 5.8** through **5.11**, an accurate relative combined standard uncertainty budget can be obtained for the process under consideration, in this case the calibration of an integrating sphere.

5.1.10 Integrating Sphere with Spectrometer Detector. Very few integrating sphere systems use a photometer as the detection device. Array spectrometers have virtually replaced photometers as the preferred detection device because, in addition to total luminous flux data, chromatic data are also collected. Two methods of calibrating an integrating sphere with an array spectrometer for lumen measurements exist. The first is using a total spectral radiant flux standard lamp to calibrate the array spectrometer for spectral responsivity, as described in **Section 5.2**. For these data, the total luminous flux is calculated relying on the uncertainty of the spectral responsivity calibration. The second method is to use a total spectral radiant flux standard lamp that has been calibration for total spectral radiant flux (W/nm^2) and total luminous flux (lm). Typically, a standard lamp calibrated for total luminous flux will have a lower uncertainty than the total luminous flux determined by combining the individual wavelengths of the total spectral radiant flux calibration. The reason for this difference is due to the calibration methods and the calibration chain leading to the SI units and is discussed in more detail in **Section 6.3.1**.

5.1.10.1 Model Equation for Spectrally Integrated Signal. When an array spectrometer is used as the detection device calibrated by a total luminous flux reference lamp, it is treated as a spectrally integrating detector; therefore, the array detector acts as a series of silicon detectors where the stored photo-induced charge over a period is read out, weighted by the spectral luminous efficiency function, divided by the calibration factors, and

summed. The channel detection covers the entire visible region. The photometer signal in **Section 5.1.1**, y_r , is replaced by:

$$y_r = \sum_{i=1}^n V(i) \cdot y_r(i) / C_{\phi}'(i), \quad (5.12)$$

where the subscript **r** refers to a standard lamp property, and:

$C_{\phi}'(i)$ **relative sphere calibration factor for a given channel:** The measurand, in units of signal per watt per millisecond for a given channel. The array spectrometers used in sphere measurement systems typically have an input optic composed of a diffusing material to cosine-correct the light input to an optical fiber. The optical fiber transmits the light to the input aperture of the array spectrometer, so there are no moving parts. The signal is with respect to a channel that correlates to a particular interval of wavelengths. The wavelength becomes an identifier or label for a particular channel or pixel. In the example presented in this section, the channels are labeled by their pixel number, i . The determination of the sphere calibration factor for a given channel, $C_{\phi}(i)$, is described in **Section 5.2**; however, in this application, only the relative sphere calibration factor is required; in effect, this becomes a scalar quantity with an insignificant uncertainty component. It is simply a weighting factor. This means the analysis presented in **Section 5.2** is used to determine the magnitude of the sphere calibration factors.

$y_r(i)$ **sphere array spectrometer signal:** An array spectro meter detects the lamp spectrum in a series of channels of a given bandwidth. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect 10 spectral measurements of the standard lamp. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal, and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9.

$V(i)$ **CIE standard photometric observer:** The function represents the ideal observer, having a relative spectral responsivity curve conforming to the spectral luminous efficiency function for photopic vision. This function is defined in ISO 23539:2005(E)/CIE S 010/E:2004, *Photometry - The CIE System of Physical Photometry*. The CIE standard photometric observer has no uncertainty.

Each channel, i , in the array spectrometer correlates to a particular interval of wavelengths. The wavelength becomes an identifier or label for a particular channel or pixel. The interpolated value of the CIE standard photometric observer function at the center wavelength of each channel should be used in this analysis.

5.1.10.2 Correlation for Spectrally Integrated Signal. Since every channel is measured simultaneously using a single source that is only dependent on a single parameter (lamp current), all the relative sphere calibration factors for a given channel are correlated with each other to some extent, and all the sphere array spectrometer signals are correlated with each other to some extent; therefore, both summations of **Equation 4.6** need to be calculated because the correlation coefficients, $r(x_i, x_j)$, are not zero. It is quite difficult to determine the level or degree of correlation between the channels. The first level of correlation comes from the fact that the channels are detecting light from a single source with a given spectral distribution. If no random components are present, the correlation coefficient between the signal components is equal to 1; but random components exist and their contribution is unknown. The next level of correlation comes from the bandpass function. Red wavelengths and blue wavelengths are not going to be correlated nearly as much as channels of red light that are within the bandpass of the array spectrometer. For the calibration coefficients, besides the fact that the calibration comes from a single source, the data for the standard lamp are highly correlated because an interpolation of the data is normally required, causing a degree of correlation.

The most conservative approach is to set the correlation coefficients, $r(x_i, x_j)$, equal to 1. Then **Equation 4.6** reduces to a sum of the component uncertainties multiplied by the component sensitivity coefficients:

$$u_c(y) = \sqrt{\left[\sum_{i=1}^m \left(\frac{\partial y}{\partial x_i} \right) u(x_i) \right]^2} = \sum_{i=1}^m \left(\frac{\partial y}{\partial x_i} \right) u(x_i) \quad (5.13)$$

5.1.10.3 Uncertainty Analysis for Spectrally Integrated Signal. The sensitivity coefficients for the spectrally integrated signal, y_r , components in **Equation 5.12** are:

$$\frac{\partial y_r}{\partial y_r(i)} = \frac{V(i)}{C'_\phi(i)}$$

Therefore, the combined standard uncertainty for the spectrally integrated signal is:

$$u_c(y_r) = \sum_{i=1}^n \left[\left(\frac{V(i)}{C'_\phi(i)} \right) u(y_r(i)) \right] \quad (5.14)$$

5.2 Calibration Using a Total Spectral Radiant Flux Standard Lamp

5.2.1 Model for an Integrating Sphere with Spectrometer. The scope of this uncertainty budget is the calibration of an integrating sphere measurement system that uses an array spectrometer as the detection device and an incandescent lamp as the calibration source, which is calibrated for watts per nanometer at a particular operating current. The quantity to be determined is the sphere calibration factor per channel, $C_\phi(\lambda)$, determined in counts per watt per millisecond for a given channel (wavelength), and its associated expanded uncertainty. The sphere calibration factor is referenced to the sphere system with the calibration lamp removed.

The model equation is written as:

$$C_\phi(\lambda) = \frac{y_r(\lambda) - y_{r_d}(\lambda)}{\Phi_{r,\lambda}(\lambda)} \cdot \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} \cdot \frac{y_E(\lambda) - y_{E_d}(\lambda)}{y_{X_r}(\lambda) - y_{X_r_d}(\lambda)} \cdot A_r(\lambda) \cdot C_{f_r} \quad (5.15)$$

where:

$C_\phi(\lambda)$ **sphere calibration factor for a given channel:** The measurand, in units of signal per watt per millisecond for a given channel. The array spectrometers used in sphere measurement systems typically have an input optic composed of a diffusing material to cosine-correct the light input to an optical fiber. The optical fiber transmits the light to the input aperture of the array spectrometer, so there are no moving parts. The signal is with respect to a channel that correlates to a particular interval of wavelengths. The wavelength becomes an identifier or label for a particular channel or pixel.

$\Phi_{r,\lambda}(\lambda)$ **standard lamp calibrated in watts per channel:** The total spectral radiant flux for the standard lamp is provided on the calibration certificate in units of watts per nanometer. These data shall be converted from wavelength data to channel data in **Section 5.2.3**. This is a Type B uncertainty because the source of standard uncertainty comes from interpolating data from a calibration certificate. In the example presented in this section, the calibration certificate states 26.585 mW, with an expanded uncertainty ($k = 2$) of 0.110 mW at a wavelength of 555 nm. These values are recorded in Line 1 of **Table 5.10**.

$y_r(\lambda)$ **sphere array spectrometer signal:** An array spectrometer detects the lamp spectrum in a series of channels with a given bandwidth. The lambda variable in this situation represents a channel in the array spectrometer. The only source of wavelength uncertainty is converting the standard lamp data to channel data, discussed in **Section 5.2.3**. Each channel is measured in counts per millisecond, cts/ms.

To determine the signal and standard uncertainty for this component, typically, 10 spectral measurements of the standard lamp are collected. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal

and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to $10 - 1 = 9$. For this example, the signal is the average of the 10 measurements, which is $y_r(\lambda) = 484.593$ cts/ms, with a standard deviation of $u(y_r(\lambda)) = 0.023$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 2 of **Table 5.10**.

$y_{rd}(\lambda)$ **sphere array spectrometer dark signal:** The array spectrometer signal measured when no light is present in the sphere. For array spectrometers this is commonly more of an electronic dark signal than a stray light signal. Each channel is measured in counts per millisecond. This calculation may automatically occur within the software. If this is the case, then $y_r(\lambda) - y_{rd}(\lambda)$ is simplified to $y_r(\lambda)$, and the average signal and standard uncertainty are determined using the method described in the previous definition.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, 10 spectral measurements are typically collected with the standard lamp turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the sphere array spectrometer signal, and averaging if selected. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. For this example, the dark signal is the average of the 10 measurements, which is $y_{rd}(\lambda) = 0.607$ cts/ms, with a standard deviation of $u(y_{rd}(\lambda)) = 0.007$ cts/ms at a wavelength of 555 nm. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 3 of **Table 5.10**.

$$\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}$$

spectral radiance: Uncertainty exists in the operation of the standard lamp. It is expressed as the ratio of spectral radiance for the standard lamp temperature specified in the calibration certificate, compared to the spectral radiance for the actual operating temperature of the standard lamp. This uncertainty is due to the lamp current measurement. The ratio of the spectral radiance and the uncertainty template is presented in **Section 5.2.3**. This is a Type B uncertainty because it is based on combination of Type A and Type B components.

$y_E(\lambda)$ **channel-dependent self-absorption factor for “empty” sphere:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the array spectrometer signal. The recommended method for correcting this error relies on use of an auxiliary lamp to determine the magnitude of the self-absorption. This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the standard lamp is removed from the sphere. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect 10 spectral measurements with the auxiliary lamp on and the standard lamp removed from the sphere. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. For this example, the “empty” sphere signal is the average of the 10 measurements, which is $y_E(\lambda) = 149.261$ cts/ms, with a standard deviation of $u(y_E(\lambda)) = 0.015$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 5 of **Table 5.10**. This is a Type A uncertainty, with degrees of freedom equal to 9.

$y_{Ed}(\lambda)$ **channel-dependent self-absorption factor for “empty” sphere dark signal:** The array spectrometer signal measured when the auxiliary lamp is off and the standard lamp is removed from the sphere. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect 10 spectral measurements with the auxiliary lamp off and the standard lamp removed. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the channel-dependent self-absorption

factor for the “empty” sphere, and averaging if selected. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. For this example, the “empty” sphere signal is the average of the 10 measurements, which is $y_E(\lambda) = 0.610$ cts/ms, with a standard deviation of $u(y_{E_d}(\lambda)) = 0.009$ cts/ms at a wavelength of 555 nm. The signal and the standard uncertainty are recorded in Line 6 of **Table 5.10**. This is a Type A uncertainty with degrees of freedom equal to 9.

$y_{x_r}(\lambda)$ **channel-dependent self-absorption factor for standard lamp:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the array spectrometer signal. The recommended method for correcting this error relies on use of an auxiliary lamp to determine the magnitude of the self-absorption. This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the standard lamp is mounted in the sphere but turned off. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect 10 spectral measurements with the auxiliary lamp on and the standard lamp mounted in the sphere but turned off. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. For this example, the empty sphere signal is the average of the 10 measurements, which is $y_{x_r}(\lambda) = 143.934$ cts/ms, with a standard deviation of $u(y_{x_r}(\lambda)) = 0.013$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 7 of **Table 5.10** (see **Section 5.2.4**). This is a Type A uncertainty, with degrees of freedom equal to 9.

$y_{x_{rd}}(\lambda)$ **channel-dependent self-absorption factor for the standard lamp dark signal:** The array spectrometer signal measured when the auxiliary lamp is off and the standard lamp is mounted in the sphere and turned off. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect 10 spectral measurements with the auxiliary lamp off and the standard lamp mounted in the sphere and turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the channel-dependent self-absorption factor for the standard lamp, and averaging if selected. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. For this example, the empty sphere signal is the average of the 10 measurements, which is $y_{x_{rd}}(\lambda) = 0.598$ cts/ms, with a standard deviation of $u(y_{x_{rd}}(\lambda)) = 0.008$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 8 of **Table 5.10** (see **Section 5.2.4**). This is a Type A uncertainty, with degrees of freedom equal to 9.

$A_r(\lambda)$ **standard lamp age or burning time:** **Figure 5.1** shows a typical reduction in nominal total spectral radiant flux for halogen lamps. The percentage is the reduction per 24 hours of use. Not all lamps that are used as total spectral radiant flux standard lamps meet this condition, and very few are predictable. Many calibration laboratories will provide a statement approximating this dependency on the calibration certificates for the standard lamps they issue. If this information is not available, one should assume that the decay rate is as shown in **Figure 5.1**. This is a Type B uncertainty because it is based on industry knowledge or data from a calibration certificate.

The value of A_r is set to 1. It is difficult and risky to apply a correction factor for the aging of the standard lamp without assuming knowledge. Increasing the standard uncertainty with the operation of the standard lamp is a more conservative approach. For this example, assuming the lamp has been used for 2 hours, the standard uncertainty is estimated at 0.4% divided by 24 hours at a wavelength of 555 nm, then multiplied by the 2 hours of use, resulting in a value of 0.00033. These values are recorded in Line 9 of **Table 5.10** (see **Section 5.2.4**).

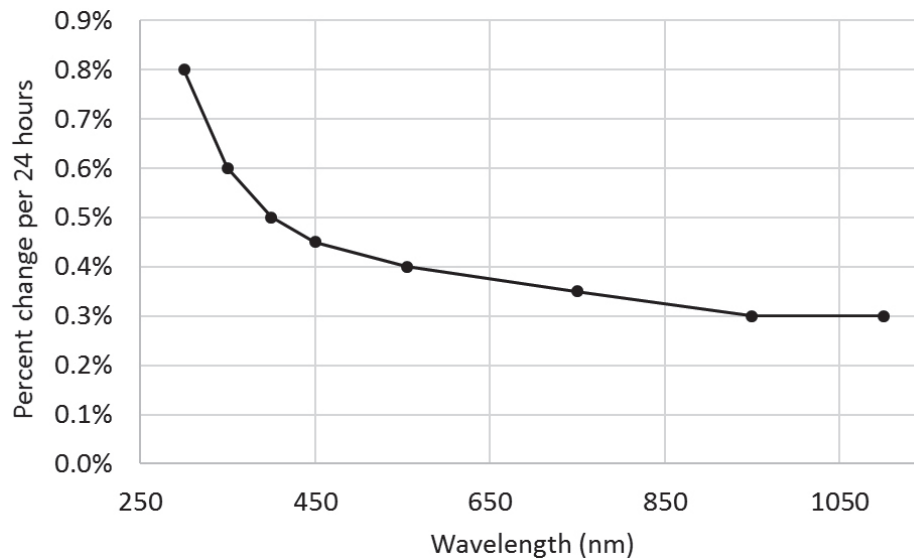


Figure 5.1. The aging rate for a nominal halogen lamp used as a total spectral radiant flux standard lamp.

C_{fr} **additional uncertainty components:** Discussed in **Section 5.2.5**.

5.2.2 Wavelength to Channel Data Conversion. To calibrate an array spectrometer, a calibration file from the data presented in the calibration certificate associated with the standard lamp is created. A sample set of calibration data is presented in **Table D.1** in **Annex D**. For the array spectrometer used in this example, the calibration data in **Table D.1** need to be interpolated. The interpolated data are a column of 1,026 total spectral radiant flux data points that correspond to a wavelength range from 347.87 nm to 1,017.18 nm. Due to the nonlinearities of the wavelength dispersion in the array spectrometer, the wavelength difference between the data points starts at 0.69 nm and becomes smaller until it is 0.62 nm in the near-infrared region. Using a cubic-spline interpolation is one of the better methods to determine the interpolated total spectral radiant flux data points. **Table D.2** shows the interpolated data for the first 20 points, and **Table D.3** shows the interpolated data for the last 20 points.

The standard uncertainty for the interpolated data points can be approximated by interpolating the standard uncertainties from the calibration certificate and accounting for the largest source of uncertainty, which is the wavelength accuracy of the array spectrometer. Since the spectral irradiance scale, and thus spectral radiance flux standard lamps, are calibrated by using a scale that is determined from the radiance temperature of a blackbody at a small interval of wavelengths, the assumption that all the uncertainties at all wavelengths are correlated holds true. A straightforward method of estimating the uncertainty due to wavelength is to shift the original wavelength data by the standard uncertainty of the wavelength scale for the array spectrometer and re-interpolate the total spectral radiant flux data. The manufacturer should provide a wavelength scale uncertainty. For this example, the standard uncertainty of the wavelength scale is 0.25 nm; therefore, 300 nm is now 300.25 nm and 310 nm is 310.25 nm. The interpolated total spectral radiant flux for the shifted wavelength data for the data point labeled *wavelength* 347.87 nm is 1.716 mW/nm instead of 1.718 mW/nm, as shown in **Table D.2**. This is a difference of 0.13%, which needs to be combined with the standard uncertainty interpolated from the certificate. **Table D.2** shows the combined certificate and wavelength uncertainty as the total uncertainty. **Table D.3** shows the results for the last 20 data points. The change in the total spectral radiant flux in the near-infrared becomes flatter; therefore, the wavelength calibration becomes less sensitive. **Figure 5.2** shows the interpolated certificate standard uncertainty, the standard uncertainty due to the wavelength uncertainty, and the total standard uncertainty for the total spectral radiant flux data used to calibrate the array spectrometer.

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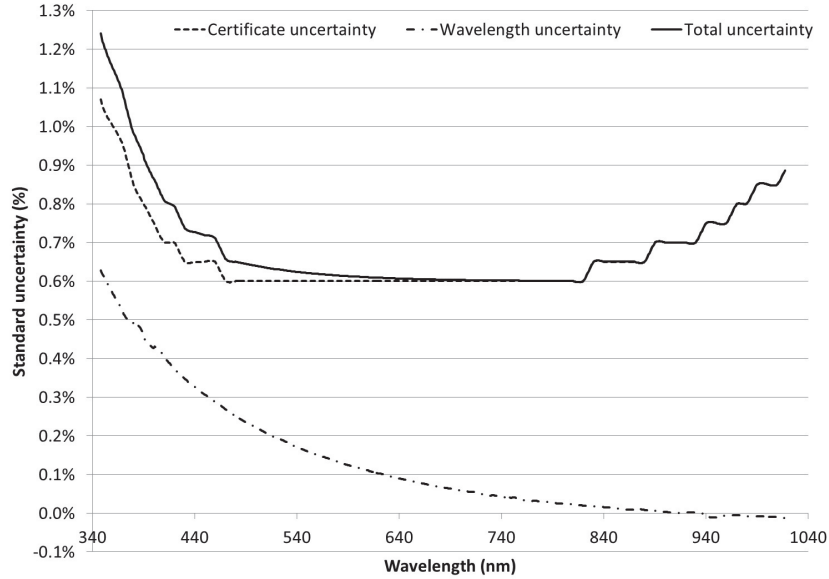


Figure 5.2. The interpolated certificate standard uncertainty, the standard uncertainty due to the wavelength uncertainty, and the total standard uncertainty, each for the total spectral radiant flux data used to calibrate the array spectrometer.

5.2.3 Uncertainty in Spectral Radiance

5.2.3.1 Model Equation. The operation of the standard lamp involves setting the current of the lamp as close as possible to the current specified in the calibration certificate. Deviations in the current cause a change in two correlated components – the overall flux of the standard lamp and the spectral shape of the flux. In order to avoid correlated components, the uncertainty in the current is accounted for by calculating the ratio of the spectral radiance for the standard lamp temperature specified in the calibration certificate, to the spectral radiance for the actual operating temperature of the standard lamp. It should be noted that the lamp is assumed to be approximated by a blackbody, when in reality it is a graybody. An alternative experimental approach based on laboratory measurements of the standard lamp is presented in **Section 5.2.9**. This ratio is described by:

$$\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} = \frac{\exp \left[a \left(\frac{U_R \cdot c_U}{J_R \cdot R} \right)^{n_{J_R}} \right] - 1}{\exp[a] - 1} \quad (5.16)$$

and

$$a = \frac{hc/\lambda}{kT_0} \quad (5.17)$$

where:

h **Plank's constant:** $6.62606957 \times 10^{-34} \text{ m}^2 \cdot \text{kg/s}$, no uncertainty

c **the speed of light:** $2.99792458 \times 10^{17} \text{ nm/s}$, no uncertainty

λ **label or identifier of the array spectrometer channel**, usually correlated to a certain interval of wavelengths

k **Boltzmann constant:** $1.3806488 \times 10^{-23} \text{ m}^2 \cdot \text{kg/s}^2 \cdot \text{K}$, no uncertainty

T_0 **correlated color temperature of the standard lamp:** Usually stated in the report informally. While there is uncertainty in this parameter, no uncertainty is listed this component, because it is negligible. For this example, the correlated color temperature for the data presented in **Table D.1** is 3068 K.

T actual correlated color temperature of the standard lamp: The correlated color temperature of the standard lamp as it is operated.

J_r reference current for the standard lamp: Provided on the calibration certificate. This component does not have uncertainty, because it is stated as a given value on the calibration certificate. In this example, the calibration certificate states 5.825 A.

R resistance of the shunt resistor: The resistor value and uncertainty are listed on the resistor calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 0.100042 Ω , with an expanded uncertainty ($k = 2$) of 0.000012 Ω . The standard uncertainty is the expanded uncertainty divided by 2, or 0.000006 Ω .

U_r lamp current measurement: The lamp current measurement is assumed to be done by measurement of voltage across a calibrated shunt resistor, R . This is a Type A uncertainty because several measurements are collected.

In this example, 30 measurements were made integrating over 10 power-line cycles for each measurement. The average was determined to be 0.582743 V, and the standard uncertainty was 0.000045 V, which is the standard deviation of the 30 points measured.

The degrees of freedom for this measurement are equal to the number of measurements made minus 1, or $30 - 1 = 29$.

c_U voltmeter calibration factor: The voltmeter has a center value close to the nominal reading, with a given uncertainty provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this case, the calibration certificate states 1.000032 with an expanded uncertainty ($k = 2$) of 0.000082. The standard uncertainty is the expanded uncertainty divided by 2, or 0.000041.

n_{J_r} sensitivity of the correlated color temperature with respect to current: For incandescent lamps, including halogen, the change in correlated color temperature with current can be expressed as a ratio of the actual lamp current divided by the desired lamp current raised to an exponent of about 0.648. This is a typical industry standard value. The standard uncertainty of the exponent is 0.005. This is a Type B uncertainty based on industry knowledge.

5.2.3.2 Sensitivity Coefficients. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 5.16** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**. The equations for the partial derivatives follow, along with the line number in **Table 5.8** where the result is recorded. It is important to note that if the ratio of currents is exactly 1, Line 7 goes to zero; therefore, a small factor that is approximately one-tenth of the resolution should be added to this ratio.

$$\frac{\partial \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}}{\partial U_r} = \frac{\exp \left[a \left(\frac{U_r \cdot c_U}{J_r \cdot R} \right)^{n_{J_r}} \right]}{\exp[a] - 1} \cdot \frac{a \cdot n_{J_r}}{U_r} \cdot \left(\frac{U_r \cdot c_U}{J_r \cdot R} \right)^{n_{J_r}} \quad \text{Line 3}$$

$$\frac{\partial \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}}{\partial c_U} = \frac{\exp \left[a \left(\frac{U_r \cdot c_U}{J_r \cdot R} \right)^{n_{J_r}} \right]}{\exp[a] - 1} \cdot \frac{a \cdot n_{J_r}}{c_U} \cdot \left(\frac{U_r \cdot c_U}{J_r \cdot R} \right)^{n_{J_r}} \quad \text{Line 4}$$

$$\frac{\partial \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}}{\partial R} = \frac{-\exp\left[a\left(\frac{U_r \cdot c_U}{J_r \cdot R}\right)^{n_{J_r}}\right]}{\exp[a] - 1} \cdot \frac{a \cdot n_{J_r}}{R} \cdot \left(\frac{U_r \cdot c_U}{J_r \cdot R}\right)^{n_{J_r}} \quad \text{Line 6}$$

$$\frac{\partial \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}}{\partial n_{J_r}} = \frac{\exp\left[a\left(\frac{U_r \cdot c_U}{J_r \cdot R}\right)^{n_{J_r}}\right]}{\exp(a) - 1} \cdot a \ln\left(\frac{U_r \cdot c_U}{J_r \cdot R}\right) \cdot \left(\frac{U_r \cdot c_U}{J_r \cdot R}\right)^{n_{J_r}} \quad \text{Line 7}$$

Table 5.8. Uncertainty Budget for the Spectral Radiance Correction.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	T_0	3,068	K					
2	$a(555nm)$	8.449775	K-1					
3	U_r	0.582743	V	0.000045	A	29	9.400	0.00042
4	c_U	1.000032		0.000041	B	∞	5.477	0.00022
5	J_r	5.825	A					
6	R	0.100042	Ω	0.000006	B	∞	-54.753	-0.00033
7	n_{J_r}	0.648		0.005	B	∞	0.00025	0.00000
	$\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}$	1.00016				103	$u_c \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}$	0.00058

5.2.3.3 Example Uncertainty Budgets. Provided in **Table 5.8** is an uncertainty budget using the **Equation 5.16** as the model and typical values for the uncertainty in spectral radiance.

5.2.3.4 Measurand and Combined Standard Uncertainty. Using **Equation 5.16** and the estimated values for x_i provided in **Table 5.8**, the measurand is calculated: $\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} = 1.00016$. The combined standard uncertainty is the square root of the sum of squares: $u_c\left(\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}\right) = 0.00058$. The combined standard uncertainty in this example is dominated by the standard uncertainty of the standard lamp current measurement. An increase in the number of measurements for the Type A components would reduce the combined standard uncertainty, but the uncertainty is rather small. The combined standard uncertainty does not change with wavelength.

5.2.4 Additional Uncertainty Components. In order to simplify the presentation of the model equation, a separate table of uncertainty components is presented where the components do not change the magnitude of the measurand, $C_\phi(\lambda)$, but do add uncertainty, and the contribution of their uncertainty is relatively small (less than 1/5 that of the largest contributor). If the contribution is large, it should be listed in the model equation as a separate factor. **Table 5.9** lists a few examples. The uncertainty contributions are estimated and the combined standard uncertainty is estimated by the square root of the sum of the squares. These are all Type B uncertainties. The relative standard uncertainty contributions presented in **Table 5.9** are reasonable estimates for industry standard uncertainties. The estimation of C_{f_r} is equal to 1, and the standard uncertainty for C_{f_r} is the combined standard uncertainty of the components listed in **Table 5.9**. The estimation of C_{f_r} and the standard uncertainty for C_{f_r} are recorded in Line 14 of **Table 5.10**.

5.2.4.1 Linearity of the Spectroradiometer (CCD Electron Well). This type of linearity is defined as the efficiency of the detection device turning photons into captured charge. As the detector well fills with charge, the efficiency of the

Table 5.9. Additional Uncertainty Components for the Model of an Integrating Sphere with Spectroradiometer.

Uncertainty Component	Relative Standard Uncertainty Contribution
Linearity of the spectroradiometer (CCD electron well)	0.05 %
Linearity of the spectroradiometer (CCD timing)	0.02 %
Near-field absorption	0.02 %
Sphere and lamp ambient temperature	0.02 %
Stray light from the room	0.02 %
Spectral stray light within spectroradiometer	0.05 %
Combined standard uncertainty	0.08 %

Table 5.10. Uncertainty Budget for an Integrating Sphere Using an Array Spectroradiometer with a High-Quality Total Spectral Radiant Flux Standard Lamp for $\lambda = 555$ nm.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	$\Phi_{r,d}(\lambda)$	26.585	mW	0.110	B	∞	-0.710	-0.0781
2	$y_r(\lambda)$	484.593	cts/ms	0.023	A	9	0.039	0.0009
3	$y_{r,d}(\lambda)$	0.607	cts/ms	0.007	A	9	-0.039	-0.0003
4	$\frac{L_E(\lambda, T_0)}{L_E(\lambda, T)}$	1.00016		0.00058	B	∞	18.880	0.0109
5	$y_E(\lambda)$	149.261	cts/ms	0.015	A	9	0.127	0.0019
6	$y_{E,d}(\lambda)$	0.610	cts/ms	0.009	A	9	-0.127	-0.0011
7	$y_{x_r}(\lambda)$	143.934	cts/ms	0.013	A	9	-0.132	-0.0017
8	$y_{x_{r,d}}(\lambda)$	0.598	cts/ms	0.008	A	9	0.132	0.0011
9	A_r	1.00000		0.00033	B	∞	18.883	0.0062
10	C_{fr}	1.0000		0.0008	B	∞	18.883	0.0151
	$C_\phi(\lambda)$	18.880	cts/ms·mW			large	$u_c(C_\phi)$	0.081

photon-to-charge conversion changes non-linearly. Ideally, any change in the light flux launched into the spectrometer should lead to a proportional change in the detector signal; otherwise, the system is not suitable for radiometric measurements. A spectrograph that is a charge-coupled device (CCD) is inherently non-linear. It is recommended that the manufacturer's non-linearity correction coefficients be used if available. For the calibration of the spectroradiometer, the standard uncertainty due to this component is very small because the signal level is constant and does not significantly change (on the order of 0.05%). This component becomes more significant when test lamps are measured.

5.2.4.2 Linearity of the spectroradiometer (CCD timing). Integration time is the period that the spectrometer detector is exposed to light. An integration time of 1 ms is the typical lower limit for most CCD-based spectrometers, while 10 μ s is the typical lower limit for on-chip electronic shuttering. A shorter integration time means faster on-line device test processing, as well as greater dynamic range for measuring high-brightness sources. The integration time shall be chosen to ensure that the detector does not become saturated. In addition, the integration time needs to be long enough to ensure that for devices not having a constant optical output (i.e., those with flicker) several cycles are sampled.

This type of linearity is associated with the signal versus the integration time. The uncertainty of the integration time will be larger for short integration times compared to longer integration times. The effect is referred to as “smearing.” The best method to avoid smearing effects is to use integration times similar to the calibration integration time. For the calibration of the spectroradiometer, the standard uncertainty due to this component is very small because the integration time does not significantly change (on the order of 0.02%). This component becomes more significant when test lamps are measured.

5.2.4.3 Near-Field Absorption. Near-field absorption is described in ANSI/IES LM-78-20 Section 7.3.4. A well designed system should have a small contribution from near-field absorption. As the sphere diameter decreases, self-absorption of the reference lamp becomes near-field absorption, which is not corrected by the auxiliary lamp measurement. Near-field absorption can be wavelength dependent because the absorption curve for the sphere reflectivity is not spectrally flat. For spheres larger than 1 m in diameter, the near-field absorption should be small. For this example, the standard uncertainty due to near-field absorption is approximated as 0.05%. For spheres smaller than 1 m, the near-field absorption error can be large and is difficult to assess. Using a series of calibrated lamps that differ by size (surface area) is a method of approximating the near-field absorption for smaller spheres with respect to the surface area of the standard lamp.

5.2.4.4 Lamp and Sphere Ambient Temperature. For an incandescent lamp, including halogen, the ambient temperature does not normally affect the operation, unless the sphere is unusually small. If the sphere is small enough to affect the operation of the lamp, then the sphere coating reflectance most likely will be affected by the ambient temperature. The lamp ambient temperature may be wavelength dependent because the lamp would produce a slightly different correlated color temperature. (Refer to ANSI/IES LM-78-20, Section 7.3.5 for more details concerning sphere coating reflectivity and ambient temperature.) For typical measurement systems and this example, the standard uncertainty due to lamp and sphere ambient temperature is approximately 0.05%.

5.2.4.5 Stray Light from the Room. Stray light from the room is typically very small or non-existent. Possible sources of room light include open ports in the sphere, the seam where the sphere halves close, and gaps around detectors or a wire feed-through. The stray light from the room is typically wavelength dependent because the room light is not spectrally flat. If the measurements are normally taken with the room lights on, then the stray light contribution can be approximated by measuring a dark signal with the lights on and off. For typical measurement systems and this example, the standard uncertainty due to room stray light is approximately 0.02%.

5.2.4.6 Spectral Stray Light within the Spectroradiometer. Stray light within the spectroradiometer is the measured quantity of light that reaches the detector from a wavelength other than that intended or selected. Therefore, stray light can cause measurement results that are erroneously high. Stray light is more of a problem with spectrometers than with monochromators, due to size constraints. In general, stray light is evenly distributed along the entire length of the array. However, due to the fact that the peak of the spectral distribution of the most common calibration source (the QTH lamp) and the peak of the spectral responsivity of the array are at about 900 nm, the stray light appears to be concentrated in the UV. If a source is then measured that does not have a similar spectrum, the blue signals will have large stray-light errors. The best solution is to have a cutoff filter in the system that eliminates the near-infrared light for the calibration source and any source to be measured.

For the calibration of the spectroradiometer, the standard uncertainty due to this component is very small because the spectral stray light within the spectroradiometer is constant for a given standard lamp (on the order of 0.05%). This component may become much more significant when test lamps are measured.

5.2.5 Integrating Sphere with Spectroradiometer Sensitivity Coefficients. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 5.15** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**. The equations for the partial derivatives follow, along with the line number in **Table 5.10** where the result is recorded.

$\frac{\partial C_{\phi(\lambda)}}{\partial \Phi_{r,\lambda}(\lambda)} = \frac{-C_{\phi(\lambda)}}{\Phi_{r,\lambda}(\lambda)}$	Line 1	$\frac{\partial C_{\phi(\lambda)}}{\partial y_r(\lambda)} = \frac{C_{\phi(\lambda)}}{y_r(\lambda) - y_{r_d}(\lambda)}$	Line 2
$\frac{\partial C_{\phi(\lambda)}}{\partial y_{r_d}(\lambda)} = \frac{-C_{\phi(\lambda)}}{y_r(\lambda) - y_{r_d}(\lambda)}$	Line 3	$\frac{\partial C_{\phi(\lambda)}}{\partial \left[\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} \right]} = \frac{-C_{\phi(\lambda)}}{\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}}$	Line 4
$\frac{\partial C_{\phi(\lambda)}}{\partial y_E(\lambda)} = \frac{C_{\phi(\lambda)}}{y_E(\lambda) - y_{E_d}(\lambda)}$	Line 5	$\frac{\partial C_{\phi(\lambda)}}{\partial y_{E_d}(\lambda)} = \frac{-C_{\phi(\lambda)}}{y_E(\lambda) - y_{E_d}(\lambda)}$	Line 6
$\frac{\partial C_{\phi(\lambda)}}{\partial y_{x_r}(\lambda)} = \frac{C_{\phi(\lambda)}}{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}$	Line 7	$\frac{\partial C_{\phi(\lambda)}}{\partial y_{x_{r_d}}(\lambda)} = \frac{-C_{\phi(\lambda)}}{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}$	Line 8
$\frac{\partial C_{\phi(\lambda)}}{\partial A_r(\lambda)} = \frac{-C_{\phi(\lambda)}}{A_r(\lambda)}$	Line 9	$\frac{\partial C_{\phi(\lambda)}}{\partial C_{f_r}(\lambda)} = \frac{-C_{\phi(\lambda)}}{C_{f_r}(\lambda)}$	Line 10

5.2.6 Example Uncertainty Budget. Provided in **Table 5.10** is an uncertainty budget using **Equation 5.15** as the model and typical values (see **Section 5.2.1**) for a high-quality sphere system with an array spectroradiometer using a high-quality total spectral radiant flux standard lamp.

5.2.7 Measurand and Combined Standard Uncertainty. Using **Equation 5.15** and the estimated values for x_i provided in **Table 5.10**, the measurand is calculated: $C_{\phi}(\lambda) = 18.880$ cts/ms·mW at 555 nm.

The combined standard uncertainty is the square root of the sum of squares, $u_c[C_{\phi}(\lambda)] = 0.081$ cts/ms·mW, or 0.43% as a relative combined standard uncertainty. The combined standard uncertainty in this example is dominated by the standard uncertainty of the standard lamp at all the wavelength channels.

5.2.8 Degrees of Freedom and Expanded Uncertainty. Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), the effective degrees of freedom for this uncertainty budget is extremely large, which leads to a coverage factor approximately equal to 2, resulting in an expanded uncertainty of $U[C_{\phi}(\lambda)] = 0.16$ cts/ms·mW, or 0.86% as a relative expanded uncertainty at 555 nm. The final statement of uncertainty would be a table that shows the results for all the channels.

5.2.9 Model Modifications. The model **Equation 5.15** is a starting point. Based on the equipment used by the laboratory, the model equation should be modified to represent the laboratory. In many cases, the dark signal is automatically measured with each measurement, which is a good practice. In this case, uncertainty in the dark signal is convolved into the uncertainty of the signal when it is measured as stated above. The model equation then becomes:

$$C_{\phi}(\lambda) = \frac{y_r(\lambda)}{\Phi_r(\lambda)} \cdot \frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} \cdot \frac{y_E(\lambda)}{y_{x_r}(\lambda)} \cdot C_{f_r} \quad (5.18)$$

Another potential model modification is an alternative approach to accounting for the uncertainty in the spectral radiance involving laboratory measurements of the standard lamp at different currents. From the spectral data collected at different currents, the exponential relationship between the reference current and the set current is determined. The model equation is:

$$C_{\phi}(\lambda) = \frac{y_r(\lambda) - y_{r_d}(\lambda)}{\Phi_{r,d}(\lambda)} \cdot \left(\frac{J_r}{J \cdot c_J} \right)^{S_r(\lambda)} \cdot \frac{y_E(\lambda) - y_{E_d}(\lambda)}{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)} \cdot A_r(\lambda) \cdot C_{f_r} \quad (5.19)$$

where:

$S_r(\lambda)$ **sensitivity of the spectral data with respect to current and wavelength:** For incandescent lamps, including halogen, the change in spectral radiance with current can be expressed as a ratio of the reference current to the actual lamp current raised to an exponent that is dependent on wavelength. The sensitivity of spectral data with respect to current and wavelength is determined experimentally by adjusting current (increase and decrease) around the set point and measuring the spectral power distribution.

J_r **reference current for the standard lamp:** The reference current for the standard lamp is provided on the calibration certificate. This component does not have uncertainty because it is stated as a given value on the calibration certificate.

J **measured current for the standard lamp:** The lamp current measurement for this model is done by measuring the current using a calibrated ammeter. This is a Type A uncertainty because several measurements are collected.

c_J **ammeter calibration factor:** The ammeter has a center value close to the nominal reading, with a given uncertainty provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

Figure 5.3 shows the spectral radiance of a standard lamp operated at a variety of currents. **Figure 5.4** shows the sensitivity of spectral data with respect to current and wavelength, and **Figure 5.5** shows the uncertainty of $S_r(\lambda)$.

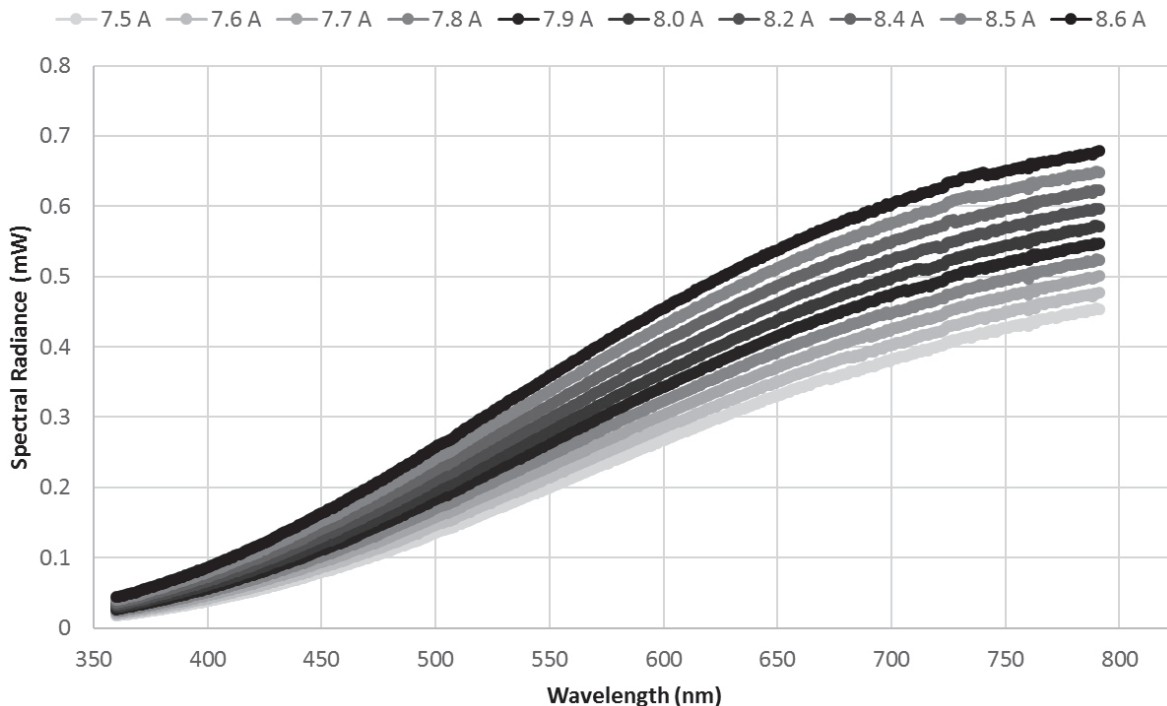


Figure 5.3. Spectral radiance of a standard lamp operated at a variety of currents.

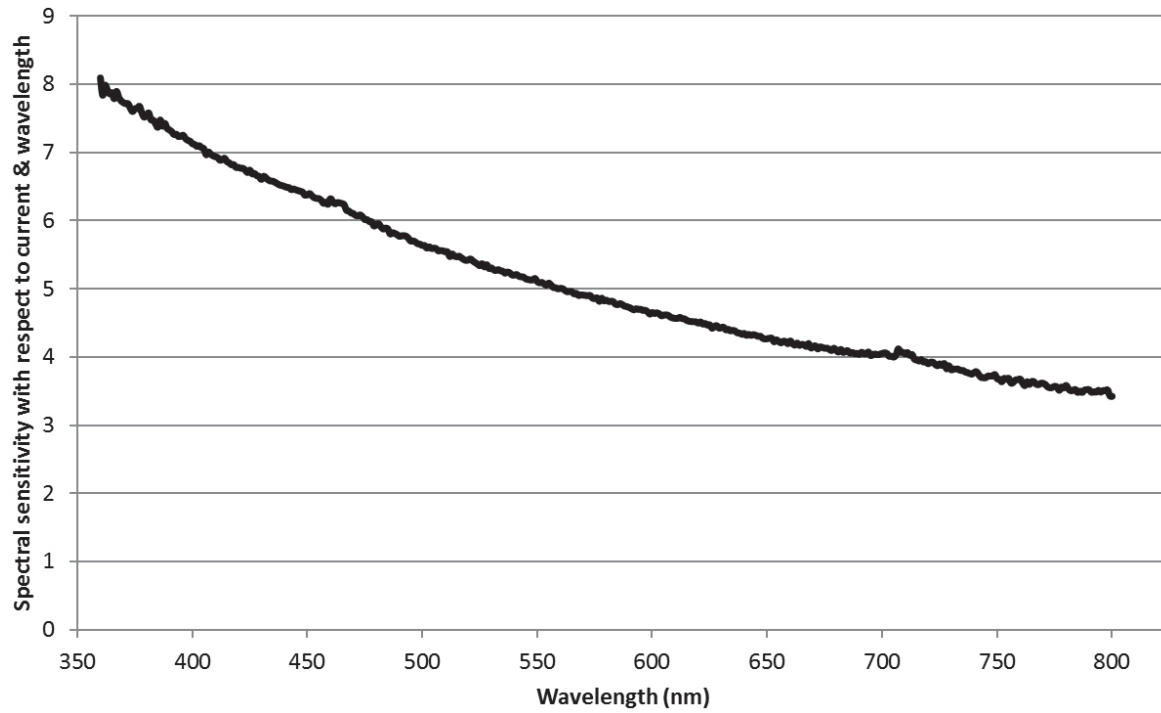


Figure 5.4. Determination of $S_r(\lambda)$ based on a fit of the experimental data shown in Figure 5.3.

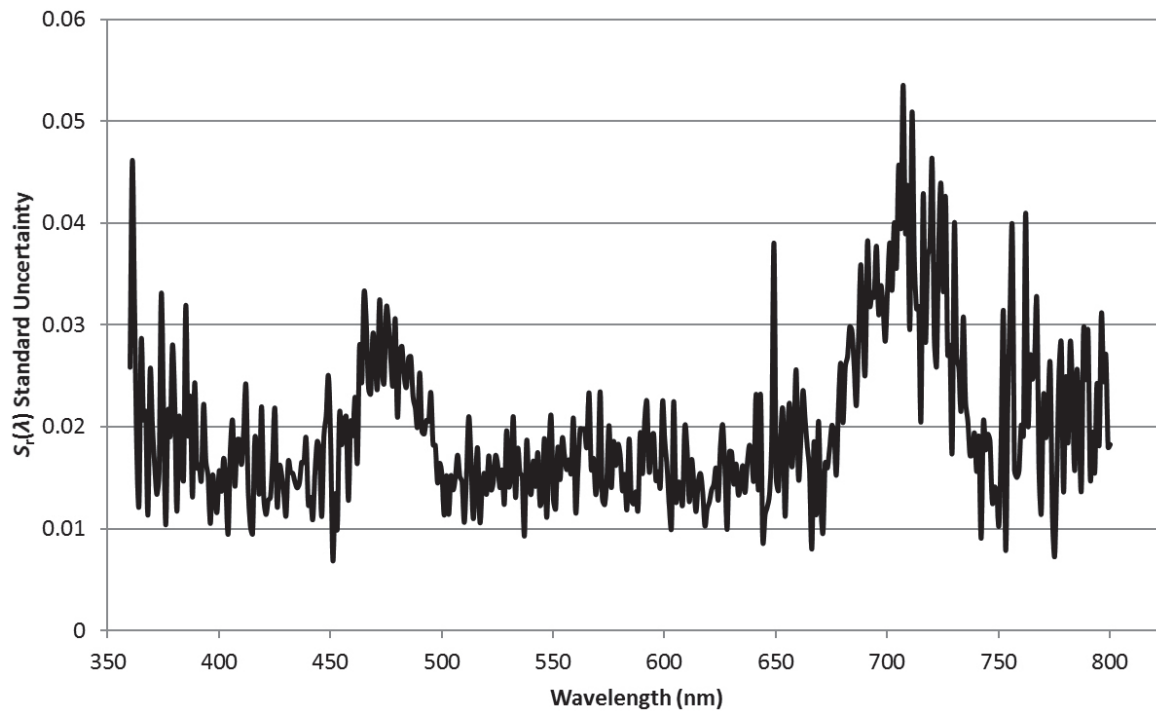


Figure 5.5. Standard uncertainty of $S_r(\lambda)$ determined from the fit of experimental data.

6.0 Calibration of Working Standard Lamps Using an Integrating Sphere

6.1 Creating Total Luminous Flux Working Standard Lamps

6.1.1 Model for an Integrating Sphere with Photometer. The scope of this uncertainty budget is the calibration of a working standard lamp using an integrating sphere measurement system that uses a photometer as the detection device and a primary standard lamp that is similar in current, correlated color temperature, shape, size, and luminous intensity distribution to the working standard lamp and is calibrated for total luminous flux (lumens) at a particular operating current. The quantity to be determined is the luminous flux (lumens) of the working standard lamp, Φ , and its associated expanded uncertainty, $U(\Phi)$.

The model equation that describes how to calculate the total luminous flux, Φ , and the uncertainty components affecting the determination of the total luminous flux for a working standard lamp is:

$$\Phi = \frac{(y - y_d)}{C_\phi} \cdot \left(\frac{U \cdot c_U}{J \cdot R} \right)^{m_J} \cdot \frac{(y_E - y_{E_d})}{(y_X - y_{X_d})} \cdot C_f \quad (6.1)$$

For many measurement systems, C_ϕ is hidden in the software parameters after a calibration cycle occurs. The working standard lamp is measured, and $(y - y_d)$ is simply the displayed lumen value. The operating current of the working standard lamp and the self-absorption correction still need to be considered as far as uncertainty. The potential to eliminate a few uncertainty components exist by substituting **Equation 5.1** for the sphere calibration factor, C_ϕ .

The result is:

$$\Phi = \frac{\Phi_r (y - y_d)}{(y_r - y_{r_d})} \cdot \left(\frac{J_r \cdot R}{U_r \cdot c_U} \right)^{m_{J_r}} \cdot \left(\frac{U \cdot c}{J \cdot R} \right)^{m_J} \cdot \frac{(y_{X_r} - y_{X_{r_d}})}{(y_X - y_{X_d})} \cdot A_r \cdot \frac{C_f}{C_{f_r}} \quad (6.2)$$

If the working standard lamp does not have a specific current set, the component involving the measurement of the working standard lamp current can be removed from the equation. Many times the laboratory will set the working standard lamp current in order to ensure that the correlated color temperature of the lamp is a fixed known value, such as 2856 K. If the laboratory has not fixed the working standard lamp current for other reasons, there is no uncertainty the first time the working standard lamp is measured because the lamp current is arbitrary. The second time the working standard lamp is measured, setting the lamp current does cause uncertainty because it needs to be set to the lamp current conditions stated during the first measurement.

If the laboratory uses exactly the same equipment to transfer the scale of the primary standard lamp to the new working standard lamp, many components can be eliminated. This situation only applies if the measurements are done in a timely fashion (usually the same day) such that the equipment has not had time to drift. If the new working standard lamp is the same model of lamp as the primary standard lamp, the sensitivity of luminous flux with respect to current is the same for both lamps ($m_{J_r} = m_J$). The model equation can be reduced to **Equation 6.3**, where the uncertainty in the voltmeter calibration and reference shunt calibration cancel.

$$\Phi = \Phi_r \cdot \frac{(y - y_d)}{(y_r - y_{r_d})} \cdot \left(\frac{J_r \cdot U}{U_r \cdot J} \right)^{m_{J_r}} \cdot \frac{(y_{X_r} - y_{X_{r_d}})}{(y_X - y_{X_d})} \cdot A_r \cdot \frac{C_f}{C_{f_r}}, \quad (6.3)$$

where subscript **r** refers to a primary standard lamp property, and:

Φ **total luminous flux of the working standard lamp:** The measurand, in lumens.

Φ_r **total luminous flux of the primary standard lamp:** The total luminous flux for the primary standard lamp, in lumens, is provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In the example presented in this section, the calibration certificate states 2,800.0 lm, with an expanded uncertainty ($k = 2$) of 16.4 lm. Because the coverage factor is equal to 2, the standard uncertainty is the expanded uncertainty divided by 2, or 8.2 lm. These values are recorded in Line 1 of **Table 6.1**.

y_r **sphere photometer signal for the primary standard lamp:** The photometer signal is a photocurrent, which may be measured using various types of equipment. The photocurrent can be measured with an ammeter that is calibrated and has a certificate with stated uncertainties. Another method is to use a transimpedance amplifier to convert the photocurrent to a voltage. The transimpedance amplifier and the voltmeter are calibrated and have certificates with stated uncertainties. However, each new piece of equipment may add additional uncertainty components. A third possibility, where the user does not have access to the measurement process, is software that reports a lumen value as the raw signal.

To determine the signal and standard uncertainty for this component, 10 measurements are typically made of the primary standard lamp operated with the stated current. For this example, the signal is the average of the 10 measurements, which is $y_r = 1.7652$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_r) = 0.0007$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9 ($10 - 1$). The signal and standard uncertainty are recorded in Line 2 of **Table 6.1**.

y_{rd} **sphere photometer dark signal for the primary standard lamp:** The photometer signal measured when the primary standard lamp is mounted in the sphere but not operating. To determine the dark signal and standard uncertainty for this component, 10 measurements of the primary standard lamp are typically made while it is not operating. For this example, the signal is the average of the 10 measurements, which is $y_{rd} = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{rd}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 3 of **Table 6.1**.

y **sphere photometer signal for the working standard lamp:** The photometer signal is a photocurrent, which may be measured using various types of equipment. To determine the signal and standard uncertainty for this component, 10 measurements of the working standard lamp operated at the stated or determined current are typically made. For this example, the signal is the average of the 10 measurements, which is $y = 1.7481$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y) = 0.0008$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 4 of **Table 6.1**.

y_d **sphere photometer dark signal for the working standard lamp:** The photometer signal measured when the working standard lamp is mounted in the sphere but not operating. To determine the dark signal and standard uncertainty for this component, 10 measurements of the working standard lamp are typically made while it is not operating. For this example, the signal is the average of the 10 measurements, which is $y_d = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_d) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 5 of **Table 6.1**.

J_r **reference current for the primary standard lamp:** The reference current (in amps) for the primary standard lamp is provided on the calibration certificate. This component does not have uncertainty, because it is stated as a given value on the calibration certificate. In this example, the calibration certificate states 5.825 A. This value is recorded in Line 6 of **Table 6.1**.

- U_r **lamp current measurement for the primary standard lamp:** The lamp current measurement for this model is done by measuring the voltage across a calibrated shunt resistor, R . The model equation also applies if the primary standard lamp current is measured directly in amps. This is a Type A uncertainty because several measurements are collected. In this example, 30 measurements are made, integrated over 10 power-line cycles. The average of the 30 measurements is 0.582743 V, and the standard deviation is 0.000015 V, which is the standard uncertainty, $u(U_r)$. The degrees of freedom for this measurement are 29 (30 – 1). The average, standard uncertainty, and degrees of freedom are recorded in Line 7 of **Table 6.1**.
- J **reference current for the working standard lamp:** The reference current (in amps) for the working standard lamp is determined by measuring the spectral output of the lamp and setting the current to provide the desired correlated color temperature. This is a Type B uncertainty determination based on a previous measurement. In this example, the previous measurement determined the current to be 5.814 A. This value is recorded in Line 8 of **Table 6.1**.
- U **lamp current measurement for the working standard lamp:** The lamp current measurement for this model is done by measuring the voltage across a calibrated shunt resistor, R . The model equation also applies if the working standard lamp current is measured directly in amps. This is a Type A uncertainty because several measurements are collected. In this example, 30 measurements are made, integrated over 10 power-line cycles. The average of the 30 measurements is 0.581643 V, and the standard deviation is 0.000014 V, which is the standard uncertainty, $u(U)$. The degrees of freedom for this measurement are 29. The average, standard uncertainty, and degrees of freedom are recorded in Line 9 of **Table 6.1**.
- m_{J_r} **sensitivity of luminous flux with respect to current:** For incandescent lamps, including halogen, the change in lumen output with current can be expressed as a ratio of the actual lamp current to the desired lamp current, and the ratio then raised to an exponent of 6.5, which has a standard uncertainty of 0.2. These are typical industry standard values. This is a Type B uncertainty because it is based on industry knowledge. These values are recorded in Line 10 of **Table 6.1**.
- y_{X_r} **self-absorption signal for the primary standard lamp:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall back onto the lamp, thus reducing the integrating sphere photometer responsivity. The recommended method for correcting this error relies upon the use of an auxiliary lamp to determine the magnitude of the self-absorption.
- The photometer signal is measured when the auxiliary lamp is on and the primary standard lamp is mounted in the sphere but not operating. For this example, the signal is the average of the 10 measurements, which is $y_{X_r} = 0.5243$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{X_r}) = 0.0004$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 11 of **Table 6.1**.
- $y_{X_{rd}}$ **self-absorption dark signal for the primary standard lamp:** The photometer signal measured when the auxiliary lamp is off and the primary standard lamp is mounted in the sphere but not operating. For this example, the signal is the average of the 10 measurements, which is $y_{X_{rd}} = 0.0002$ nA. The standard deviation calculated from the 10 measurements, which is the standard uncertainty, $u(y_{X_{rd}}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 12 of **Table 6.1**.
- y_X **self-absorption factor for the working standard lamp:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall back onto the lamp, thus reducing the integrating sphere photometer responsivity. The recommended method for correcting this error relies upon the use of an auxiliary lamp to determine the magnitude of the self-absorption.
- The photometer signal is measured when the auxiliary lamp is on and the working lamp is mounted in the sphere but not operating. For this example, the signal is the average of the 10 measurements, which is $y_X = 0.5286$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty,

$u(y_X) = 0.0004$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 13 of **Table 6.1**.

y_{X_d} **self-absorption dark signal for the working standard lamp:** The photometer signal measured when the auxiliary lamp is off and the working standard lamp is mounted in the sphere but not operating. For this example, the signal is the average of the 10 measurements, which is $y_{X_d} = 0.0002$ nA. The standard deviation calculated from the 10 measurements is the standard uncertainty, $u(y_{X_d}) = 0.0002$ nA. This is a Type A uncertainty, with degrees of freedom equal to 9. The signal and standard uncertainty are recorded in Line 14 of **Table 6.1**.

A_r **primary standard lamp age or burning time:** The operation of a good incandescent (including halogen) standard lamp typically reduces the lumen output by 0.4% over a period of 24 hours. Not all lamps meet this condition, and very few are predictable. Many calibration laboratories will provide a statement approximating this dependency on the calibration certificates for the standard lamps they issue. If this information is not available, the decay rate is assumed to be 0.4% per 24 hours. This is a Type B uncertainty because it is based on industry knowledge or data from a calibration certificate.

The value of A_r is set to 1. It is difficult and risky to apply a correction factor for the aging of the standard lamp without assuming knowledge. Increasing the standard uncertainty with the operation of the primary standard lamp is a more conservative approach. For this example, assuming the lamp has been used for 2 hours, the standard uncertainty is estimated at 0.4% divided by 24 hours, then multiplied by the 2 hours of use, resulting in a value of 0.00033. These values are recorded in Line 15 of **Table 6.1**.

$\frac{C_f}{C_{f_r}}$ **additional uncertainty components:** Discussed in **Section 6.1.2**.

6.1.2 Additional Uncertainty Components. Since the working standard lamp is similar in current, correlated color temperature, shape, size, and luminous intensity distribution to the primary standard lamp, many of the additional uncertainty components are highly negatively correlated, such that they cancel. The only additional component that may remain is the spatial responsivity of the sphere. Spheres have a spatial response to the initial ray of light from the lamp. Even though the primary standard lamp and working standard lamp have very similar luminous intensity distributions, it is unlikely the distributions will be aligned with the sphere. The estimation of $\frac{C_f}{C_{f_r}}$ is equal to 1, and the standard uncertainty for the additional uncertainty component—which is the spatial responsivity of the sphere, $u\left(\frac{C_f}{C_{f_r}}\right)$ —is estimated to be 0.10 %. The magnitude of this uncertainty component can be estimated by rotating the working standard lamp within the sphere and measuring the signal.

6.1.3 Integrating Sphere with Photometer Sensitivity Coefficients for Working Standard Lamps. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 6.3** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**. The equations for the partial derivatives follow, along with the line number in **Table 6.1** where the result is recorded.

$\frac{\partial \Phi}{\partial \Phi_r} = \frac{\Phi}{\Phi_r}$	Line 1	$\frac{\partial \Phi}{\partial y_r} = \frac{-\Phi}{(y_r - y_{rd})}$	Line 2
$\frac{\partial \Phi}{\partial y_{rd}} = \frac{-\Phi}{(y_r - y_{rd})}$	Line 3	$\frac{\partial \Phi}{\partial y} = \frac{\Phi}{(y - y_d)}$	Line 4
$\frac{\partial \Phi}{\partial y_d} = \frac{-\Phi}{(y - y_d)}$	Line 5	$\frac{\partial \Phi}{\partial U_r} = \frac{-m_{J_r} \cdot \Phi}{U_r}$	Line 7
$\frac{\partial \Phi}{\partial U} = \frac{m_{J_r} \cdot \Phi}{U}$	Line 9	$\frac{\partial \Phi}{\partial m_{J_r}} = \Phi \cdot \ln \left(\frac{J_r \cdot U}{U_r \cdot J} \right)$	Line 10
$\frac{\partial \Phi}{\partial y_{x_r}} = \frac{\Phi}{y_{x_r} - y_{x_{rd}}}$	Line 11	$\frac{\partial \Phi}{\partial y_{x_{rd}}} = \frac{-\Phi}{y_{x_r} - y_{x_{rd}}}$	Line 12
$\frac{\partial \Phi}{\partial y_x} = \frac{\Phi}{y_x - y_{x_d}}$	Line 13	$\frac{\partial \Phi}{\partial y_{x_d}} = \frac{-\Phi}{y_x - y_{x_d}}$	Line 14
$\frac{\partial \Phi}{\partial A_r} = \frac{\Phi}{A_r}$	Line 15	$\frac{\partial \Phi}{\partial C_f / C_{f_r}} = \frac{\Phi}{C_f / C_{f_r}}$	Line 16

6.1.5 Example Uncertainty Budget. Provided in **Table 6.1** is an uncertainty budget using **Equation 6.3** as the model and typical values (see **Section 6.1.1**) for a high-quality sphere system with a photometer detector using a high-quality luminous flux primary standard lamp to calibrate a working standard lamp.

6.1.6 Measurand and Combined Uncertainty. Using **Equation 6.3** and the estimated values for x_i provided in **Table 6.1**, the measurand is calculated: $\Phi = 2750.3$ lm. The combined standard uncertainty is the square root of the sum of squares: $u_c(\Phi) = 9.4$ lm, or 0.34% as a relative combined standard uncertainty.

6.1.7 Degrees of Freedom and Expanded Uncertainty. Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), the effective degrees of freedom, ν_{eff} , for this uncertainty budget is 1,546. Using an effective degrees of freedom of 1,546 and the Student's t approximation for coverage factor (**Equation 4.12**, **Section 4.5**), the coverage factor is approximately 2, resulting in an expanded uncertainty of $U(\Phi) = 19$ lm, or 0.68% as a relative expanded uncertainty. The uncertainty community typically rounds the coverage factor to 2.00 if the calculated value is within 0.05 of 2.00. The final statement of uncertainty would be:

The total luminous flux of the working standard lamp calibrated on [date] was determined, traceable to the candela SI unit, with a magnitude of 2,750.3 lm and an associated expanded uncertainty of 19 lm, or a relative expanded uncertainty of 0.68%. The stated uncertainty was calculated according to the GUM by a multiplication of the standard uncertainty with a coverage factor of $k = 2$.

Table 6.1. Uncertainty Budget for an Integrating Sphere Using a Photometer Detector and a Total Luminous Flux Primary Standard Lamp to Calibrate a Working Standard Lamp.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	Φ_r	2,800.0	lm	8.2	B	∞	0.9823	8.055
2	y_r	1.7652	nA	0.0007	A	9	-1,558.3	-1.091
3	y_{rd}	0.0002	nA	0.0002	A	9	1,558.3	0.312
4	y	1.7481	nA	0.0008	A	9	1,573.5	1.259
5	y_d	0.0002	nA	0.0002	A	9	-1,573.5	-0.315
6	J_r	5.825	A					
7	U_r	0.582743	V	0.000015	A	9	-30,677	-0.460
8	J	5.814	A					
9	U	0.581643	V	0.000014	A	9	30,735	0.430
10	m_{Jr}	6.5		0.2	B	∞	0.0022	0.000
11	y_{xr}	0.5243	nA	0.0004	A	9	5,247.7	2.099
12	y_{xrd}	0.0002	nA	0.0002	A	9	-5,247.7	-1.050
13	y_x	0.5286	nA	0.0004	A	9	-5,205.0	-2.082
14	y_{xd}	0.0002	nA	0.0002	A	9	5,205.0	1.041
15	A_r	1.0000		0.00033	B	∞	2,750.3	0.825
16	C_f/C_{fr}	1.0000		0.0010	B	∞	2,750.3	2.750
	Φ	2,750.3	lm				$u_c(\Phi)$	9.4

6.1.8 Model Modifications. The model **Equation 6.3** is a starting point. Based on the equipment used by the laboratory, the model equation should be modified to represent the laboratory. The example presented in this section is for a laboratory that uses an integrating sphere with a photometer to transfer the scale from a primary standard lamp operated on DC current to a working standard lamp operated on AC voltage. The AC-voltage lamp is then used to calibrate a goniophotometer system that only has AC power available. The model equation describing this example is:

$$\Phi = \frac{\Phi_r(y - y_d)}{(y_r - y_{rd})} \cdot \left(\frac{J_r \cdot R}{U_r \cdot c} \right)^{m_{Jr}} \cdot \left(\frac{U \cdot c_U}{U_w} \right)^{m_{Uw}} \cdot \frac{(y_{xr} - y_{xrd})}{(y_x - y_{xd})} \cdot A_r \cdot \frac{C_f}{C_{fr}}, \quad (6.4)$$

where many of the components are described in **Section 6.1.1**, and additional components are:

- c** **voltmeter calibration factor:** The voltmeter has a center value close to the nominal reading, with a given uncertainty provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 1.000032, with an expanded uncertainty ($k = 2$) of 0.000008. The standard uncertainty is the expanded uncertainty divided by 2, or 0.000004. These values are recorded in line 8 of **Table 6.2**.

- R** **resistance of the shunt resistor:** The resistor value and uncertainty are listed on the resistor calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate.

In this example, the calibration certificate states 0.100045 Ω , with an expanded uncertainty ($k = 2$) of 0.000006 Ω . The standard uncertainty is the expanded uncertainty divided by 2, or 0.000003 Ω . These values are recorded in Line 9 of **Table 6.2**.

- U_w **reference voltage for the working standard lamp:** The reference voltage (in volts) for the working standard lamp is determined prior to the test. The reference voltage is set by determining a voltage that causes the lamp to produce a specific correlated color temperature, or it may be an arbitrary set voltage, such as 120.00 V, the set voltage for this example. This component does not have uncertainty because it is stated as a given value. If the working standard lamp voltage is set to produce a given correlated color temperature, the uncertainty in that measurement is in the realization of the correlated color temperature, not the voltage. The set voltage is recorded in Line 11 of **Table 6.2**.
- U **lamp voltage measurement for the working standard lamp:** The lamp voltage measurement for this model is done by using an AC voltmeter. This is a Type A uncertainty because several measurements are collected. In this example, 10 measurements are made, each integrated over 10 power-line cycles. The average of the 10 measurements is 120.01 V, and the standard deviation is 0.01 V, which is the standard uncertainty, $u(U)$. The degrees of freedom for this measurement are $10 - 1 = 9$. The average, standard uncertainty, and degrees of freedom are recorded in Line 12 of **Table 6.2**.
- c_U **AC voltmeter calibration factor:** The AC voltmeter may have a center value close to the nominal reading, with a given uncertainty provided on the calibration certificate. This is a Type B uncertainty because the source of the standard uncertainty information is a calibration certificate. Another possibility is that the laboratory has a calibration certificate stating that the AC voltmeter is within manufacturer-specified tolerances. The AC voltmeter calibration factor, c_U , is 1.000, and the tolerance is modeled by a uniform distribution, which is converted to a standard uncertainty. The specifications of a typical AC voltmeter states that the tolerance for AC voltage measurements at 60 Hz is 0.1% of the reading plus 0.1% of the range plus 10 mV. Measuring a 120-V lamp on a range of 150 V gives a tolerance of ± 0.28 V for a reading of 120.00 V. The tolerance is modeled by a uniform distribution, where $a_- = 120.00 \text{ V} - 0.28 \text{ V} = 119.72 \text{ V}$, and $a_+ = 120.00 \text{ V} + 0.28 \text{ V} = 120.28 \text{ V}$; therefore, $u(c_U) = 0.16 \text{ V}$. (**Section 4.4**) These values are recorded in Line 13 of **Table 6.2**.
- m_{U_w} **sensitivity of luminous flux with respect to voltage:** For incandescent lamps, including halogen, the change in lumen output with voltage can be expressed as a ratio of the actual lamp voltage divided by the desired lamp voltage, and the ratio then raised to an exponent of 3.5, which has a standard uncertainty of 0.2. These are typical industry standard values. This is a Type B uncertainty because it is based on industry knowledge. These values are recorded in Line 14 of **Table 6.2**.

Since the working standard lamp calibrated to a fixed voltage is likely to have a different correlated color temperature than the primary standard lamp, the uncertainty of this difference in correlated color temperature is noted in the additional uncertainty components. In this example the standard uncertainty due to differences in correlated color temperature is 0.10%. Taking the square root of the sum of the squares for the 0.10% in correlated color temperature difference and 0.10% for the spatial responsivity gives the standard uncertainty for the additional uncertainty components, $u\left(\frac{C_f}{C_{fr}}\right)$, estimated to be 0.14%. This value is recorded in Line 20 of **Table 6.2**.

The equations for the partial derivatives of the new components presented in **Equation 6.4** follow, along with the line number in **Table 6.3** where the result is recorded (see **Section 6.2.4**).

$$\begin{array}{llll} \frac{\partial \Phi}{\partial U_r} = \frac{-m_{J_r} \cdot \Phi}{U_r} & \text{Line 7} & \frac{\partial \Phi}{\partial c} = \frac{-m_{J_r} \cdot \Phi}{c} & \text{Line 8} & \frac{\partial \Phi}{\partial R} = \frac{m_{J_r} \cdot \Phi}{R} & \text{Line 9} \\ \frac{\partial \Phi}{\partial U} = \frac{m_{U_w} \cdot \Phi}{U} & \text{Line 12} & \frac{\partial \Phi}{\partial c_U} = \frac{m_{U_w} \cdot \Phi}{c_U} & \text{Line 13} & \frac{\partial \Phi}{\partial m_{U_w}} = \Phi \cdot \ln\left(\frac{U \cdot c_U}{U_w}\right) & \text{Line 14} \end{array}$$

Provided in **Table 6.2** is an uncertainty budget using **Equation 6.4** as the model and typical values (see **Section 6.1.1** and **Section 6.1.8**) for a high-quality sphere system with a photometer detector using a high-quality luminous flux primary standard lamp to calibrate a working standard lamp.

Table 6.2. Uncertainty Budget for an Integrating Sphere Using a Photometer Detector and a Total Luminous Flux Primary Standard Lamp to Calibrate a Working Standard Lamp.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	Φ_r	2,800.0	lm	8.2	B	∞	0.9825	8.06
2	y_r	1.7652	nA	0.0007	A	9	-1,558.7	-1.09
3	y_{rd}	0.0002	nA	0.0002	A	9	1,558.7	0.31
4	y	1.7481	nA	0.0008	A	9	1,574.0	1.26
5	y_d	0.0002	nA	0.0002	A	9	-1,574.0	-0.31
6	J_r	5.825	A					
7	U_r	0.582743	V	0.000015	A	9	-30,686	-0.46
8	c	1.000032		0.000004	B	∞	-17,872	-0.07
9	R	0.100045	Ω	0.000003	B	∞	178,743	0.54
10	m_{Jr}	6.5		0.2	B	∞	-0.00225	0.00
11	U_w	120.00	V					
12	U	120.01	V	0.01	A	9	80.23	0.80
13	c_u	1.0000		0.0013	B	∞	9,628.9	12.52
14	m_{u_w}	3.5		0.2	B	∞	0.229	0.05
15	y_{x_r}	0.5243	nA	0.0004	A	9	5,249.2	2.10
16	y_{x_r}	0.0002	nA	0.0002	A	9	-5,249.2	-1.05
17	y_x	0.5286	nA	0.0004	A	9	-5,206.5	-2.08
18	y_{x_d}	0.0002	nA	0.0002	A	9	5,206.5	1.04
19	A_r	1.0000		0.00033	B	∞	2,751.1	0.91
20	C_f/C_{fr}	1.0000		0.0014	B	∞	2,751.1	3.85
	Φ	2,751.1	lm				$u_c(\Phi)$	16

Using **Equation 6.4** and the estimated values for x_i provided in **Table 6.2**, the measurand is calculated: $\Phi = 2,751.1$ lm. The combined standard uncertainty is the square root of the sum of squares: $u_c(\Phi) = 16$ lm, or 0.58% as a relative combined standard uncertainty.

Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), the effective degrees of freedom for this uncertainty budget are $\nu_{eff} = 12,054$. Using an effective degrees of freedom value of 12,054 and the Student's t approximation for coverage factor (**Equation 4.12**, presented in **Section 4.5**), the coverage factor is approximately 2, resulting in an expanded uncertainty of $U(\Phi) = 32$ lm, or 1.2% as a relative expanded uncertainty. The uncertainty community typically rounds the coverage factor to 2.00 if the calculated value is within 0.05 of 2.00. The final statement of uncertainty would be:

The total luminous flux of the working lamp calibrated on [date] was determined, traceable to the candela SI unit, with a magnitude of 2,751.1 lm and an associated expanded uncertainty of 32 lm, or a relative expanded uncertainty of 1.2 %. The stated uncertainty was calculated according to the GUM by a multiplication of the standard uncertainty with a coverage factor of $k = 2$.

6.1.9 Integrating Sphere with an Array Spectrometer. **Section 5.1.9** describes the uncertainty analysis for using an array spectrometer as a spectrally integrating device. Since every channel in an array spectrometer is measured

simultaneously using a single source that is only dependent on a single parameter, lamp current, all the relative sphere calibration factors for a given channel are correlated with each other to some extent, and all the sphere array spectrometer signals are correlated with each other to a significant extent. Therefore, both summations of **Equation 4.6** need to be calculated because the correlation coefficients, $r(x_i, x_j)$, are not zero. It is quite difficult to determine the level or degree of correlation between the channels. The most conservative approach is to set the correlation coefficients equal to 1. **Equation 5.13** (see **Section 5.1.10.2**) is used to calculate the standard uncertainty in the y_{R_i} , y_{R_d} , y_i , y_d , $y_{X_{R_i}}$, $y_{X_{R_d}}$, y_{X_i} , and y_{X_d} signals.

6.2 Creating Total Spectral Radiant Flux Working Standard Lamps

6.2.1 Model for an Integrating Sphere with Array Spectrometer. The scope of this uncertainty budget is the calibration of a working standard lamp using an integrating sphere measurement system that uses an array spectrometer as the detection device and an incandescent primary standard lamp that is similar in current, correlated color temperature, shape, size, and luminous intensity distribution to the working standard lamp and is calibrated for watts per nanometer at a particular operating current. The quantity to be determined is the total spectral radiant flux of the working lamp, $\Phi_\lambda(\lambda)$, in units of watts per nanometer, and its associated expanded uncertainty, $U(\Phi_\lambda(\lambda))$.

The model equation that describes how to calculate the total spectral radiant flux, $\Phi_\lambda(\lambda)$, and the uncertainty components affecting the determination of the total spectral radiant flux for a working standard lamp is written as:

$$\Phi_\lambda(\lambda) = \frac{y(\lambda) - y_d(\lambda)}{C_\Phi(\lambda)} \cdot \frac{L_e(\lambda, T)}{L_e(\lambda, T_0)} \cdot \frac{y_E(\lambda) - y_{E_d}(\lambda)}{y_X(\lambda) - y_{X_d}(\lambda)} \cdot C_f \quad (6.5)$$

Substituting **Equation 5.15** for the sphere calibration factor, $C_\Phi(\lambda)$, the result is:

$$\Phi_\lambda(\lambda) = \frac{\Phi_{r,\lambda}(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r,d}(\lambda)} \cdot \frac{L_e(\lambda, T_{r,0})}{L_e(\lambda, T_r)} \cdot \frac{L_e(\lambda, T)}{L_e(\lambda, T_0)} \cdot \frac{y_{X_r}(\lambda) - y_{X_{r,d}}(\lambda)}{y_X(\lambda) - y_{X_d}(\lambda)} \cdot A_r \cdot \frac{C_f}{C_{f_r}}, \quad (6.6)$$

where:

$\Phi_\lambda(\lambda)$ **total spectral radiant flux for a given wavelength:** The measurand, in watts per nanometer. The array spectrometers used in sphere measurement systems typically have an input optic composed of a diffusing material to cosine-correct the light input to an optical fiber. The optical fiber transmits the light to the input aperture of the array spectrometer, so there are no moving parts. The signal is with respect to a channel that correlates to a particular interval of wavelengths. The wavelength becomes an identifier or label for a particular channel or pixel. Since a primary standard lamp is used that is calibrated in watts per nanometer, and the operating conditions are very similar—specifically, the correlated color temperature—the relative uncertainties associated with the channel to a particular interval of wavelengths are minimized.

$\Phi_{r,\lambda}(\lambda)$ **primary standard lamp calibrated in watts per channel:** The total spectral radiant flux for the primary standard lamp is provided on the calibration certificate in units of watts per nanometer. These data shall be converted from wavelength data to channel data using the process described in **Section 5.2.2**. This is a Type B uncertainty because the source of standard uncertainty comes from interpolating data from a calibration certificate. In the example presented in this section, the calibration certificate states 26.585 mW, with an expanded uncertainty ($k = 2$) of 0.110 mW at a wavelength of 555 nm. These values are recorded in Line 1 of **Table 6.3**.

$y(\lambda)$ **sphere array spectrometer signal for the working standard lamp:** An array spectrometer detects the lamp spectrum in a series of channels of a given bandwidth. The lambda variable in this situation represents a channel in the array spectrometer. The only source of wavelength uncertainty is in

converting the primary standard lamp data to channel data, as discussed in **Section 5.2.2**. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect a series of spectral measurements for the working standard lamp (for example, 10). A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the signal is the average of the 10 measurements, which is $y(\lambda) = 475.943$ cts/ms, with a standard deviation of $u[y(\lambda)] = 0.025$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 2 of **Table 6.3**.

$y_d(\lambda)$

sphere array spectrometer dark signal for the working standard lamp: The array spectrometer signal measured when no light is present in the sphere. For array spectrometers this is commonly more of an electronic dark signal than a stray light signal. Each channel is measured in counts per millisecond. This calculation may automatically occur within the software. If this is the case, then $y(\lambda) - y_d(\lambda)$ is simplified to $y(\lambda)$, and the average signal and standard uncertainty are determined using the method described in the previous definition.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect a series of spectral measurements with the working standard lamp turned off (for example, 10). A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the sphere array spectrometer signal, and averaging if selected. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the signal is the average of the 10 measurements, which is $y_d(\lambda) = 0.546$ cts/ms, with a standard deviation of $u[y_d(\lambda)] = 0.007$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 3 of **Table 6.3**.

$y_r(\lambda)$

sphere array spectrometer signal for the primary standard lamp: An array spectrometer detects the lamp spectrum in a series of channels of a given bandwidth. The lambda variable in this situation represents a channel in the array spectrometer. The only source of wavelength uncertainty is in converting the primary standard lamp data to channel data, as discussed in **Section 5.2.2**. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect a series of spectral measurements for the primary standard lamp (for example, 10). A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80 % of full signal and may include averaging done by the control software. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the signal is the average of the 10 measurements, which is $y_r(\lambda) = 484.593$ cts/ms, with a standard deviation of $u[y_r(\lambda)] = 0.023$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 4 of **Table 6.3**.

$y_{rd}(\lambda)$

sphere array spectrometer dark signal for the primary standard lamp: The array spectrometer signal measured when no light is present in the sphere. For array spectrometers, this is commonly more of an electronic dark signal than a stray light signal. Each channel is measured in counts per millisecond. This calculation may automatically occur within the software. If this is the case, then $y_r(\lambda) - y_{rd}(\lambda)$ is simplified to $y_r(\lambda)$, and the average signal and standard uncertainty are determined using the method described in the previous definition.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect a series of spectral measurements with the primary standard lamp turned off (for example, 10). A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the sphere array spectrometer signal, and averaging if selected. The standard deviation calculated from the 10 measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the signal is the average of the 10 measurements, which is $y_d(\lambda) = 0.607$ cts/ms, with a standard deviation of $u[y_d(\lambda)] = 0.007$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 5 of

Table 6.3.

$$\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}$$

spectral radiance of the working standard lamp: Uncertainty exists in the operation of the working standard lamp. If the working standard lamp does not have a specific current set, the component involving the measurement of the working standard lamp current that affects the spectral radiance can be removed from the equation. Many times the laboratory will set the working standard lamp current in order to ensure that the correlated color temperature of the lamp is a fixed known value, such as 3100 K. If the laboratory has not fixed the working standard lamp current for other reasons, there is no uncertainty the first time the working standard lamp is measured, because the lamp current is arbitrary. The second time the working standard lamp is operated and measured, setting the lamp current does cause uncertainty in the spectral radiance because it needs to be set to the lamp current conditions stated during the first measurement. The ratio of the spectral radiance and the uncertainty template is presented in **Section 5.2.3** for the primary standard lamp and will be repeated as an approximation for the working standard lamp. This is a Type B uncertainty because it is based on combination of Type A and Type B components. For this example, the spectral radiance correction for the working standard lamp is $\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)} = 0.99988$, with a standard deviation of $u_c\left(\frac{L_e(\lambda, T_0)}{L_e(\lambda, T)}\right) = 0.00058$ at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 6 of **Table 6.3**.

$$\frac{L_e(\lambda, T_{r,0})}{L_e(\lambda, T_r)}$$

spectral radiance of the primary standard lamp: Uncertainty exists in the operation of the primary standard lamp. It is expressed as the ratio of spectral radiance for the primary standard lamp temperature specified in the calibration certificate, compared to the spectral radiance for the actual operating temperature of the primary standard lamp. This uncertainty is due to the lamp current measurement. The ratio of the spectral radiance and the uncertainty template is presented in **Section 5.2.3**. This is a Type B uncertainty because it is based on combination of Type A and Type B components. For this example, the spectral radiance correction for the primary standard lamp is $\frac{L_e(\lambda, T_{r,0})}{L_e(\lambda, T_r)} = 1.00016$, with a standard deviation of $u_c\left(\frac{L_e(\lambda, T_{r,0})}{L_e(\lambda, T_r)}\right) = 0.00058$ at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 7 of **Table 6.3**.

$$y_s(\lambda)$$

channel-dependent self-absorption factor for the working standard lamp: A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the array spectrometer signal. The recommended method for correcting this error relies on use of an auxiliary lamp to determine the magnitude of the self-absorption. This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the working standard lamp is mounted in the sphere and is turned off. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect a series of spectral measurements (for example, 10) with the auxiliary lamp on and the working standard lamp mounted in the sphere but turned off. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the working standard lamp signal is the average of the 10 measurements, which is $y_x(\lambda) = 142.753$ cts/ms, with a standard deviation of $u[y_x(\lambda)] = 0.014$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 8 of **Table 6.3**.

 $y_{xd}(\lambda)$

channel-dependent self-absorption factor for the working standard lamp dark signal: The array spectrometer signal measured when the auxiliary lamp is off and the working standard lamp is mounted in the sphere and is turned off. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect a series of spectral measurements (perhaps ten) with the auxiliary lamp off and the working standard lamp mounted in the sphere and turned off. A single measurement consists of a set of data that is collected under normal conditions which is the integration time used to measure the channel-dependent self-absorption factor for the working standard lamp and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the working standard lamp signal is the average of the 10 measurements, which is $y_{xd}(\lambda) = 0.610$ cts/ms, with a standard deviation of $u[y_{xd}(\lambda)] = 0.009$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 9 of **Table 6.3**.

 $y_{xr}(\lambda)$

channel-dependent self-absorption factor for primary standard lamp: A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the array spectrometer signal. The recommended method for correcting this error relies upon use of an auxiliary lamp to determine the magnitude of the self-absorption. This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the primary standard lamp mounted in the sphere but turned off. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would typically collect a series of spectral measurements (for example, 10) with the auxiliary lamp on and the primary standard lamp mounted in the sphere but turned off. A single measurement consists of a set of data that is collected under normal conditions which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the working standard lamp signal is the average of the 10 measurements, which is $y_{xr}(\lambda) = 143.934$ cts/ms, with a standard deviation of $u[y_{xr}(\lambda)] = 0.013$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 8 of **Table 6.3**.

 $y_{xrd}(\lambda)$

channel-dependent self-absorption factor for the primary standard lamp dark signal: The array spectrometer signal measured when the auxiliary lamp is off and the primary standard lamp is mounted in the sphere and is turned off. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would typically collect a series of spectral measurements (for example, 10) with the auxiliary lamp off and the primary standard lamp mounted in the sphere and turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the channel-dependent self-absorption factor for the primary standard lamp, and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to 9 if a series of 10 measurements were collected. For this example, the working standard lamp signal is the average of the 10 measurements, which is $y_{x_{rd}}(\lambda) = 0.598$ cts/ms, with a standard deviation of $u[y_{x_{rd}}(\lambda)] = 0.008$ cts/ms at a wavelength of 555 nm. The signal and standard uncertainty are recorded in Line 11 of **Table 6.3**.

$A_r(\lambda)$ **primary standard lamp age or burning time:** **Figure 5.1** (see **Section 5.2.1**) shows a typical reduction in nominal total spectral radiant flux for halogen lamps. The percentage is the reduction per 24 hours of use. Not all lamps that are used as total spectral radiant flux standard lamps meet this condition, and very few are predictable. Many calibration laboratories will provide a statement approximating this dependency on the calibration certificates for the primary standard lamps they issue. If this information is not available, it should be assumed that the decay rate is as shown in **Figure 5.1**. This is a Type B uncertainty because it is based on industry knowledge or data from a calibration certificate.

The value of A_r is set to 1. It is difficult and risky to apply a correction factor for the aging of the standard lamp without assuming knowledge. Increasing the standard uncertainty with the operation of the primary standard lamp is a more conservative approach. For this example, assuming the lamp has been used for 2 hours, the standard uncertainty is estimated at 0.4% divided by 24 hours at a wavelength of 555 nm, then multiplied by the 2 hours of use, resulting in a value of 0.00033. These values are recorded in Line 12 of **Table 6.3**.

$$\frac{C_f}{C_{fr}}$$

additional uncertainty components: Discussed in **Section 6.2.2**.

6.2.2 Additional Uncertainty Components. Since the working standard lamp is similar in current, correlated color temperature, shape, size, and luminous intensity distribution to the primary standard lamp, many of the additional uncertainty components are highly negatively correlated, such that they cancel. The only additional component that may remain is the spatial responsivity of the sphere. Spheres have a spatial response to the initial ray of light from the lamp. Even though the primary standard lamp and working standard lamp have very similar luminous intensity distributions, it is unlikely the distributions will be aligned with the sphere. The estimation of $\frac{C_f}{C_{fr}}$ is equal to 1, and the standard uncertainty for the additional uncertainty component, which is the spatial responsivity of the sphere, $u\left(\frac{C_f}{C_{fr}}\right)$, is estimated to be 0.10%. The magnitude of this uncertainty component can be estimated by rotating the working standard lamp within the sphere and measuring the signal.

6.2.3 Integrating Sphere with Array Spectrometer Sensitivity Coefficients for Working Standard Lamps. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 6.6** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**. The equations for the partial derivatives follow, along with the line number in **Table 6.3** where the result is recorded.

$\frac{\partial \Phi_\lambda(\lambda)}{\partial \Phi_{r,\lambda}(\lambda)} = \frac{\Phi_\lambda(\lambda)}{\Phi_{r,\lambda}(\lambda)}$	Line 1	$\frac{\partial \Phi_\lambda(\lambda)}{\partial y(\lambda)} = \frac{\Phi_\lambda(\lambda)}{y(\lambda) - y_d(\lambda)}$	Line 2
$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_d(\lambda)} = \frac{-\Phi_\lambda(\lambda)}{y(\lambda) - y_d(\lambda)}$	Line 3	$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_r(\lambda)} = \frac{-\Phi_\lambda(\lambda)}{y_r(\lambda) - y_{r,d}(\lambda)}$	Line 4
$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_{r,d}(\lambda)} = \frac{\Phi_\lambda(\lambda)}{y_r(\lambda) - y_{r,d}(\lambda)}$	Line 5	$\frac{\partial \Phi_\lambda(\lambda)}{\partial [L_e(\lambda, T)/L_e(\lambda, T_0)]} = \frac{\Phi_\lambda(\lambda)}{L_e(\lambda, T)/L_e(\lambda, T_0)}$	Line 6
$\frac{\partial \Phi_\lambda(\lambda)}{\partial [L_e(\lambda, T_{r,0})/L_e(\lambda, T_r)]} = \frac{\Phi_\lambda(\lambda)}{L_e(\lambda, T_{r,0})/L_e(\lambda, T_r)}$	Line 7	$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_x(\lambda)} = \frac{-\Phi_\lambda(\lambda)}{y_x(\lambda) - y_{x,d}(\lambda)}$	Line 8
$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_{x,d}(\lambda)} = \frac{\Phi_\lambda(\lambda)}{y_x(\lambda) - y_{x,d}(\lambda)}$	Line 9	$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_{x_r}(\lambda)} = \frac{\Phi_\lambda(\lambda)}{y_{x_r}(\lambda) - y_{x_r,d}(\lambda)}$	Line 10
$\frac{\partial \Phi_\lambda(\lambda)}{\partial y_{x_{r,d}}(\lambda)} = \frac{-\Phi_\lambda(\lambda)}{y_{x_r}(\lambda) - y_{x_{r,d}}(\lambda)}$	Line 11	$\frac{\partial \Phi_\lambda(\lambda)}{\partial A_r(\lambda)} = \frac{\Phi_\lambda(\lambda)}{A_r(\lambda)}$	Line 12
$\frac{\partial \Phi_\lambda(\lambda)}{\partial C_f/C_{fr}} = \frac{\Phi_\lambda(\lambda)}{C_f/C_{fr}}$	Line 13		

6.2.4 Example Uncertainty Budget. Provided in **Table 6.3** is an uncertainty budget using the **Equation 6.6** as the model and typical values (see **Section 6.1.1**) for a high-quality sphere system with an array spectroradiometer using a high-quality total spectral radiant flux primary standard lamp to calibrate a working standard lamp.

6.2.5 Measurand and Combined Uncertainty. Using **Equation 6.6** and the estimated values for x_i provided in **Table 6.3**, the measurand is calculated: $\Phi(\lambda) = 25.897$ mW at 555 nm.

Table 6.3. Uncertainty Budget for an Integrating Sphere Using an Array Spectroradiometer with a High-Quality Total Spectral Radiant Flux Primary Standard Lamp to Calibrate a Working Standard Lamp for $\lambda = 555$ nm.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	$\Phi_{r,\lambda}(\lambda)$	26.585	mW	0.110	B	∞	0.9741	0.107
2	$y(\lambda)$	475.943	cts/mw	0.025	A	9	0.0545	0.00136
3	$y_d(\lambda)$	0.546	cts/mw	0.007	A	9	-0.0545	-0.000381
4	$y_r(\lambda)$	484.593	cts/ms	0.023	A	9	-0.0535	-0.00123
5	$y_{r,d}(\lambda)$	0.607	cts/ms	0.007	A	9	0.0535	0.000375
6	$\frac{L_e(\lambda, T)}{L_e(\lambda, T_0)}$	0.99988		0.00058	B	∞	25.900	0.0150
7	$\frac{L_e(\lambda, T_{r,0})}{L_e(\lambda, T_r)}$	1.00016		0.00058	B	∞	25.893	0.0150
8	$y_x(\lambda)$	142.753	cts/ms	0.014	A	9	-0.1822	-0.00255
9	$y_{x,d}(\lambda)$	0.610	cts/ms	0.009	A	9	0.1822	0.00164
10	$y_{x_r}(\lambda)$	143.934	cts/ms	0.013	A	9	0.1807	0.00235
11	$y_{x_{r,d}}(\lambda)$	0.598	cts/ms	0.008	A	9	-0.1807	-0.00145
12	A_r	1.0000		0.00033	B	∞	25.897	0.00855
13	C_f/C_{fr}	1.0000		0.0010	B	∞	25.897	0.0259
	$\Phi(\lambda)$	25.897	mW			large	$u_c [\Phi(\lambda)]$	0.113

The combined standard uncertainty is the square root of the sum of squares, $u_c[\Phi(\lambda)] = 0.113 \text{ mW}$, or 0.44% as a relative combined standard uncertainty. The combined standard uncertainty in this example is dominated by the standard uncertainty of the primary standard lamp at all the wavelength channels.

6.2.6 Degrees of Freedom and Expanded Uncertainty. Using the Welch-Satterthwaite formula (Equation 4.11, presented in Section 4.5), the effective degrees of freedom for this uncertainty budget are extremely large, which leads to a coverage factor approximately equal to 2, resulting in an expanded uncertainty of $U[\Phi(\lambda)] = 0.225 \text{ mW}$, or 0.88% as a relative expanded uncertainty at 555 nm. The final statement of uncertainty would be a table that shows the results for all the channels.

6.3 Creating a Luminous Flux Working Standard Lamp from a Total Spectral Radiant Flux Standard Lamp

6.3.1 SI Unit Traceability: Total Spectral Radiant and Luminous Flux Scales. Currently (2017), a calibration laboratory is able to issue a standard lamp that is calibrated for total luminous flux (lumens) and total spectral radiant flux (W/nm). Following the procedures described in this Section, the testing laboratory will conclude that when the total spectral radiant flux calibration is used to create working lumen standards, the uncertainty is approximately twice that obtained from a calibration of a working lumen standard based on a total luminous flux standard lamp. The difference is real and stems back to the scale realization that is performed at the U.S. National Institute of Standards and Technology (NIST).

Until 1991, the NIST unit for luminous intensity was derived from the NIST spectral irradiance scale,^{5,6} which was based on a blackbody at the freezing point of gold, and therefore dependent on the temperature scale. In 1990, the international temperature scale was revised,⁷ and the gold freezing point temperature changed from 1337.58 K to 1337.33 K. Due to the change in the temperature scale, the NIST luminous intensity unit increased in magnitude by 0.35%. In 1992 at NIST, a new luminous intensity unit (candela) was realized based on the absolute responsivity of detectors (using 100% Q.E. silicon detectors,⁸ and subsequently a cryogenic electrical substitution radiometer⁹). The old luminous intensity unit was replaced with the new unit in 1992.

The new candela is realized and maintained on a group of NIST standard photometers, which are calibrated for illuminance responsivity in A/lx. These NIST standard photometers also embody the NIST illuminance unit and allow luminous intensity to be determined from measured illuminance and distance. The realization and maintenance of the photometric units at NIST are shown in Figure 6.1. The NIST reference cryogenic radiometer, the Primary Optical Watt Radiometer (POWR),¹⁰ maintains the optical power unit (Watt), which serves as the basis for all radiometric and photometric units and scales realized at NIST. The radiometer is cooled by liquid helium to 5 K and works on the principle of electrical substitution. Based on laser-beam power measurements with the POWR at several wavelengths, the NIST detector spectral responsivity scale is maintained on silicon photodiode light-trapping detectors. The radiometer is designed to provide optical power measurements with expanded uncertainties of 0.02% ($k = 2$), which has been verified by an intercomparison with two other cryogenic radiometer facilities at NIST.

The spectral responsivity scale is transferred to other detectors using the Spectral Comparator Facility (SCF),¹¹ where the absolute spectral responsivity $s(\lambda)$ (A/W) of each of the NIST standard photometers is determined. The illuminance responsivity [A/lx] of each photometer is then calculated from $s(\lambda)$, the area of the aperture, and other correction factors. The relative expanded uncertainty of the illuminance responsivity determination is 0.39%. The NIST standard photometers are recalibrated annually utilizing the detector spectral responsivity scale.

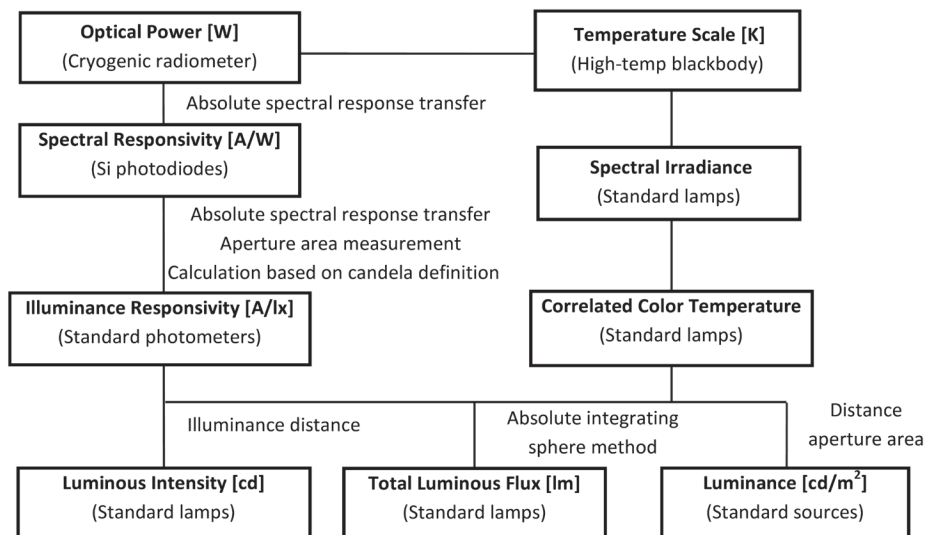


Figure 6.1. Realization and maintenance of the NIST photometric units.

In 1995, a new NIST luminous flux unit was derived, based on the detector-based candela introduced in 1992, with a new method using an integrating sphere and an external source. The basic principle of the Absolute Integrating Sphere Method (AIS) is to measure the total flux of a lamp inside the sphere and compare it to a known amount of flux introduced into the sphere from a source outside the sphere. Calibrations are based on the illuminance measurement of an external source with NIST standard photometers, thereby introducing a detector-based procedure for the total luminous flux calibration. A higher accuracy is achieved since the calibration chain is shortened with the standard lamps and the transfer process eliminated. The AIS method provides an advantage of automatically correcting for the self-absorption of the test lamps and the drift of the sphere responsivity during the measurement session. The process of the realization of the lumen and the subsequent recalibration of many working standard lamps are no longer necessary. Instead, the NIST standard photometers are to be calibrated periodically to maintain the unit.

The NIST spectral irradiance scale is generated with filter radiometers calibrated for absolute spectral power responsivity traceable to the POWR instead of with a gold freezing point blackbody. The calibrated filter radiometers are then used to establish the radiance temperature of a high-temperature blackbody operating near 3000 K. The spectral irradiance of the high-temperature blackbody is then determined with knowledge of the geometric factors and is used to assign the spectral irradiances of a group of 1,000-W FEL* lamps. The spectral irradiance lamps are used to calibrate the CCD-array spectrometer on the NIST goniometer that is used in the measurement of the total spectral radiant flux. This chain of calibration produces an uncertainty that is approximately twice as large as that of the lumen detector based calibration. NIST research is working toward calibrating the CCD-array spectrometer with a laser based system that calibrates the CCD-array spectrometer directly against the NIST silicon trap detectors calibrated against POWR. This calibration chain will have a smaller uncertainty, making the total spectral radiant flux scale nearly the same as the total luminous flux scale with respect to uncertainty.

6.3.2 Integrating-Sphere Transfer Model: Total Spectral Radiant Flux to Luminous Flux. The scope of this uncertainty budget is the calibration of a working standard lamp for total luminous flux (lumens) at a particular operating current with an integrating sphere measurement system that uses an array spectrometer as the detection device and a primary standard lamp calibrated for watts per nanometer at a particular operating current that is similar

* A type of tungsten-halogen lamp.

in current, correlated color temperature, shape, and size to the working standard lamp. The quantity to be determined is the luminous flux of the working standard lamp, Φ , in lumens, and its associated expanded uncertainty, $U(\Phi)$.

The total luminous flux is calculated by:

$$\Phi = K_m \int_{\lambda} \Phi(\lambda) V(\lambda) d\lambda, \quad (6.7)$$

where K_m is the maximum spectral luminous efficacy, 683 lm/W, $\Phi(\lambda)$ is the total spectral radiant flux in W/nm, and $V(\lambda)$ the CIE standard photometric observer function, or luminous efficiency function, which describes the spectral sensitivities of the human visual system in the photopic range. The model equation that describes how to calculate the total luminous flux and the uncertainty components affecting the determination of the total luminous flux, Φ , for a working standard lamp is:

$$\Phi = \frac{K_m}{A_r} \left[\sum_{360 \text{ nm}}^{830 \text{ nm}} \left(\frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r,d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r,d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right) \Delta\lambda(\lambda) \right] \cdot \left(\frac{J_r \cdot R}{U_r \cdot c_U} \right)^{m_{J_r}} \cdot \left(\frac{U \cdot c_U}{J \cdot R} \right)^{m_J} \cdot \frac{C_f}{C_{f_r}} \quad (6.8)$$

If the working standard lamp does not have a specific current set, the component involving the measurement of the working standard lamp current can be removed from the equation. Many times the laboratory will set the working standard lamp current in order to ensure that the correlated color temperature of the lamp is a fixed known value, such as 2856 K. If the laboratory has not fixed the working standard lamp current for other reasons, there is no uncertainty the first time the working standard lamp is measured, because the lamp current is arbitrary. The second time the working standard lamp is measured, setting the lamp current does cause uncertainty because it needs to be set to the lamp current conditions recorded during the first measurement.

If the laboratory uses exactly the same equipment to transfer the scale of the primary standard lamp to the new working standard lamp, a few of the uncertainty components can be eliminated. This situation only applies if the measurements are done in a timely fashion (usually same day), such that the equipment has not had time to drift. If the new working standard lamp is the same model of lamp as the primary standard lamp, the sensitivity of luminous flux with respect to current is the same for both lamps ($m_{J_r} = m_J$). The model equation can be reduced to **Equation 6.9**, where the uncertainty in the voltmeter calibration and reference shunt calibration cancel.

$$\Phi = \frac{K_m}{A_r} \left[\sum_{360 \text{ nm}}^{830 \text{ nm}} \left(\frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r,d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r,d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right) \Delta\lambda(\lambda) \right] \cdot \left(\frac{J_r \cdot U}{J \cdot U_r} \right)^{m_{J_r}} \cdot \frac{C_f}{C_{f_r}}, \quad (6.9)$$

where subscript **r** refers to a primary standard lamp property, and:

Φ **total luminous flux of the working standard lamp:** The measurand, in units of lumens.

K_m **maximum spectral luminous efficacy:** This quantity is defined in the definition of the candela, 683 lm/W, and it has no uncertainty.

$\Phi_{r,\lambda}(\lambda)$ **primary standard lamp calibrated in watts per channel:** The total spectral radiant flux for the primary standard lamp is provided on the calibration certificate in units of watts per nanometer (W/nm). These data shall be converted from wavelength data to channel data according to the process described in **Section 5.2.2**. This is a Type B uncertainty because the source of standard uncertainty comes from interpolating data from a calibration certificate and understanding the correlated components contributing to the overall uncertainty. The standard uncertainty for the reference lamp is discussed further in **Section 6.3.2**.

$V(\lambda)$ **CIE standard photometric observer:** The function representing the ideal observer with a relative spectral responsivity curve that conforms to the spectral luminous efficiency function for photopic vision. This function is defined in ISO 23539:2005(E)/CIE S 010/E:2004, *Photometry– The CIE System of Physical Photometry*, and contributes no uncertainty.

$y_r(\lambda)$ **sphere array spectrometer signal for the primary standard lamp:** An array spectrometer detects the lamp spectrum in a series of channels of a given bandwidth. The λ variable in this situation represents a channel in the array spectrometer. The only source of wavelength uncertainty is in converting the primary standard lamp data to channel data, as discussed in **Section 5.2.2**. Each channel is measured in counts per millisecond (cts/ms).

To determine the signal and standard uncertainty for this component, a user would collect a series of spectral measurements for the primary standard lamp. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_{rd}(\lambda)$ **sphere array spectrometer dark signal for the primary standard lamp:** The array spectrometer signal measured when no light is present in the sphere. For array spectrometers, this is commonly more of an electronic dark signal than a stray light signal. Each channel is measured in counts per millisecond. This calculation may automatically occur within the software. If this is the case, then $y_r(\lambda) - y_{rd}(\lambda)$ is simplified to $y_r(\lambda)$, and the average signal and standard uncertainty are determined using the method described in the previous definition.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would collect a series of spectral measurements with the primary standard lamp turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the sphere array spectrometer signal, and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y(\lambda)$ **sphere array spectrometer signal for the working standard lamp:** To determine the signal and standard uncertainty for this component, a user would collect a series of spectral measurements for the working standard lamp. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_d(\lambda)$ **sphere array spectrometer dark signal for the working standard lamp:** The array spectrometer signal measured when no light is present in the sphere. If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would collect a series of spectral measurements with the working standard lamp turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the sphere array spectrometer signal, and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_{xr}(\lambda)$ **channel-dependent self-absorption factor for the primary standard lamp:** A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the array spectrometer signal. The recommended method for correcting this error relies on use of an auxiliary lamp to determine the magnitude of the self-absorption. This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the primary standard lamp is mounted in the sphere but turned off. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would collect a series of spectral measurements with the auxiliary lamp on and the primary standard lamp mounted in the sphere but turned off. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_{x_{rd}}(\lambda)$ **channel-dependent self-absorption factor for the primary standard lamp dark signal:** The array spectrometer signal measured when the auxiliary lamp is off and the primary standard lamp is mounted in the sphere and turned off. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would collect a series of spectral measurements with the auxiliary lamp off and the primary standard lamp mounted in the sphere and turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the channel-dependent self-absorption factor for the primary standard lamp, and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_x(\lambda)$ **channel-dependent self-absorption factor for working standard lamp:** This measurement is the array spectrometer signal measured when the auxiliary lamp is on and the working standard lamp is mounted in the sphere but turned off. Each channel is measured in counts per millisecond.

To determine the signal and standard uncertainty for this component, a user would collect a series of spectral measurements with the auxiliary lamp on and the working standard lamp mounted in the sphere but turned off. A single measurement consists of a set of data collected under normal conditions, which is an integration time that results in a count signal that is approximately 80% of full signal and may include averaging done by the control software. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$y_{x_d}(\lambda)$ **channel-dependent self-absorption factor for the working standard lamp dark signal:** The array spectrometer signal measured when the auxiliary lamp is off and the working standard lamp is mounted in the sphere and is turned off. Each channel is measured in counts per millisecond.

If the dark signal is not subtracted automatically, then to determine the signal and standard uncertainty for the dark signal component, a user would collect a series of spectral measurements with the auxiliary lamp off and the working standard lamp mounted in the sphere and turned off. A single measurement consists of a set of data collected under normal conditions, which is the integration time used to measure the channel-dependent self-absorption factor for the working standard lamp, and averaging if selected. The standard deviation calculated from the series of measurements for each channel is the standard uncertainty. This is a Type A uncertainty, with degrees of freedom equal to the number of measurements in the series minus 1.

$\Delta\lambda$ **wavelength interval:** The array spectrometer reports intervals of wavelength data as a particular channel identified by a single wavelength. The difference between the channels is the wavelength interval, which may not be consistent across the entire spectrum due to the nonlinearities of the wavelength dispersion in the array spectrometer. In the example presented in this section, the wavelength difference between the data points starts at 0.69 nm in the ultraviolet region and becomes smaller until it is 0.62 nm in the near-infrared region. The uncertainty of the wavelength interval is approximated as the wavelength scale uncertainty divided by the number of channels in the array spectrometer. For this example, the standard

uncertainty of the wavelength scale is 0.25 nm and there are 1,025 intervals, giving a standard uncertainty of 0.00024 nm. The standard uncertainty is most likely smaller than this approximation because the ratio of the signal from the primary standard lamp to that of the working standard lamp is correlated and tends to cancel out wavelength interval uncertainties.

- U_r **primary standard lamp current measurement:** The primary standard lamp current measurement for this model is performed by measuring the voltage across a calibrated shunt resistor, R . This is a Type A uncertainty because several measurements are collected. In this example, 30 measurements are made, integrated over 10 power-line cycles. The average of the 30 measurements is 0.582743 V, and the standard deviation is 0.000015 V, which is the standard uncertainty, $u(U_r)$. The degrees of freedom for this measurement are $30 - 1 = 29$. The average, standard uncertainty, and degrees of freedom are recorded in Line 3 of **Table 6.5**.
- J_r **reference current for the primary standard lamp:** The reference current for the primary standard lamp, in amps, is provided on the calibration certificate. This component does not have uncertainty, because it is stated as a given value on the calibration certificate.
- In this example, the calibration certificate states 5.825 A. This value is recorded in Line 4 of **Table 6.5**.
- U **working standard lamp current measurement:** The working standard lamp current measurement for this model is performed by measuring the voltage across a calibrated shunt resistor, R , which is the same shunt resistor used for the primary standard lamp current measurement. This is a Type A uncertainty because several measurements are collected. In this example, 30 measurements are made, integrated over 10 power-line cycles. The average of the 30 measurements is 0.581643 V, and the standard deviation is 0.000014 V, which is the standard uncertainty, $u(U)$. The degrees of freedom for this measurement are 29. The average, standard uncertainty, and degrees of freedom are recorded in Line 5 of **Table 6.5**.
- J **reference current for the working standard lamp:** The reference current for the working standard lamp, in amps, is determined by measuring the spectral output of the lamp and setting the current to provide the desired correlated color temperature. This is a Type B uncertainty determination based on a previous measurement. In this example, the previous measurement determined the current to be 5.814 A. This value is recorded in Line 6 of **Table 6.5**.
- m_{J_r} **sensitivity of luminous flux with respect to current:** For incandescent lamps, including halogen, the change in lumen output with current can be expressed as the ratio of the actual lamp current to the desired lamp current, with the ratio raised to an exponent of 6.5, which has a standard uncertainty of 0.2. These are typical industry standard values. This is a Type B uncertainty because it is based on industry knowledge. These values are recorded in Line 7 of **Table 6.5**.
- A_r **primary standard lamp age or burning time:** The operation of an incandescent (including halogen) primary standard lamp typically reduces the lumen output by 0.4% over 24 hours. Not all lamps meet this condition, and very few are predictable. Many calibration laboratories will provide a statement approximating this dependency on the calibration certificates for the primary standard lamps they issue. If this information is not available, the decay rate is assumed to be 0.4% per 24 hours. This is a Type B uncertainty because it is based on industry knowledge or data from a calibration certificate.
- The value of A_r is set to 1. It is difficult and risky to apply a correction factor for the aging of the primary standard lamp without assuming knowledge. Increasing the standard uncertainty with the operation of the primary standard lamp is a more conservative approach. For this example, assuming the lamp has been used for 2 hours, the standard uncertainty is estimated at 0.4% divided by 24 hours, then multiplied by 2 hours of use, resulting in a value of 0.00033. These values are recorded in Line 8 of **Table 6.5** (see **Section 6.2.4**).

6.3.3 Additional Uncertainty Components. In order to simplify the presentation of the model equation, a separate table of uncertainty components is presented, where the components do not change the magnitude of the measurand, Φ , but do add uncertainty, and the contribution of their uncertainty is relatively small (less than 1/5 of the largest contributor). If the contribution is large, it should be listed in the model equation as a separate factor. The estimation of $\frac{C_f}{C_r}$ is equal to 1, and the standard uncertainty for $u\left(\frac{C_f}{C_r}\right)$ is the combined standard uncertainty of the components.

Since the working standard lamp is similar in current, correlated color temperature, shape, size, and luminous intensity distribution to the primary standard lamp, many of the additional uncertainty components are highly negatively correlated, such that they cancel. **Section 5.2.4** discusses additional uncertainty components related to spectral sphere measurements. The only additional component that may remain is the spatial responsivity of the sphere. Spheres have a spatial response to the initial ray of light from the lamp. Even though the primary standard lamp and working standard lamp have very similar luminous intensity distributions, it is unlikely the distributions will be aligned with the sphere. The standard uncertainty for the additional uncertainty component that is the spatial responsivity of the sphere, $\left(\frac{C_f}{C_r}\right)$, is estimated to be 0.10%. The magnitude of this uncertainty component can be estimated by rotating the working standard lamp within the sphere and measuring the signal.

Occasionally, the working standard lamp is similar in many aspects to the primary standard lamp but is operated at a different current in order to create a different correlated color temperature, such as 2856 K, compared to the primary standard lamp operated to produce 3100 K. The different correlated color temperatures have different spectral stray light components and linearity properties in the array spectrometer, which may cause additional uncertainties and need to be accounted for.

6.3.4 Array Spectrometer Data Summation Uncertainty. As stated in **Section 4.4**, the partial derivative of the model equation for each component of uncertainty describes the dependence of the measurand on the uncertainty components. In this case, the sensitivity coefficients are determined from the partial derivatives of **Equation 6.9** with respect to each uncertainty component. A brief description of the techniques used to determine the partial derivatives is described in **Annex B**.

Since every channel is measured simultaneously using a single source that is only dependent on a single parameter, lamp current, all the relative sphere calibration factors for a given channel are correlated with each other to some extent and all the sphere array spectrometer signals are correlated with each other to some extent. Therefore, both summations of **Equation 4.6** need to be calculated because the correlation coefficients, $r(x_i, x_j)$, are not zero. It is quite difficult to determine the level or degree of correlation between the channels. The first level of correlation comes from the fact that the channels are detecting light from a single source with a given spectral distribution. If no random components were present, the correlation coefficient between the signal components would be equal to 1. However, random components exist, which work to reduce the correlation coefficients to zero. Estimating the change in the correlation coefficient is difficult.

The next level of correlation comes from the bandpass function. Red wavelengths and blue wavelengths are not going to be correlated nearly as much as channels of red light that are within the bandpass of the array spectrometer. For the calibration coefficients, in addition to the fact that the calibration comes from a single source, the data for the primary standard lamp are highly correlated because normally an interpolation of the data is required, causing a degree of correlation.

The most conservative approach is to set the correlation coefficients, $r(x_i, x_j)$, equal to 1. With the correlation coefficients equal to 1, **Equation 4.6** reduces to a sum of the component uncertainties multiplied by the component sensitivity coefficients:

$$u_c(y) = \sum_{i=1}^m \left[\left(\frac{\partial y}{\partial x_i} \right) \cdot u(x_i) \right] \quad (6.10)$$

Taking this approach, the uncertainty of the summation, $u(s)$, in **Equation 6.9** is calculated from:

$$s = \sum_{360 \text{ nm}}^{830 \text{ nm}} \left(\frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right) \Delta\lambda(\lambda) \quad (6.11)$$

It is then entered into the overall uncertainty budget shown in **Table 6.5** (in **Section 6.3.6**) for **Equation 6.9** using the sensitivity coefficients shown in **Section 6.3.5**.

The sensitivity coefficient for the primary standard lamp calibrated in watts per channel, $\Phi_{r,\lambda}(\lambda)$, is:

$$\frac{\partial s}{\partial \Phi_{r,\lambda}(\lambda)} = \left[\frac{V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right] \cdot \Delta\lambda(\lambda) \quad (6.12)$$

Therefore, the combined standard uncertainty component for the primary standard lamp calibrated in watts per channel is:

$$\begin{aligned} u_c(\Phi_r) &= \sum_{\lambda=360 \text{ nm}}^{830 \text{ nm}} \left\{ \left[\frac{\partial s}{\partial \Phi_{r,\lambda}(\lambda)} \right] \cdot u[\Phi_{r,\lambda}(\lambda)] \right\} \\ &= \sum_{\lambda=360 \text{ nm}}^{830 \text{ nm}} \left\{ \left[\left(\frac{V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right) \cdot \Delta\lambda(\lambda) \right] \cdot u[\Phi_{r,\lambda}(\lambda)] \right\} \end{aligned} \quad (6.13)$$

The sensitivity coefficient for the sphere array spectrometer signal for the primary standard lamp, $y_r(\lambda)$ —including the dark signal, which is automatically subtracted in the array spectrometer software—is:

$$\frac{\partial s}{\partial [y_r(\lambda) - y_{r_d}(\lambda)]} = \left\{ \frac{-\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{[y_r(\lambda) - y_{r_d}(\lambda)]^2} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right\} \cdot \Delta\lambda(\lambda) \quad (6.14)$$

The sensitivity coefficient for the sphere array spectrometer signal for the working standard lamp, $y(\lambda)$ —including the dark signal, which is automatically subtracted in the array spectrometer software—is:

$$\frac{\partial s}{\partial [y(\lambda) - y_d(\lambda)]} = \left[\frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda)}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)}{y_x(\lambda) - y_{x_d}(\lambda)} \right] \cdot \Delta\lambda(\lambda) \quad (6.15)$$

The sensitivity coefficient for the channel-dependent self-absorption factor for the primary standard lamp, $y_{x_r}(\lambda)$ —including the dark signal, which is automatically subtracted in the array spectrometer software—is:

$$\frac{\partial s}{\partial [y_{x_r}(\lambda) - y_{x_{r_d}}(\lambda)]} = \left\{ \frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{1}{y_x(\lambda) - y_{x_d}(\lambda)} \right\} \cdot \Delta\lambda(\lambda) \quad (6.16)$$

The sensitivity coefficient for the channel-dependent self-absorption factor for the working standard lamp, $y_X(\lambda)$ —including the dark signal, which is automatically subtracted in the array spectrometer software—is:

$$\frac{\partial S}{\partial [y_X(\lambda) - y_{X_d}(\lambda)]} = \left\{ \frac{-\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{X_r}(\lambda) - y_{X_{r_d}}(\lambda)}{[y_X(\lambda) - y_{X_d}(\lambda)]^2} \right\} \cdot \Delta\lambda(\lambda) \quad (6.17)$$

The sensitivity coefficient for the channel-dependent self-absorption factor for the working standard lamp dark signal, $\Delta\lambda(\lambda)$, is:

$$\frac{\partial S}{\partial \Delta\lambda(\lambda)} = \left\{ \frac{\Phi_{r,\lambda}(\lambda) \cdot V(\lambda) \cdot [y(\lambda) - y_d(\lambda)]}{y_r(\lambda) - y_{r_d}(\lambda)} \cdot \frac{y_{X_r}(\lambda) - y_{X_{r_d}}(\lambda)}{y_X(\lambda) - y_{X_d}(\lambda)} \right\} \quad (6.18)$$

The combined standard uncertainty for S is the square root of the sum of the squares of all the standard uncertainties determined for the components in the summation presented in **Equation 6.11**. **Table 6.4** shows the results of the component calculations for the sample data presented in **Annex E**.

Table 6.4. Uncertainty Budget for the Array Spectrometer Summation.

	Symbol	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν
1	$\Phi_{r,d}(\lambda)$	W/nm	0.0164	B	∞
2	$y_r(\lambda) - y_{r_d}(\lambda)$	cts	-0.0055	A	2
3	$y(\lambda) - y_d(\lambda)$	cts	0.0055	A	2
4	$y_{X_r}(\lambda) - y_{X_{r_d}}(\lambda)$	cts	0.0044	A	2
5	$y_X(\lambda) - y_{X_d}(\lambda)$	cts	-0.0044	A	2
6	$\Delta\lambda(\lambda)$	nm	0.0010	B	∞
	$S = 2.7647$	w	0.0191	$u_c(S)$	103

6.3.5 Lumen Sensitivity Coefficients for Integrating Sphere with Array Spectrometer. Through the analysis in **Section 6.3.4** of the summation in **Equation 6.11**, **Equation 6.9** has been reduced to:

$$\Phi = \frac{K_m \cdot S}{A_r} \cdot \left(\frac{J_r \cdot U}{J \cdot U_r} \right)^{m_{J_r}} \cdot \frac{C_f}{C_{f_r}} \quad (6.19)$$

The equations for the partial derivatives of **Eq. 6.19** follow, along with the line number in **Table 6.5** where the results are recorded.

$\frac{\partial \Phi}{\partial S} = \frac{\Phi}{S}$	Line 1	$\frac{\partial \Phi}{\partial A_r} = \frac{\Phi}{A_r}$	Line 2	$\frac{\partial \Phi}{\partial U_r} = \frac{-m_{J_r} \cdot \Phi}{U_r}$	Line 4
$\frac{\partial \Phi}{\partial U} = \frac{m_{J_r} \cdot \Phi}{U}$	Line 6	$\frac{\partial \Phi}{\partial m_{J_r}} = \Phi \cdot \ln \left(\frac{J_r \cdot U}{J \cdot U_r} \right)$	Line 7	$\frac{\partial \Phi}{\partial C_f/C_{f_r}} = \frac{\Phi}{C_f/C_{f_r}}$	Line 8

6.3.6 Example Uncertainty Budget. Provided in **Table 6.5** is an uncertainty budget using **Equation 6.19** as the model and typical values (see **Annex E**, **Section 6.3.1**, and **Section 6.3.4**) for a high-quality sphere system with an array spectrometer using a high-quality total spectral radiant flux primary standard lamp.

Table 6.5. Uncertainty Budget for an Integrating Sphere Using an Array Spectrometer with a High-Quality Total Spectral Radiant Flux Primary Standard Lamp to Create a Working Standard Lumen Lamp.

	Symbol	Value x_i	Units	Standard Uncertainty $u(x_i)$	Type	Deg. of Freedom ν	Sensitivity c_i	Contribution $u_i(y)$
1	S	2.7647	W	0.0191	A	103	683.00	13.05
2	K_m	683.0	lm/W					
3	U_r	0.582743	V	0.000015	A	29	-21,059	-0.32
4	J_r	5.825	A					
5	U	0.581643	V	0.000014	A	29	21,102	0.30
6	J	5.814	A					
7	m_{J_r}	6.5		0.2	B	∞	0.00	0.00
8	A_r	1.00000		0.00033	B	∞	-1,888	-0.62
9	C_f/C_{f_r}	1.0000		0.0010	B	∞	1,888	1.89
	Φ	1,888	lm			108	$u_c(\Phi)$	13

6.3.7 Measurand and Combined Standard Uncertainty. Using **Equation 6.19** and the estimated values for x_i provided in **Table 6.5**, the measurand is calculated: $\Phi = 1,888$ lm.

The combined standard uncertainty is the square root of the sum of squares, $u_c(\Phi) = 13$ lm, or 0.70% as a relative combined standard uncertainty.

6.3.8 Degrees of Freedom and Expanded Uncertainty. Using the Welch-Satterthwaite formula (**Equation 4.11**, presented in **Section 4.5**), there are effectively 108 degrees of freedom for this uncertainty budget, which leads to a coverage factor approximately equal to 2.03, resulting in an expanded uncertainty of $U(\Phi) = 26$ lm, or 1.4% as a relative expanded uncertainty.

Annex A – Tolerance, Uncertainty, and Confidence Intervals

A manufacturer's tolerance is the limit of acceptable unintended deviation from a nominal or theoretical measurement determination. Therefore, a pair of tolerances, upper and lower, defines a range within which a true value may fall while still being acceptable. The manufacturer's tolerance is based on the quality of the components used in the product and the quality of the assembly process. The term "tolerance interval" is defined in JCGM 106:2012, *Evaluation of Measurement Data – The Role of Measurement Uncertainty in Conformity Assessment*. A tolerance interval is an acceptable range of the true value of the parameter (not the range of readings of an instrument).

To ensure that the true value of the parameter is within the tolerance interval at 95% confidence level, the range to be accepted for the readings of an instrument measuring the parameter shall be in a range smaller than the tolerance interval, reduced by the expanded uncertainty ($k = 2$, nominally) of the measurement of the parameter on both tolerance limits.

As an example, an AC voltmeter is stated to have a tolerance interval of plus and minus [0.1% of the reading + 0.1% of the range + 4 mV + (0.02•frequency)%] of the reading for a range setting of 215 V. For a reading of 100 V of 60-Hz AC electricity, the tolerance interval would be plus and minus 0.33 V. In order to validate that the instrument complies with the stated tolerance, a voltage, 100 V in magnitude, is measured by the instrument. The 100-V source has been calibrated by another method, perhaps another instrument that has a much smaller tolerance interval or calibration factor. In this example, the 100-V source has been calibrated with an expanded uncertainty of 0.05%. The acceptance interval is defined as the tolerance interval minus twice the expanded uncertainty, as shown in **Figure A.1**. In this case if the instrument reading is within the acceptance interval, then it meets the specified tolerance interval with a confidence of 95%. The test uncertainty ratio (TUR) is the tolerance interval divided by twice the expanded uncertainty of the calibration of the source, which becomes 6.6:1 in this example. To achieve the same level of confidence with a test uncertainty ratio of 4:1 means that the acceptance interval would be much smaller.

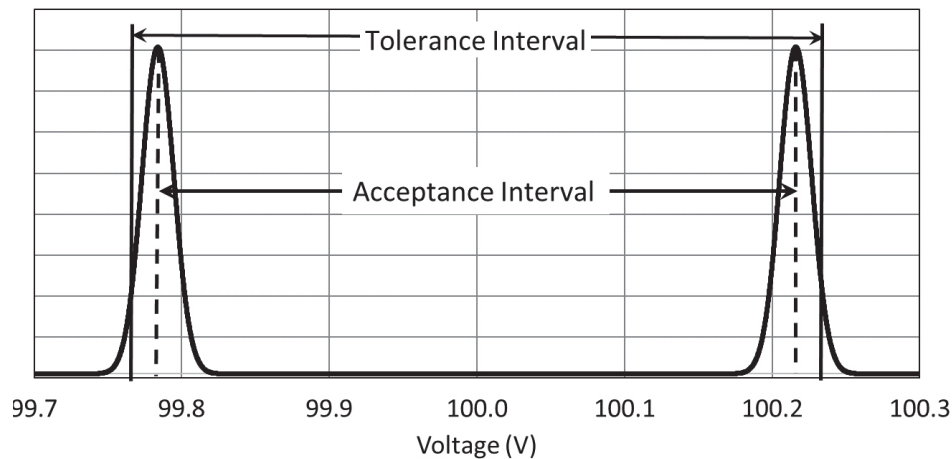


Figure A.1. AC voltmeter conformity assessment demonstrating tolerance interval and acceptance interval.

To determine the standard uncertainty of the measurement of voltage using the instrument, the tolerance interval is divided by the square root of three; this is because the tolerance interval is simply a uniform probability distribution. In this case, the standard uncertainty of the AC voltmeter on a range setting of 215 V for the measurement of 100-V, 60-Hz AC electricity is 0.19 V.

In many cases, the instrument is very repeatable; i.e., the repeatability range is much smaller than the tolerance interval. Such instruments can be characterized by measuring several calibrated voltages that span the range of the measurements. By using a weighted fit, a calibration factor with an offset can be determined.* The calibration factor and offset may have a smaller uncertainty than the tolerance interval method, if the TUR for the voltage calibrations is very large (i.e., the voltage calibration is small) and the instrument is stabilized in order to maintain the calibration factor and offset.

Table A.1 provides a set of measurement data for the calibration method described previously. **Equations A.1** through **A.4** provide the method for calculating a weighted least-squares fit to the data provided in **Table A.1**. The summations with indexes i, j range from the first data point to the last data point.

* Press WH, Teukolsky SA, Vetterling WT, Flannery BP. Numerical Recipes in C – The Art of Scientific Computing, 2nd ed., Cambridge University Press; 1992. Section 15.2, p. 664.

Table A.1. Output Voltage from 6.5-Digit and 8.5-Digit Voltmeters Measuring a Voltage Source.

6.5-Digit voltmeter x_i (V)	8.5-Digit Voltmeter y_i (V)	Uncertainty of 8.5-Digit Voltmeter σ_i (V)	6.5-Digit Voltmeter x_i (V)	8.5-Digit Voltmeter y_i (V)	Uncertainty of 8.5-Digit Voltmeter σ_i (V)
9.99981	9.999840	0.000016	-0.499992	-0.4999978	0.0000032
8.99983	8.999854	0.000014	-0.999981	-0.9999895	0.0000035
7.99985	7.999872	0.000013	-1.99993	-1.999943	0.0000045
6.99987	6.999890	0.000011	-2.99995	-2.999964	0.0000057
5.99990	5.999910	0.0000098	-3.99993	-3.999938	0.0000070
4.99992	4.999929	0.0000084	-4.99991	-4.999917	0.0000084
3.99994	3.999946	0.0000070	-5.99988	-5.999901	0.0000098
2.99997	2.999967	0.0000057	-6.99986	-6.999881	0.000011
1.99997	1.999981	0.0000045	-7.99984	-7.999861	0.000013
0.999988	0.9999860	0.0000035	-8.99982	-8.999843	0.000014
0.500002	0.4999960	0.0000032	-9.99980	-9.999826	0.000016
0.0000120	0.00000526	0.0000030			

$$m = \left[\left(\sum \frac{1}{\sigma_i^2} \right) \cdot \sum \frac{x_i y_i}{\sigma_i^2} - \left(\sum \frac{x_i}{\sigma_i^2} \right) \cdot \sum \frac{y_i}{\sigma_i^2} \right] / \left\{ \sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{x_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{x_i}{\sigma_i^2} \right)^2 \right\} \quad (\text{A.1})$$

$$b = \left[\left(\sum \frac{x_i^2}{\sigma_i^2} \right) \cdot \sum \frac{y_i}{\sigma_i^2} - \left(\sum \frac{x_i}{\sigma_i^2} \right) \cdot \sum \frac{x_i y_i}{\sigma_i^2} \right] / \left\{ \sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{x_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{x_i}{\sigma_i^2} \right)^2 \right\} \quad (\text{A.2})$$

$$\sigma_m = \sqrt{\left(\sum \frac{1}{\sigma_i^2} \right) / \left\{ \sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{x_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{x_i}{\sigma_i^2} \right)^2 \right\}} \quad (\text{A.3})$$

$$\sigma_b = \sqrt{\left(\sum \frac{x_i^2}{\sigma_i^2} \right) / \left\{ \sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{x_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{x_i}{\sigma_i^2} \right)^2 \right\}} \quad (\text{A.4})$$

In the above equations:

x_i is the voltage measured by the 6.5-digit voltmeter

y_i is the voltage measured by the 8.5-digit voltmeter

σ_i is the standard uncertainty in the 8.5-digit voltmeter readings

$m = 1.00000259$ is the slope of the fit or the gain of the voltmeter

$\sigma_m = 0.00000039$ is the standard uncertainty of the gain of the voltmeter

$b = -0.0000041$ V is the offset of the voltmeter

$\sigma_b = 0.0000011$ V is the uncertainty in the offset

If the uncertainty in the data is not constant across all levels of the variables, using weighted least squares with weights that are inversely proportional to the variance at each level of the explanatory variables yields the most precise parameter estimates possible.

If the least-squares fit is used to determine new values, then intervals are required to assess the uncertainty of the new value. A confidence interval describes the bounds of the average determined value. A prediction interval describes the bounds of a single determined value. A confidence interval is going to be smaller than a prediction interval with the same confidence level because the average determined value is based on many observations of a single determined value.

The confidence interval, CI , is calculated using **Equation A.5**. The prediction interval, PI , is calculated using **Equation A.6**. In addition, the error in estimating the average determined value, x^* , or a single determined value, grows as x^* moves away from average x . Thus, the farther x^* is from average x , the wider the confidence and prediction intervals will be.

$$CI(x^*) = t_{\alpha, n-2} \cdot \sqrt{a^T C a} , \quad (\text{A.5})$$

where:

$t_{\alpha, n-2}$ is the Student's t distribution [in EXCEL, T.INV.2T(1- α , n-2)]

α is the probability or confidence level

n is the degrees of freedom

C is the covariance matrix

a is the vector composed of the sensitivity coefficients

$$PI(x^*) = t_{\alpha, n-2} \cdot \sqrt{\sum \frac{(y_i - \hat{y}_i)^2}{n-2} + a^T C a} , \quad (\text{A.6})$$

where $\hat{y}_i$ is the calculated value based on the weighted fit.

For a linear function, $y = m x + b$,

$$a^T C a = \begin{bmatrix} \frac{\partial y}{\partial m} & \frac{\partial y}{\partial b} \end{bmatrix} \begin{bmatrix} \sigma_m^2 & Cov(m, b) \\ Cov(m, b) & \sigma_b^2 \end{bmatrix} \begin{bmatrix} \frac{\partial y}{\partial m} \\ \frac{\partial y}{\partial b} \end{bmatrix} \quad (\text{A.7})$$

Reducing the matrix math gives:

$$a^T C a = x^{*2} \sigma_m^2 + \sigma_b^2 + 2x^* [Cov(m, b)] , \quad (\text{A.8})$$

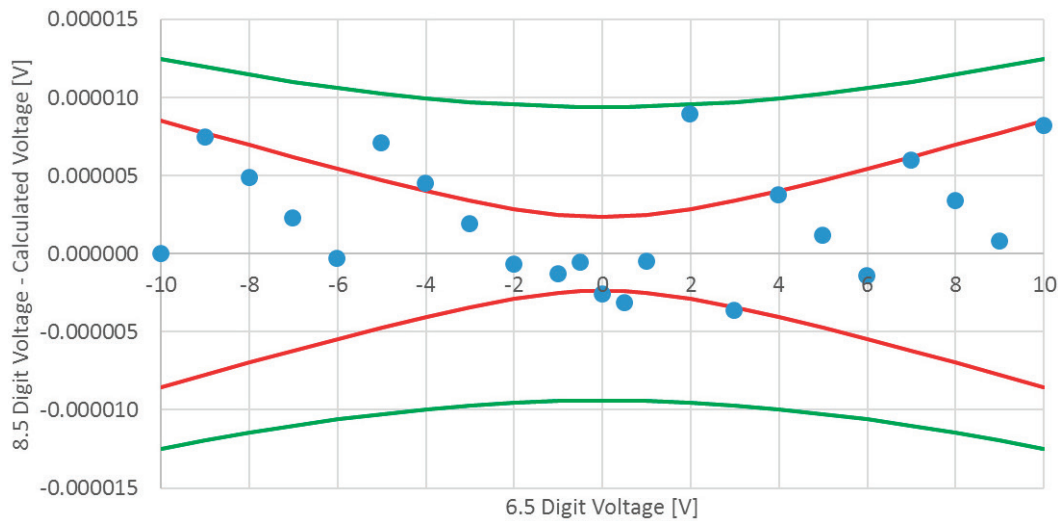
with:

$$Cov(m, b) = \frac{b \cdot \sigma_m \cdot \sigma_b \sqrt{\sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{x_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{x_i}{\sigma_i^2} \right)^2}}{\sum_j \left[\frac{1}{\sigma_j^2} \cdot \left(\sum_i \frac{y_i^2}{\sigma_i^2} \right) \right] - \left(\sum_i \frac{y_i}{\sigma_i^2} \right)^2} \quad (\text{A.9})$$

Table A.2 and **Figure A.2** show the results based on the data in **Table A.1**.

Table A.2. Confidence and Prediction Intervals ($\alpha = 95\%$ for This Example).

6.5-Digit voltmeter x_i (V)	8.5-Digit Calculation $\hat{y}_i$ (V)	Confidence Interval $CI(x^*)$ (V)	Prediction Interval $PI(x^*)$ (V)
9.99981	9.99983	0.0000085	0.0000125
8.99983	8.99985	0.0000077	0.0000120
7.99985	7.99987	0.0000070	0.0000115
6.99987	6.99988	0.0000062	0.0000110
5.99990	5.99991	0.0000055	0.0000106
4.99992	4.99993	0.0000047	0.0000103
3.99994	3.99995	0.0000040	0.0000100
2.99997	2.99997	0.0000034	0.0000097
1.99997	1.99997	0.0000029	0.0000096
0.999988	0.999986	0.0000025	0.0000094
0.500002	0.499999	0.0000024	0.0000094
0.0000120	0.0000079	0.0000024	0.0000094
-0.499992	-0.50000	0.0000024	0.0000094
-0.999981	-0.99999	0.0000025	0.0000094
-1.99993	-1.99994	0.0000029	0.0000096
-2.99995	-2.99996	0.0000034	0.0000097
-3.99993	-3.99994	0.0000040	0.0000100
-4.99991	-4.99993	0.0000047	0.0000103
-5.99988	-5.99990	0.0000055	0.0000106
-6.99986	-6.99988	0.0000062	0.0000110
-7.99984	-7.99986	0.0000070	0.0000115
-8.99982	-8.99985	0.0000077	0.0000120
-9.99980	-9.99983	0.0000085	0.0000125

**Figure A.2. The results of the calculations. The red lines are the confidence intervals ($\alpha = 95\%$), and the green lines are the prediction intervals ($\alpha = 95\%$).**

Annex B – Partial Derivatives

Partial derivatives are defined as derivatives of a function of multiple variables when all but the variable of interest are held constant during the differentiation. The partial derivative of a function $f(x_1, x_2, \dots, x_n)$ in the direction of x_i at a point $(a_1, a_2, \dots, a_n)$ is defined to be:

$$\frac{\partial f}{\partial x_i}(a_1, a_2, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h} \quad (\text{B.1})$$

The result is the slope of the tangent at the defined point. There are a few techniques required to calculate the partial derivatives used in this document.

Constant function rule:

The derivative of a constant is equal to zero. There is no change in the slope.

$$\frac{\partial \alpha}{\partial x} = 0, \text{ where } \alpha \text{ is any constant} \quad (\text{B.2})$$

Sum of functions rule:

The derivative of the sum of two functions is the sum of the derivatives of the two functions.

$$\frac{\partial(u + v)}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} \quad (\text{B.3})$$

Difference of functions rule:

The derivative of the difference of two functions is the difference of the derivatives of the two functions.

$$\frac{\partial(u - v)}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} \quad (\text{B.4})$$

Power rule:

The derivative of a function to the n th power is n times the function to the $(n-1)$ power, times the derivative of the function.

$$\frac{\partial u^n}{\partial x} = n u^{(n-1)} \cdot \frac{\partial u}{\partial x} \quad (\text{B.5})$$

Product of functions rule:

The derivative of the multiplication of two functions is the derivative of the first function times the second function, plus the derivative of the second function times the first function.

$$\frac{\partial(uv)}{\partial x} = \frac{\partial u}{\partial x} \cdot v + \frac{\partial v}{\partial x} \cdot u \quad (\text{B.6})$$

Quotient of functions rule:

The derivative of the quotient of two functions is the derivative of the first function times the second function, subtracting the derivative of the second function times the first function, and then dividing the whole quantity by the second function squared.

$$\frac{\partial(u/v)}{\partial x} = \frac{\frac{\partial u}{\partial x} \cdot v - \frac{\partial v}{\partial x} \cdot u}{v^2} \quad (\text{B.7})$$

Exponential function rule:

The derivative of the base e value taken to a functional power is the base e value raised to the functional power times the derivative of the functional power.

$$\frac{\partial e^u}{\partial x} = e^u \cdot \frac{\partial u}{\partial x}$$

The derivative of a constant taken to a functional power is the constant raised to the functional power times the natural logarithm of the constant times the derivative of the functional power.

$$\frac{\partial a^u}{\partial x} = a^u \ln a \cdot \frac{\partial u}{\partial x} \quad (\text{B.8})$$

Logarithmic function rule:

The derivative of a natural logarithm of a function is 1 divided by the function and then multiplied by the derivative of the function.

$$\frac{\partial \ln u}{\partial x} = \frac{1}{u} \cdot \frac{\partial u}{\partial x}$$

The derivative of a base a logarithm of a function is 1 divided by the quantity of the function and by the natural logarithm of a , and then multiplied by the derivative of the function.

$$\frac{\partial \log_a u}{\partial x} = \frac{1}{u \cdot \ln a} \cdot \frac{\partial u}{\partial x} \quad (\text{B.9})$$

For uncertainty components where the partial derivative is difficult to calculate, the sensitivity coefficient can be approximated by calculating the slope of the function while varying the uncertainty component. **Section 5.1.3** describes the resistance of the shunt resistor. Using **Equation 4.5**, the sensitivity coefficient for the resistance of the shunt resistor can be determined by calculating the sphere responsivity with the shunt resistor value plus the uncertainty, and with the shunt resistor value minus the uncertainty.

$$C_\phi[R + u(R)] = C(0.100048) = 6.5382 \times 10^{-4} \text{ nA/lm}$$

$$C_\phi[R - u(R)] = C(0.100042) = 6.5357 \times 10^{-4} \text{ nA/lm}$$

The sensitivity coefficient is calculated by using **Equation 4.5** restated:

$$\frac{\partial C_\phi}{\partial R} = \frac{C_\phi[R + u(R)] - C_\phi[R - u(R)]}{2 \cdot u(R)} = 4.247 \times 10^{-2}$$

The difference between the slope determination and the derivative determination is 0.30%, which is a small difference in this example.

Annex C – Sensitivity Coefficients Calculated as Slopes

For uncertainty components where the dependency cannot be modeled by an analytical equation, the sensitivity coefficient can be approximated experimentally. **Section 5.1.3** describes the sensitivity of luminous flux with respect to current. This sensitivity is determined by measuring the luminous flux at a variety of currents, accounting for the change in correlated color temperature. **Table C.1** shows the results for a halogen lamp, including a 1% standard uncertainty for the measurements.

Table C.1. Luminous Flux Data for a Halogen Lamp Operated at Various Currents.

Current (A)	Luminous Flux (lm)	Standard Uncertainty (lm)
4.222	250.6	0.25
5.256	1,256.5	1.25
6.345	4,119.8	4.12
7.323	9,481.6	9.48
7.886	14,832.1	14.8
8.711	30,032.4	30.0

By using a weighted fit to a power law, the sensitivity of luminous flux with respect to current is determined:

$$\Phi_r = A \cdot J^{m_j} \quad (\text{C.1})$$

where A is a normalization constant (one of the fitting parameters), J is the operating current, Φ_r is the measured luminous flux, and m_j is the sensitivity of luminous flux with respect to current (the second of the fitting parameters). The result of the fit is $m_j = 6.502 \pm 0.002$. The sensitivity coefficient calculated from the results gives exactly the same answer shown in **Table 5.2**. A standard uncertainty of 0.02 is shown in **Table 5.2**, which is more representative of a collection of lamps than a single result.

Annex D – Total Spectral Radiant Flux Data Measured with an Array Spectrometer and Integrating Sphere

Table D.1. Total Spectral Radiant Flux Data for a Standard Lamp with Expanded Uncertainty ($k = 2$).

Wavelength (nm)	Total Spectral Radiant Flux (mW/nm)	Expanded Uncertainty (%)	Wavelength (nm)	Total Spectral Radiant Flux (mW/nm)	Expanded Uncertainty (%)
300	0.4009	2.8	710	48.76	1.2
310	0.5772	2.7	720	49.81	1.2
320	0.7902	2.6	730	50.75	1.2
330	1.064	2.4	740	51.67	1.2
340	1.401	2.3	750	52.52	1.2
350	1.812	2.1	760	53.35	1.2
360	2.297	2.0	770	54.05	1.2
370	2.856	1.9	780	54.73	1.2
380	3.490	1.7	790	55.36	1.2
390	4.236	1.6	800	55.92	1.2
400	5.047	1.5	810	56.46	1.2
410	5.974	1.4	820	56.93	1.2
420	6.979	1.4	830	57.35	1.3
430	8.063	1.3	840	57.73	1.3
440	9.226	1.3	850	58.06	1.3
450	10.47	1.3	860	58.34	1.3
460	11.79	1.3	870	58.55	1.3
470	13.18	1.2	880	58.78	1.3
480	14.62	1.2	890	58.93	1.4
490	16.11	1.2	900	59.04	1.4
500	17.65	1.2	910	59.09	1.4
510	19.23	1.2	920	59.11	1.4
520	20.83	1.2	930	59.16	1.4
530	22.47	1.2	940	59.16	1.5
540	24.11	1.2	950	58.91	1.5
550	25.76	1.2	960	58.75	1.5
560	27.41	1.2	970	58.62	1.6
570	29.06	1.2	980	58.47	1.6
580	30.70	1.2	990	58.27	1.7
590	32.31	1.2	1,000	58.07	1.7
600	33.91	1.2	1,010	57.85	1.7
610	35.49	1.2	1,020	57.58	1.8
620	37.02	1.2	1,030	57.25	1.8
630	38.53	1.2	1,040	56.93	1.9
640	39.98	1.2	1,050	56.50	2.0
650	41.39	1.2	1,060	56.21	2.0
660	42.75	1.2	1,070	55.80	2.1
670	44.06	1.2	1,080	55.53	2.1
680	45.32	1.2	1,090	55.03	2.2
690	46.52	1.2	1,100	54.60	2.3
700	47.67	1.2			

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Table D.2. Total Spectral Radiant Flux Data for a Standard Lamp with Expanded Uncertainty ($k = 2$).

Array Wavelength (nm)	Interpolated Total Spectral Radiant Flux (mW/nm)	Interpolated Standard Uncertainty (%)	Shifted (0.25 nm) Interpolated Total Spectral Radiant Flux (mW/nm)	Standard Uncertainty Due to Wavelength (%)	Total Standard Uncertainty (%)
347.87	1.718	1.07	1.707	0.63	1.24
348.56	1.747	1.06	1.736	0.62	1.23
349.24	1.778	1.06	1.767	0.62	1.23
349.93	1.808	1.05	1.797	0.62	1.22
350.62	1.839	1.05	1.828	0.61	1.21
351.30	1.871	1.04	1.859	0.61	1.21
351.99	1.902	1.04	1.891	0.61	1.20
352.68	1.934	1.03	1.922	0.60	1.20
353.37	1.967	1.03	1.955	0.60	1.19
354.05	1.999	1.03	1.987	0.60	1.19
354.74	2.033	1.02	2.021	0.59	1.18
355.43	2.066	1.02	2.054	0.59	1.18
356.11	2.099	1.02	2.087	0.59	1.17
356.80	2.134	1.01	2.121	0.58	1.17
357.49	2.168	1.01	2.155	0.58	1.16
358.18	2.202	1.01	2.190	0.58	1.16
358.86	2.238	1.00	2.225	0.57	1.16
359.55	2.273	1.00	2.260	0.57	1.15
360.24	2.308	1.00	2.295	0.57	1.15
360.92	2.345	1.00	2.332	0.56	1.14

Table D.3. Total Spectral Radiant Flux Data for a Standard Lamp with Expanded Uncertainty ($k = 2$).

Array Wavelength (nm)	Interpolated Total Spectral Radiant Flux (mW/nm)	Interpolated Standard Uncertainty (%)	Shifted (0.25 nm) Interpolated Total Spectral Radiant Flux (mW/nm)	Standard Uncertainty Due to Wavelength (%)	Total Standard Uncertainty (%)
1,005.41	57.96	0.85	57.96	-0.01	0.85
1,006.03	57.94	0.85	57.95	-0.01	0.85
1,006.65	57.93	0.85	57.93	-0.01	0.85
1,007.27	57.91	0.85	57.92	-0.01	0.85
1,007.89	57.90	0.85	57.91	-0.01	0.85
1,008.51	57.89	0.85	57.89	-0.01	0.85
1,009.13	57.87	0.85	57.88	-0.01	0.85
1,009.75	57.86	0.85	57.86	-0.01	0.85
1,010.37	57.84	0.85	57.85	-0.01	0.85
1,010.98	57.83	0.85	57.83	-0.01	0.85
1,011.60	57.81	0.86	57.82	-0.01	0.86
1,012.22	57.79	0.86	57.80	-0.01	0.86
1,012.84	57.78	0.86	57.79	-0.01	0.86
1,013.46	57.76	0.87	57.77	-0.01	0.87
1,014.08	57.75	0.87	57.75	-0.01	0.87
1,014.70	57.73	0.87	57.74	-0.01	0.87
1,015.32	57.71	0.88	57.72	-0.01	0.88
1,015.94	57.70	0.88	57.70	-0.01	0.88
1,016.56	57.68	0.88	57.69	-0.01	0.88
1,017.18	57.66	0.89	57.67	-0.01	0.89

Annex E – Total Spectral Radiant Flux Data and Signal for Working Standard Lamp Transfer

Table E.1. Total Spectral Radiant Flux and Measurement Data for the Example in Section 6.3.

Wavelength (nm)	$V(\lambda)$	$\Phi_r(\lambda)$ (mW/nm)	$\Phi_r(\lambda)$ (Std Uncert)	$y_r(\lambda) - y_{rd}(\lambda)$ (Counts)	$y_r(\lambda) - y_{rd}(\lambda)$ (Std Uncert)
360.05	0.000004	0.002299	0.000023	1,061.1	4.1
370.41	0.000012	0.002856	0.000026	1,785.5	5.9
380.76	0.000039	0.003489	0.000030	2,779.4	7.9
391.11	0.000135	0.004235	0.000034	4,079.6	10.0
401.46	0.000434	0.005046	0.000038	5,548.9	12.5
411.73	0.001362	0.005972	0.000043	7,271.2	15.4
422.00	0.005159	0.006977	0.000048	8,784.5	17.8
432.27	0.013583	0.008059	0.000054	10,110.9	20.2
442.47	0.025610	0.009223	0.000060	11,189.4	22.3
452.68	0.041768	0.010471	0.000068	12,541.3	24.4
462.88	0.065278	0.011788	0.000074	13,736.4	26.2
473.05	0.103367	0.013177	0.000082	14,582.0	27.5
483.21	0.156462	0.014613	0.000089	15,326.2	28.7
493.37	0.236857	0.016109	0.000097	15,901.5	29.8
503.53	0.371699	0.017651	0.000106	16,754.4	31.5
513.64	0.565690	0.019229	0.000115	17,624.9	33.5
523.75	0.761969	0.020834	0.000124	18,142.4	34.9
533.86	0.895494	0.022466	0.000133	18,498.3	36.1
543.97	0.971261	0.024107	0.000143	18,747.1	37.2
554.00	0.999112	0.025761	0.000153	18,860.0	37.7
564.03	0.982724	0.027410	0.000163	18,790.9	37.5
574.06	0.923458	0.029061	0.000172	18,889.8	37.6
584.09	0.827581	0.030696	0.000181	19,150.4	38.1
594.13	0.707497	0.032311	0.000190	18,791.1	37.3
604.13	0.579638	0.033910	0.000199	18,534.4	36.8
614.13	0.453403	0.035490	0.000208	18,342.6	36.3
624.09	0.332818	0.037023	0.000216	18,084.3	35.8
634.05	0.226053	0.038524	0.000225	17,783.2	35.2
643.95	0.152278	0.039977	0.000233	17,363.3	34.3
653.85	0.091123	0.041386	0.000241	15,804.9	31.2
663.70	0.050674	0.042750	0.000249	14,532.3	28.7
673.55	0.026329	0.044065	0.000257	14,358.3	28.4
683.41	0.013811	0.045321	0.000265	14,204.8	28.1
693.26	0.006591	0.046515	0.000272	14,329.3	28.3
703.11	0.003354	0.047675	0.000280	14,104.1	27.9
712.97	0.001825	0.048757	0.000287	13,844.6	27.5
722.82	0.000911	0.049811	0.000295	12,556.2	25.0
732.68	0.000450	0.050751	0.000302	11,628.4	23.2
742.46	0.000215	0.051666	0.000309	11,349.9	22.7
752.23	0.000104	0.052524	0.000316	10,726.0	21.6
762.01	0.000052	0.053350	0.000321	9,473.5	19.3
771.73	0.000028	0.054052	0.000327	9,267.7	19.0
781.45	0.000014	0.054734	0.000332	8,775.8	18.2
791.17	0.000007	0.055364	0.000338	7,885.1	16.5
801.04	0.000003	0.055925	0.000343	7,118.5	15.1
810.92	0.000002	0.056458	0.000349	6,444.1	13.8
820.47	0.000001	0.056929	0.000355	5,832.1	12.6
830.02	0.000000	0.057349	0.000360	5,336.1	11.7

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Table E.2. Measurement Data for the Example in Section 6.3.

Wavelength (nm)	$y_r(\lambda) - y_d(\lambda)$ (Counts)	$y(\lambda) - y_d(\lambda)$ (Std Uncert)	$\Delta\lambda(\lambda)$ (nm)	$\Delta\lambda(\lambda)$ (Std Uncert)
360.05	1,167.5	4.5	10.3527	0.0035
370.41	1,964.8	6.5	10.3529	0.0035
380.76	3,057.7	8.7	10.3529	0.0035
391.11	4,488.2	11.0	10.3528	0.0035
401.46	6,103.8	13.7	10.2696	0.0035
411.73	7,999.1	16.9	10.2696	0.0035
422.00	9,663.3	19.6	10.2697	0.0035
432.27	11,122.3	22.2	10.2010	0.0035
442.47	12,309.1	24.5	10.2009	0.0035
452.68	13,795.8	26.8	10.2010	0.0035
462.88	15,110.7	28.8	10.1700	0.0035
473.05	16,040.7	30.2	10.1606	0.0035
483.21	16,858.9	31.6	10.1607	0.0035
493.37	17,492.6	32.8	10.1607	0.0035
503.53	18,430.5	34.7	10.1096	0.0035
513.64	19,387.6	36.8	10.1097	0.0035
523.75	19,957.4	38.4	10.1096	0.0035
533.86	20,348.8	39.7	10.1096	0.0035
543.97	20,622.4	41.0	10.0320	0.0035
554.00	20,746.3	41.5	10.0320	0.0035
564.03	20,670.5	41.2	10.0320	0.0035
574.06	20,779.4	41.4	10.0320	0.0035
584.09	21,065.7	41.9	10.0320	0.0035
594.13	20,671.0	41.1	10.0033	0.0035
604.13	20,388.4	40.5	10.0034	0.0035
614.13	20,177.9	40.0	9.9561	0.0035
624.09	19,893.7	39.4	9.9561	0.0035
634.05	19,562.3	38.7	9.9037	0.0035
643.95	19,100.4	37.7	9.9037	0.0035
653.85	17,385.7	34.3	9.8506	0.0035
663.70	15,986.4	31.6	9.8512	0.0035
673.55	15,794.9	31.2	9.8511	0.0035
683.41	15,626.3	30.9	9.8511	0.0035
693.26	15,763.1	31.2	9.8554	0.0035
703.11	15,515.4	30.7	9.8554	0.0035
712.97	15,229.7	30.2	9.8554	0.0035
722.82	13,812.6	27.5	9.8554	0.0035
732.68	12,791.5	25.5	9.7770	0.0035
742.46	12,485.0	25.0	9.7771	0.0035
752.23	11,799.2	23.8	9.7772	0.0035
762.01	10,421.1	21.2	9.7211	0.0035
771.73	10,195.4	20.9	9.7212	0.0035
781.45	9,653.9	20.0	9.7212	0.0035
791.17	8,673.9	18.2	9.8718	0.0035
801.04	7,831.3	16.6	9.8718	0.0035
810.92	7,089.4	15.1	9.5497	0.0035
820.47	6,416.0	13.9	9.5496	0.0035
830.02	5,870.1	12.8	9.5432	0.0035

Table E.3. Auxiliary Measurement Data for the Example in Section 6.3.

Wavelength (nm)	$y_{x_r}(\lambda) - y_{x_{rd}}(\lambda)$ (Counts)	$y_{x_r}(\lambda) - y_{x_{rd}}(\lambda)$ (Std Uncert)	$y_x(\lambda) - y_{x_r}(\lambda)$ (Counts)	$y_x(\lambda) - y_{x_r}(\lambda)$ (Std Uncert)
360.05	319.4	1.0	362.4	1.1
370.41	535.9	1.4	609.7	1.6
380.76	835.1	1.9	948.0	2.2
391.11	1,225.0	2.4	1,391.9	2.7
401.46	1,665.4	3.0	1,892.7	3.4
411.73	2,181.9	3.7	2,480.1	4.2
422.00	2,636.3	4.3	2,996.5	4.9
432.27	3,034.9	4.8	3,449.4	5.5
442.47	3,357.3	5.3	3,817.5	6.1
452.68	3,763.4	5.8	4,278.2	6.7
462.88	4,121.2	6.3	4,684.8	7.2
473.05	4,375.9	6.6	4,974.1	7.5
483.21	4,598.3	6.9	5,227.5	7.8
493.37	4,772.1	7.2	5,424.6	8.1
503.53	5,026.8	7.6	5,713.6	8.6
513.64	5,288.2	8.0	6,011.4	9.1
523.75	5,443.5	8.4	6,188.5	9.5
533.86	5,551.3	8.7	6,309.4	9.9
543.97	5,626.0	8.9	6,393.8	10.2
554.00	5,658.2	9.0	6,433.1	10.3
564.03	5,637.3	9.0	6,409.1	10.2
574.06	5,668.6	9.0	6,442.8	10.3
584.09	5,746.6	9.1	6,531.8	10.4
594.13	5,638.1	9.0	6,409.1	10.2
604.13	5,562.3	8.8	6,321.3	10.0
614.13	5,503.2	8.7	6,255.8	9.9
624.09	5,426.1	8.6	6,168.0	9.8
634.05	5,335.3	8.4	6,064.8	9.6
643.95	5,209.1	8.2	5,922.9	9.4
653.85	4,743.4	7.5	5,390.8	8.5
663.70	4,360.8	6.9	4,957.5	7.8
673.55	4,308.1	6.8	4,896.8	7.7
683.41	4,263.3	6.7	4,845.8	7.7
693.26	4,299.7	6.8	4,887.9	7.7
703.11	4,231.6	6.7	4,810.3	7.6
712.97	4,153.5	6.6	4,722.7	7.5
722.82	3,767.0	6.0	4,282.2	6.8
732.68	3,489.5	5.6	3,965.4	6.3
742.46	3,405.9	5.5	3,871.1	6.2
752.23	3,218.6	5.2	3,659.5	5.9
762.01	2,843.8	4.6	3,231.3	5.3
771.73	2,781.3	4.6	3,162.2	5.2
781.45	2,633.7	4.4	2,993.7	5.0
791.17	2,366.6	4.0	2,690.9	4.5
801.04	2,136.7	3.6	2,429.2	4.1
810.92	1,933.3	3.3	2,199.2	3.8
820.47	1,751.3	3.0	1,989.1	3.4
830.02	1,602.5	2.8	1,821.7	3.2

ACKNOWLEDGEMENTS

The Testing Procedures Committee would like to thank Heather Waugh and Bailey Shawbitz for their significant contributions to the development of this document.

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